

Image Processing and Video Analysis

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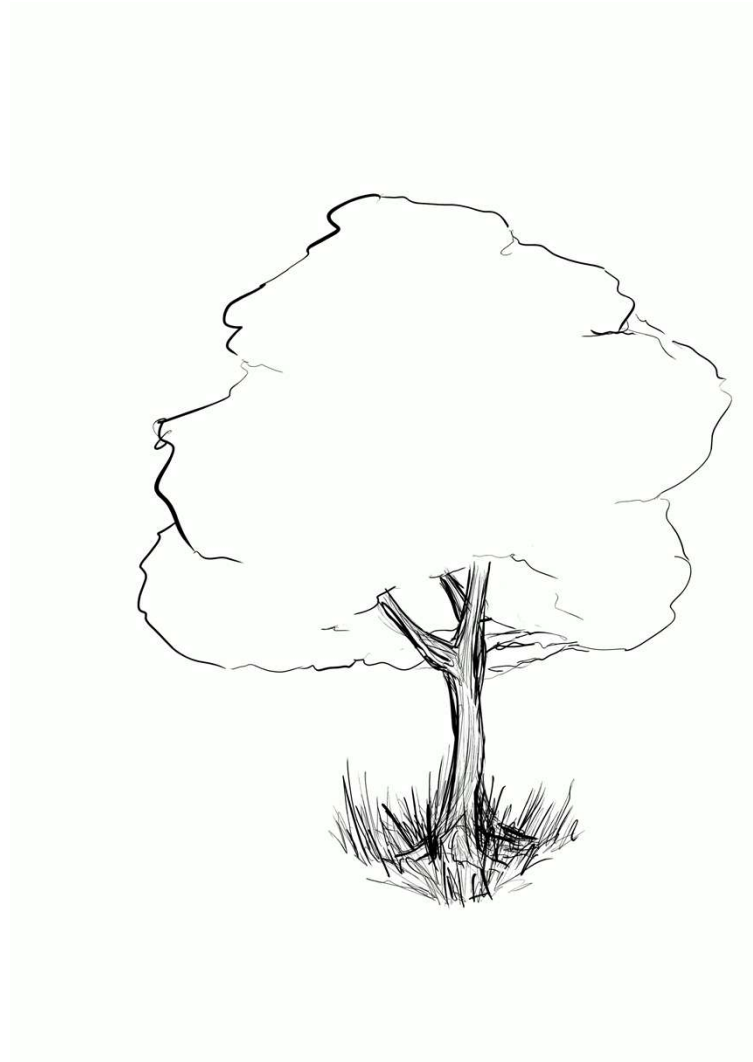
International Institute of Information Technology Bhubaneswar

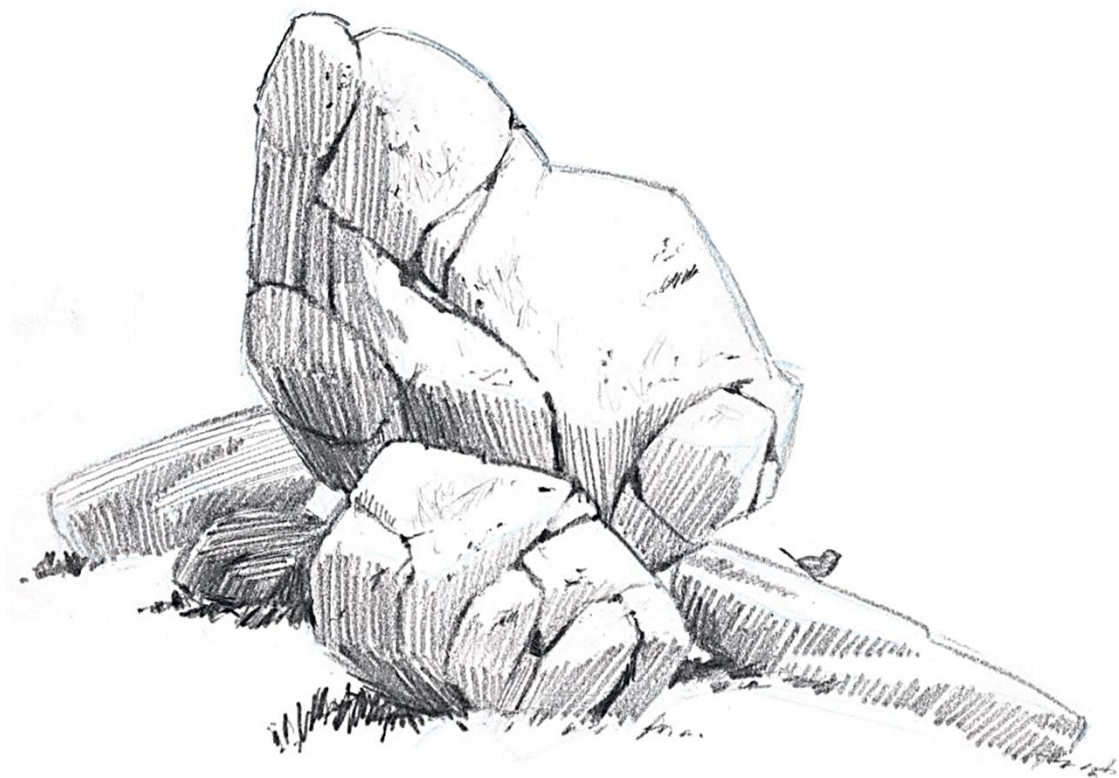
(IIIT Bhubaneswar)

deepakkumar@iiit-bh.ac.in

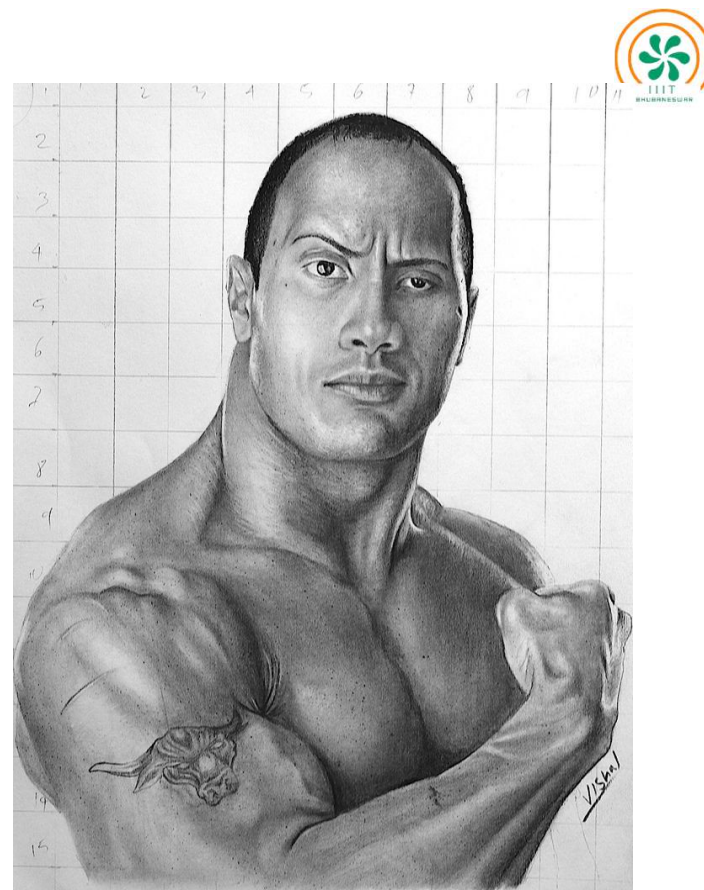
deepakrout@ieee.org

Image... Processing





HUMAN



ROCK

"A picture is worth a thousand words"



Introduction: Image

- An **image** is a **visual representation** of something.
- It can depict real objects, people, places, ideas, or abstract patterns, and it is made up of visual elements such as colors, shapes, lines, textures, etc.
- For example:
 - A photo you take with your phone is an image.
 - A company logo is an image.
 - A medical X-ray is an image.
 - A satellite picture of Earth is an image.

Introduction: Image

Digital images generally fall into two main types:

- **Raster images** consist of pixels and can lose quality when enlarged.
- **Vector images** are made from mathematical shapes and can be scaled to any size without losing quality.

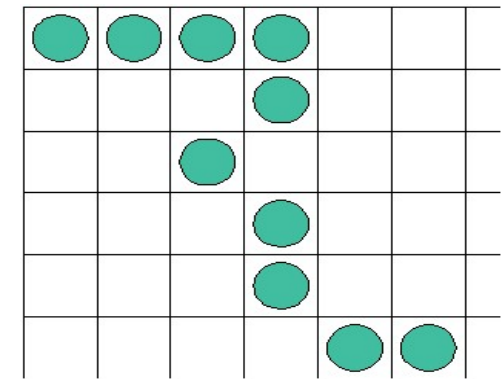
An image is any visual representation of information, whether created by a camera, drawn by an artist, or generated by a computer.

Example of Raster and Vector Image

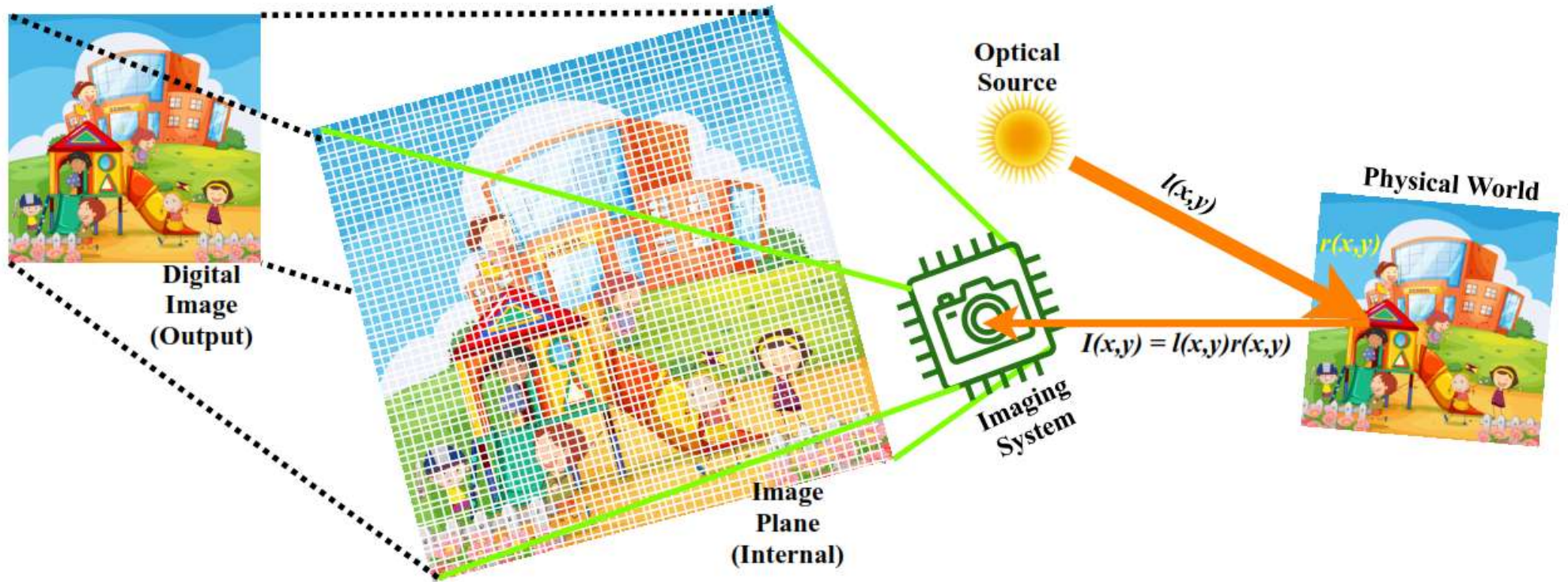


Introduction: Image

- **Image:** A **visual representation** or depiction of an object, scene, or phenomenon.
 - **Digital Image:** It is a two-dimensional array of pixels (picture elements), where each pixel corresponds to a specific location in the image and carries information about the color and intensity of that location.
- A representation of the original image by a discrete set of data points
- Each of these data points is called a picture element, commonly referred to as a “pixel”



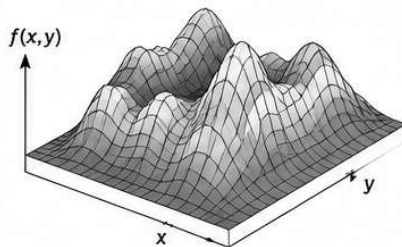
Magnified
Fragment of an
Image



SAMPLING AND QUANTIZATION OF AN IMAGE

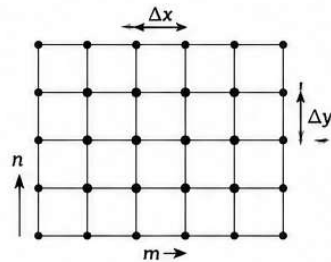


1. CONTINUOUS IMAGE



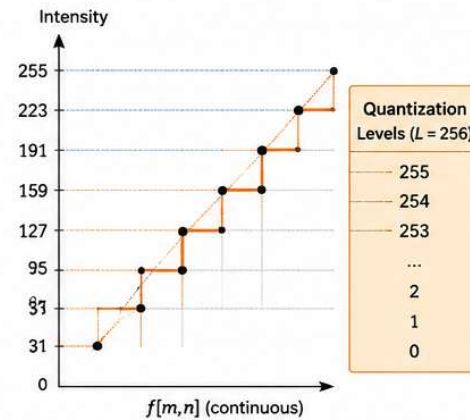
A real-world image is continuous in both space (x,y) and intensity.

2. SAMPLING (SPATIAL)

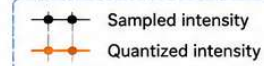


The image is sampled at intervals Δx and Δy to obtain a discrete set of points $f[m,n]$ (pixels).

3. QUANTIZATION (INTENSITY)



Each sampled intensity is mapped to one of L discrete levels.
(Here $L = 256$ for 8-bit image)



4. DIGITAL IMAGE (OUTPUT)



52	80	112	80	59	...
55	87	120	101	62	...
61	98	135	112	68	...
57	90	122	103	59	...
...	...	...	...	...	...

The result is a digital image $g[m,n]$ with discrete spatial locations and discrete intensity levels.

KEY POINTS

- **Sampling** → converts continuous coordinates (x,y) to discrete coordinates (m,n) .
- **Quantization** → converts continuous intensity to one of L discrete levels.
- **More samples** (smaller $\Delta x, \Delta y$) → higher spatial resolution.
- **More levels** (larger L or more bits) → higher intensity resolution.

EXAMPLE

- 8-bit image → $L = 2^8 = 256$ levels (0 to 255)
- 512×512 sampling → 262,144 pixels
- Each pixel needs 8 bits → total size = $512 \times 512 \times 8$ bits = 262,144 bytes (≈ 256 KB)

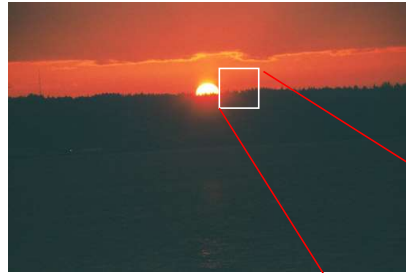
FORMULA

Sampling: $f(x,y) \xrightarrow{\text{sampling}} f[m,n] = f(m\Delta x, n\Delta y)$

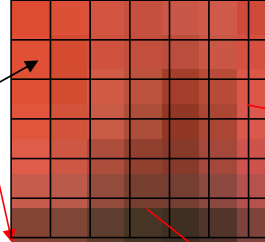
Quantization: $f[m,n] \xrightarrow{Q} g[m,n]$

Digital image: $g[m,n] = Q\{f(m\Delta x, n\Delta y)\}$

Digital Image



Digital image = a multidimensional array of numbers (such as intensity image) or vectors (such as color image)



Each component in the image called pixel associates with the pixel value (a single number in the case of intensity images or a vector in the case of color images).

10	10	16	28		
9	65	70	56	43	
15	32	99	70	56	78
32	21	60	90	96	67
	54	85	85	43	92
		32	65	87	99

Image Formation Model

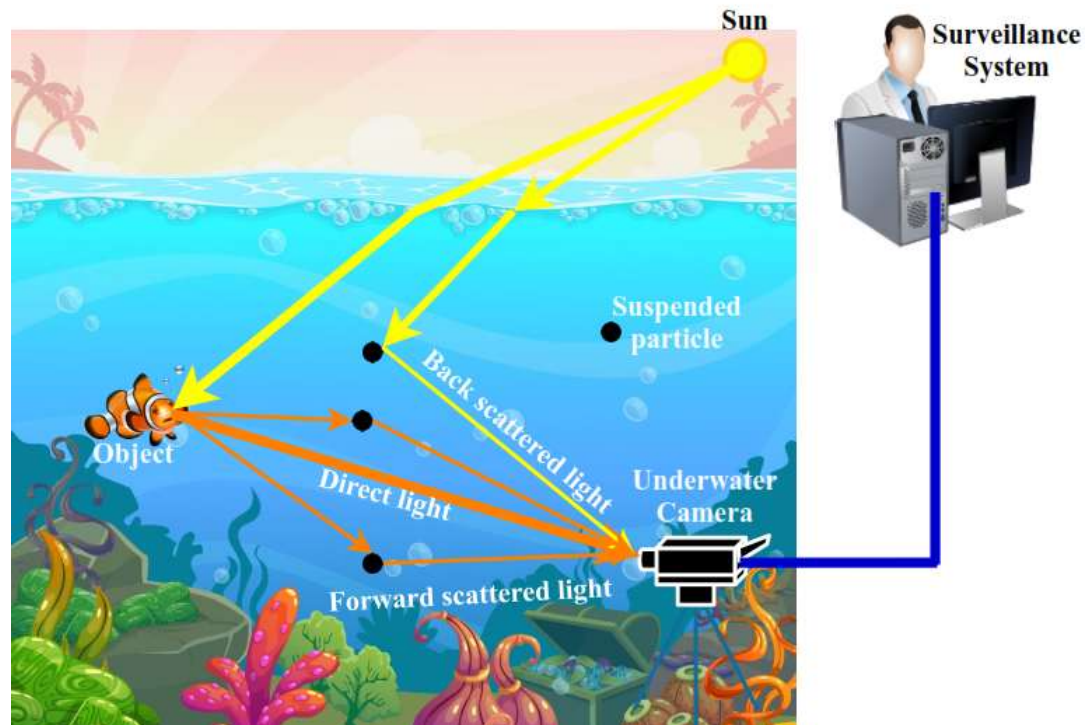
$$I_c(x, y) = J_c(x, y)t(x, y) + (1 - t(x, y))b_c$$

Acquired Image

Clear Image

Transmission Map

Surrounding Light



Example: Real World Issue



Digital image processing

- Digital image processing is the study of representation and manipulation of pictorial information by a computer.
- Improve pictorial information for better clarity (human interpretation). Automatic machine processing of scene data (interpretation by a machine/non-human, storage, transmission)

Image processing vs. computer vision

A useful distinction is:

Image Processing → Improve/transform the image

Computer Vision → Understand the image

For example, removing noise from an X-ray is **image processing**, whereas identifying a tumor from that X-ray is a **computer vision/image analysis** task.

Applications

- **Medical Imaging:** Techniques like X-rays, MRIs, CT scans, and ultrasounds rely on image processing for visualization, diagnosis, and treatment planning. Image enhancement, segmentation, and 3D reconstruction help medical professionals identify anomalies and make informed decisions.
- **Remote Sensing:** Satellite and aerial imagery aid in environmental monitoring, disaster assessment, and urban planning. Image processing allows the extraction of valuable data, such as land cover classification, vegetation health, and changes over time.
- **Computer Vision:** In fields like robotics and automation, computer vision enables machines to interpret and understand visual data. Applications include object detection, facial recognition, gesture recognition, and autonomous vehicles.
- **Entertainment and Gaming:** Special effects in movies, video games, and virtual reality are heavily reliant on image processing. Techniques like green screen compositing, motion capture, and 3D rendering enhance visual experiences.

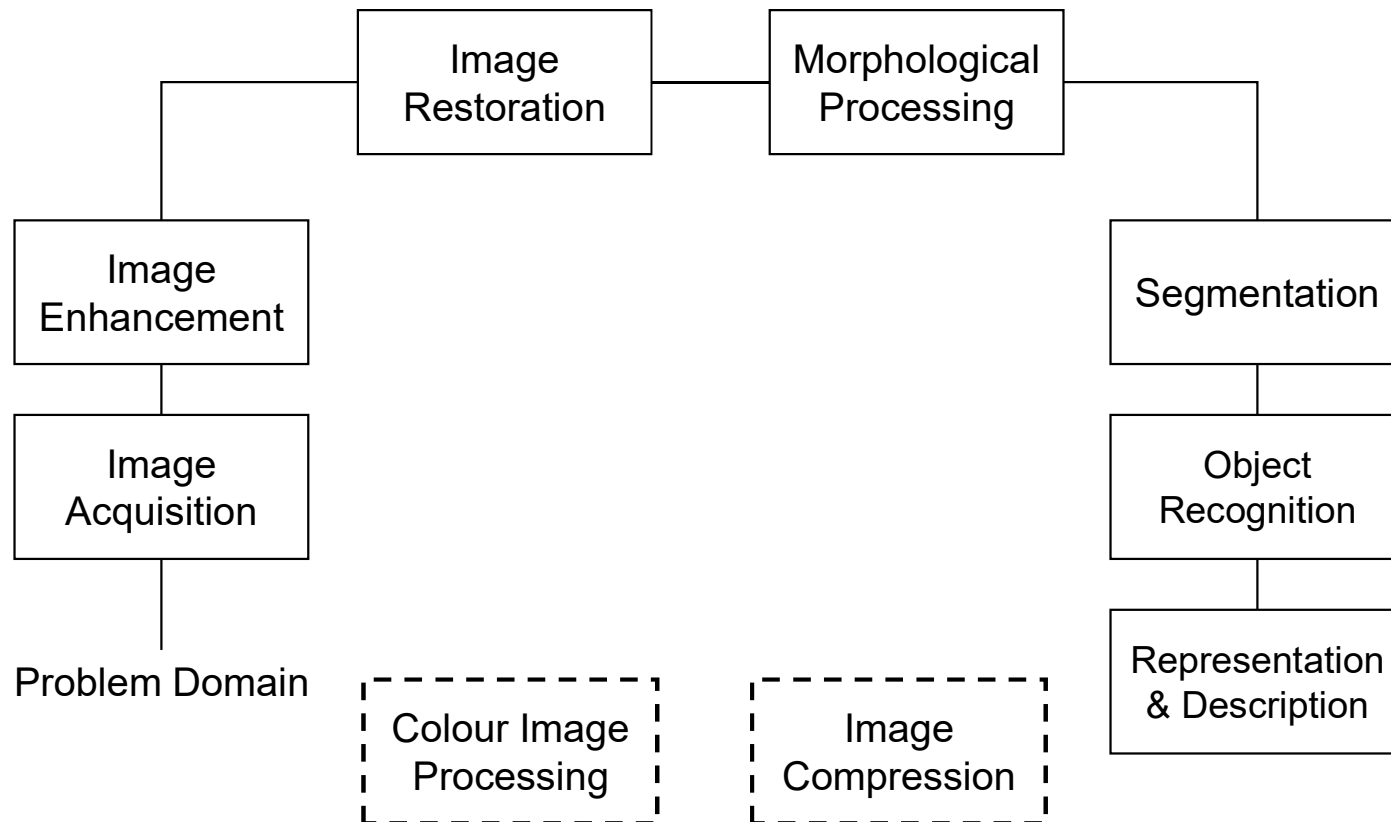
Applications

- **Biometrics:** Image processing plays a crucial role in biometric systems for personal identification, such as fingerprint recognition, iris scanning, and voice analysis.
- **Artificial Intelligence and Machine Learning:** Image analysis is integral to training and deploying machine learning models, particularly in image classification, object detection, and image generation tasks.
- **Security and Surveillance:** Image processing aids in video surveillance, detecting suspicious activities, and identifying intruders through motion detection and facial recognition.
- **Quality Control:** Industries like manufacturing and agriculture use image processing for quality assessment and defect detection in products and produce.
- **Archaeology and Cultural Heritage:** Image processing helps researchers analyze and restore old artworks, artifacts, and manuscripts, preserving cultural heritage.

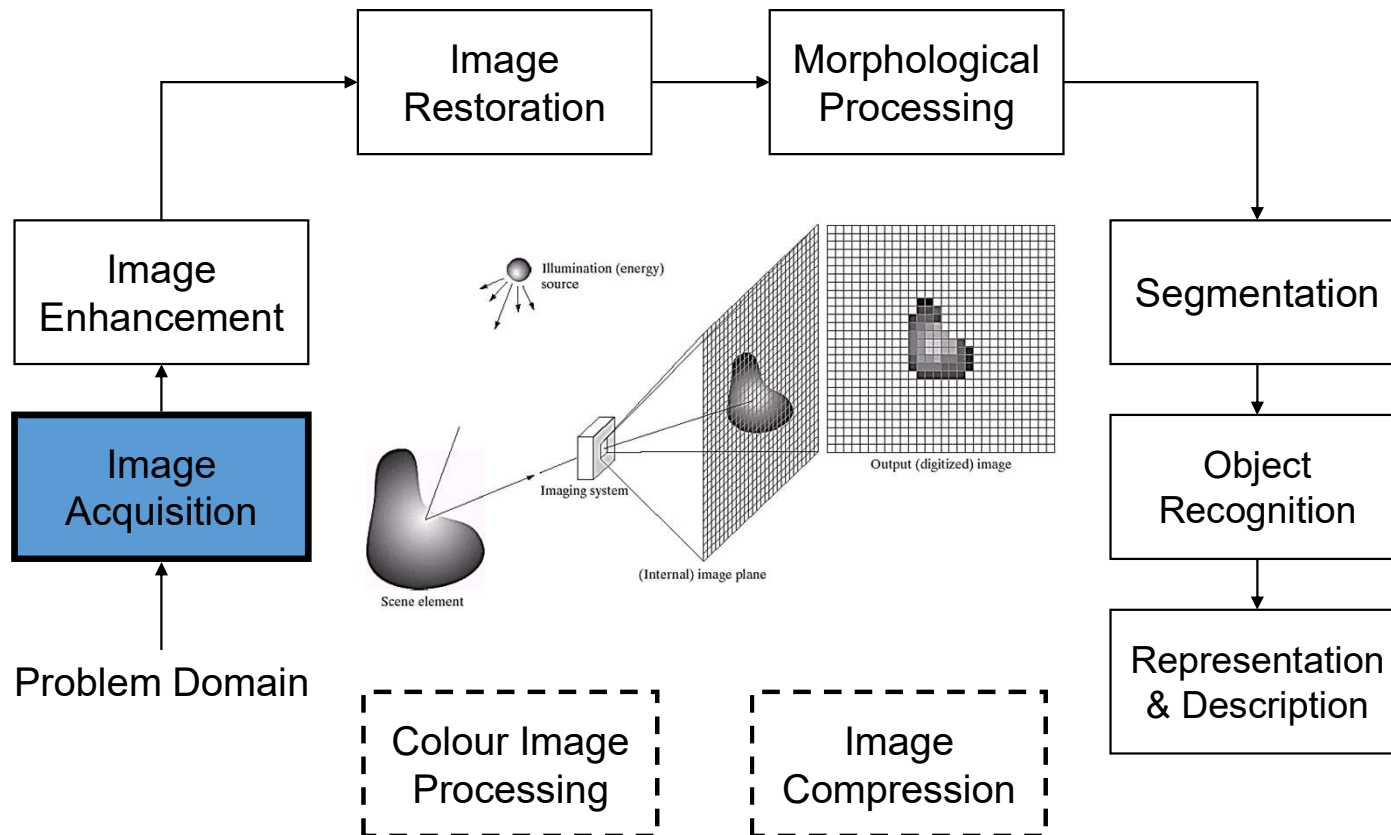
Applications

- **Geographical Information Systems (GIS):** Satellite and aerial images are used to create maps, monitor land use changes, and manage natural resources.
- **Forensics:** Image processing assists in criminal investigations by enhancing and analyzing surveillance footage, fingerprints, and facial features.
- **Medicine and Biology:** Microscopic images from biological studies benefit from techniques like cell counting, tissue analysis, and tracking of cellular movements.
- **Document Analysis:** Handwriting recognition, document classification, and text extraction from images facilitate digitization and automation in document management.
- **Meteorology:** Image processing is used to analyze and interpret satellite and radar images for weather forecasting and climate research.
- These applications highlight the immense versatility of digital image processing, making it an essential tool across various industries, sciences, and technologies.

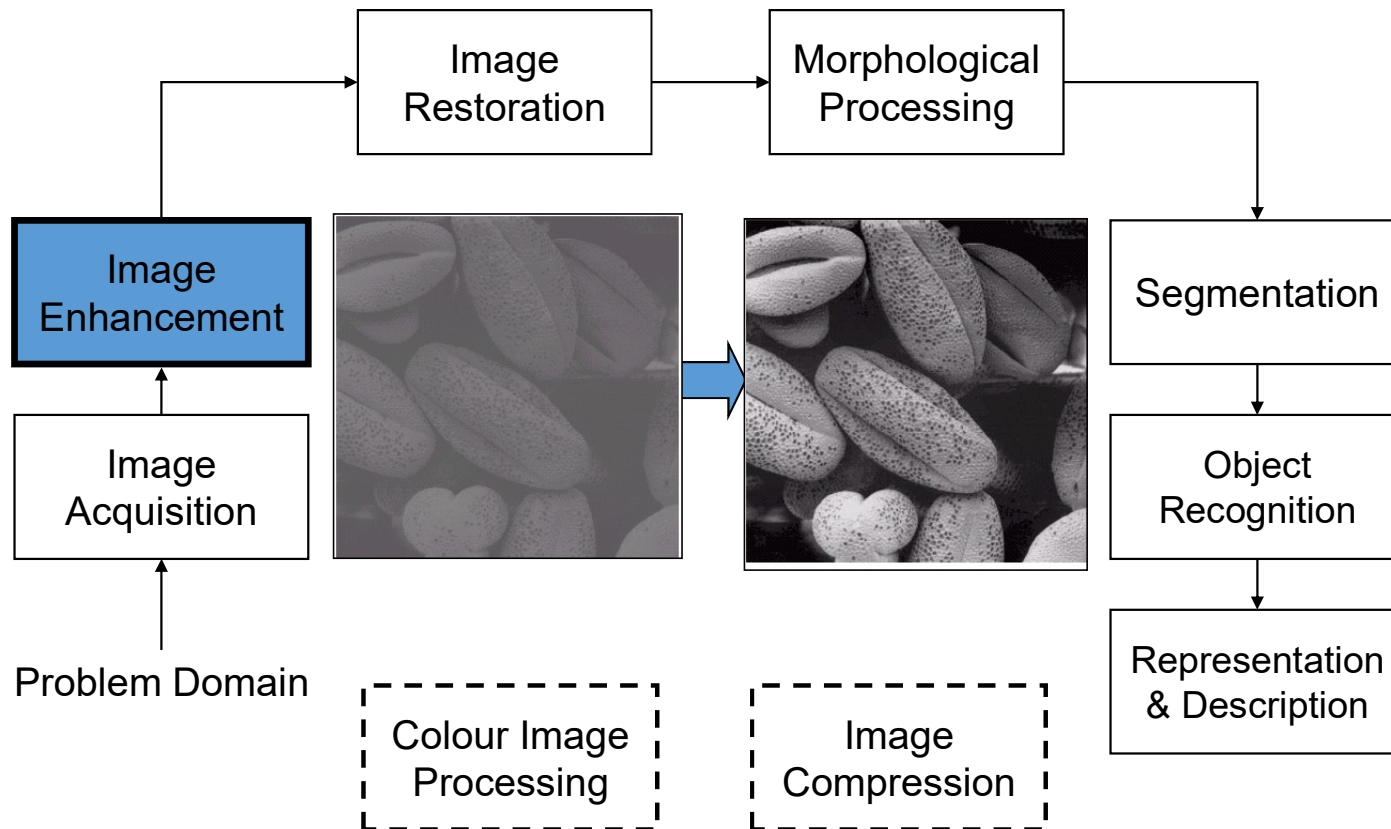
Key Stages in Digital Image Processing



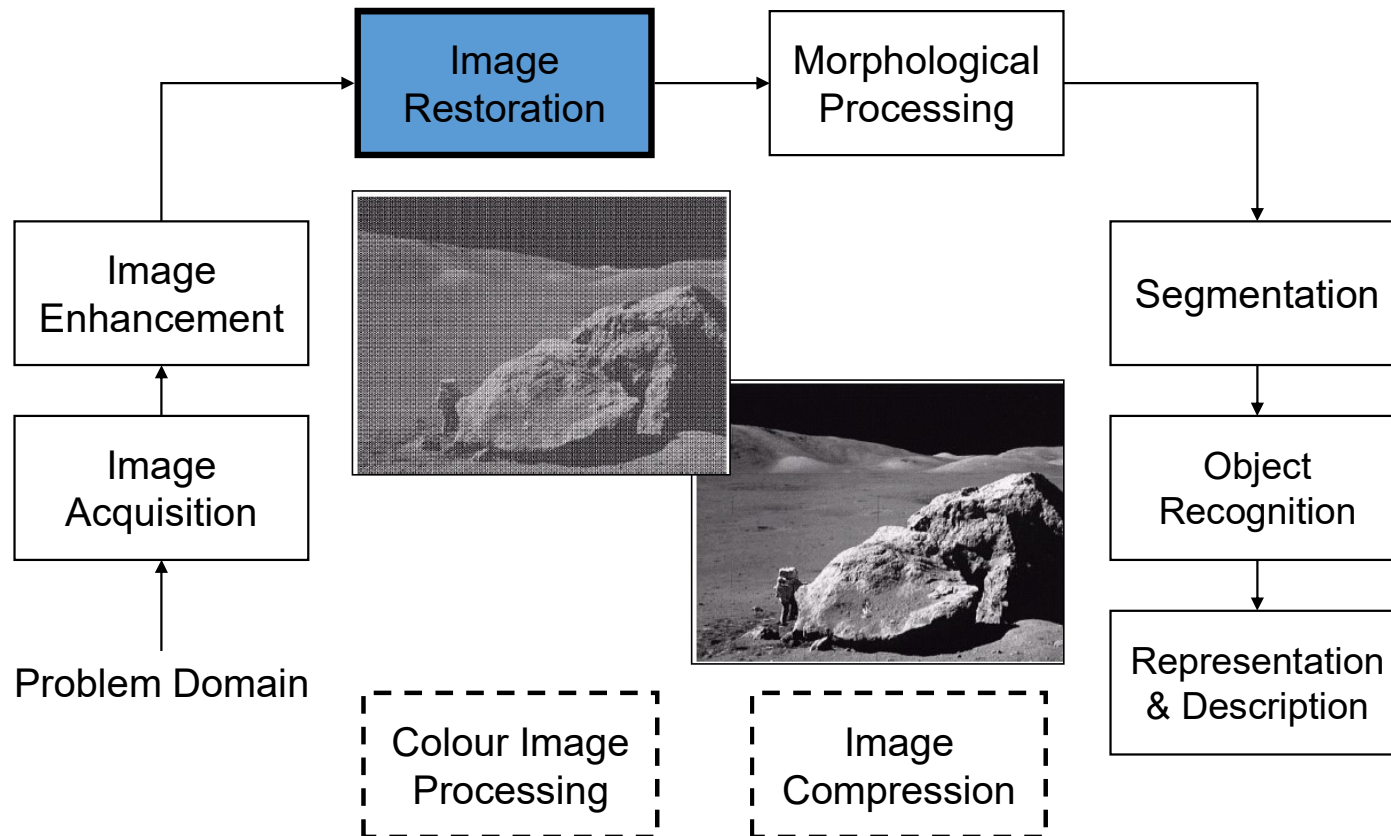
Key Stages in Digital Image Processing: Image Acquisition



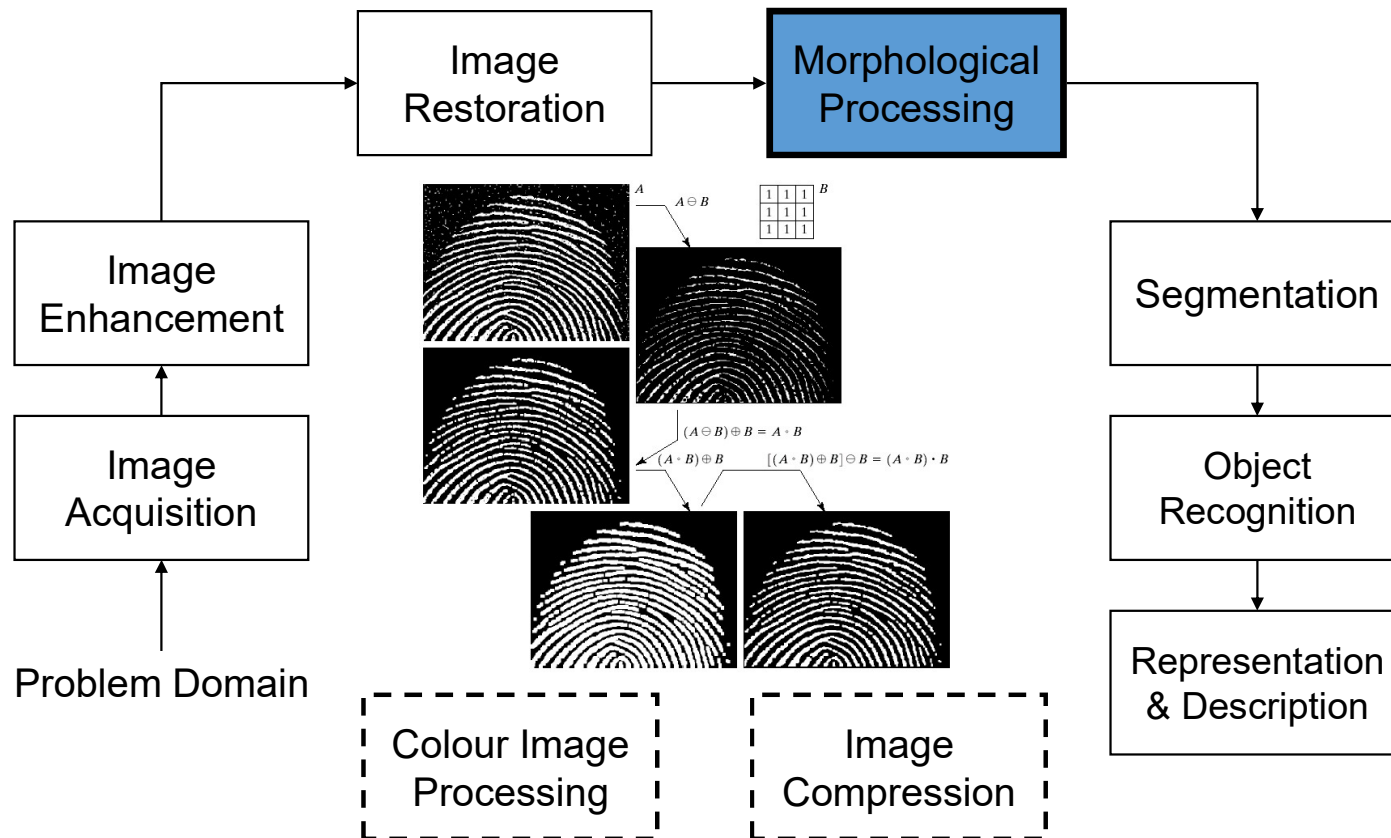
Key Stages in Digital Image Processing: Image Enhancement



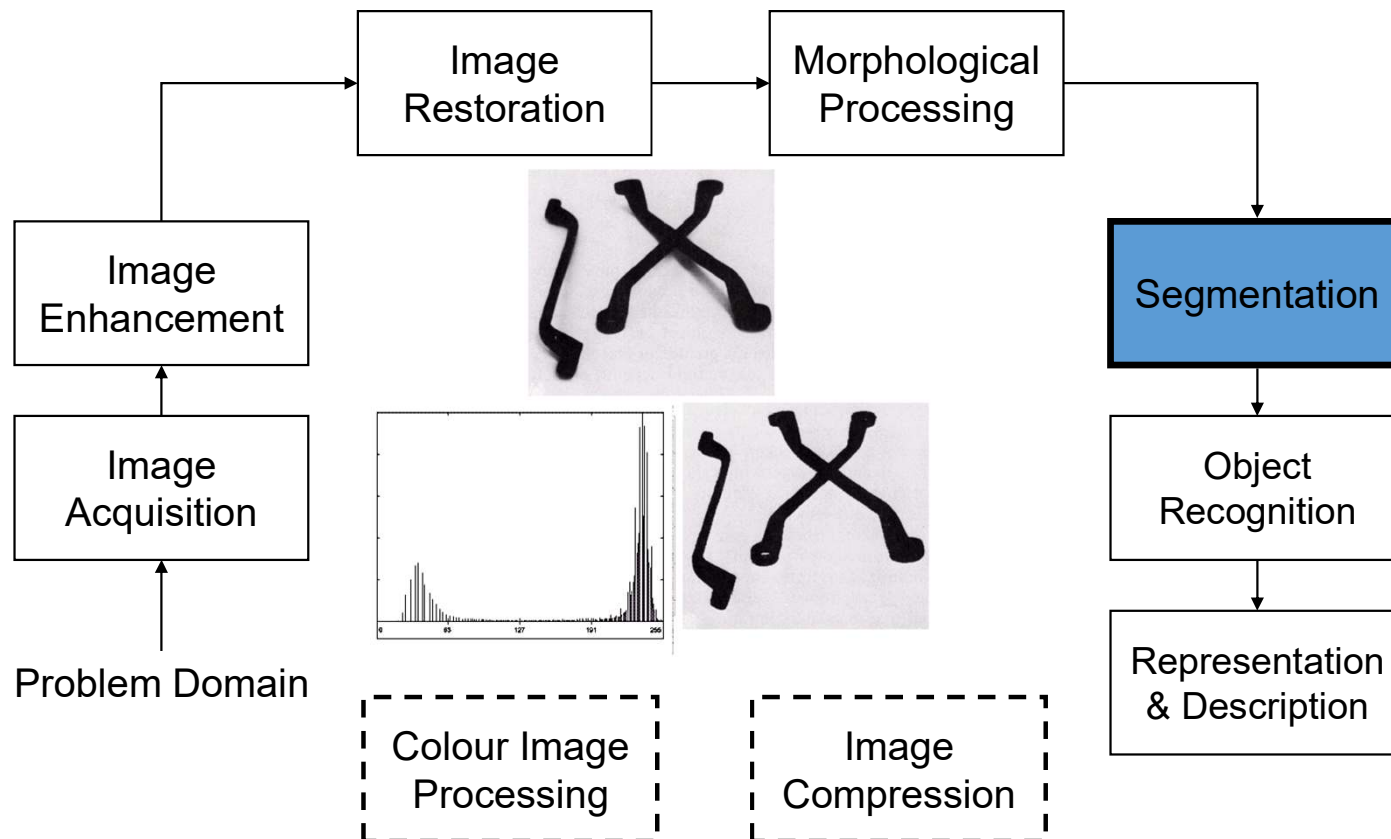
Key Stages in Digital Image Processing: Image Restoration



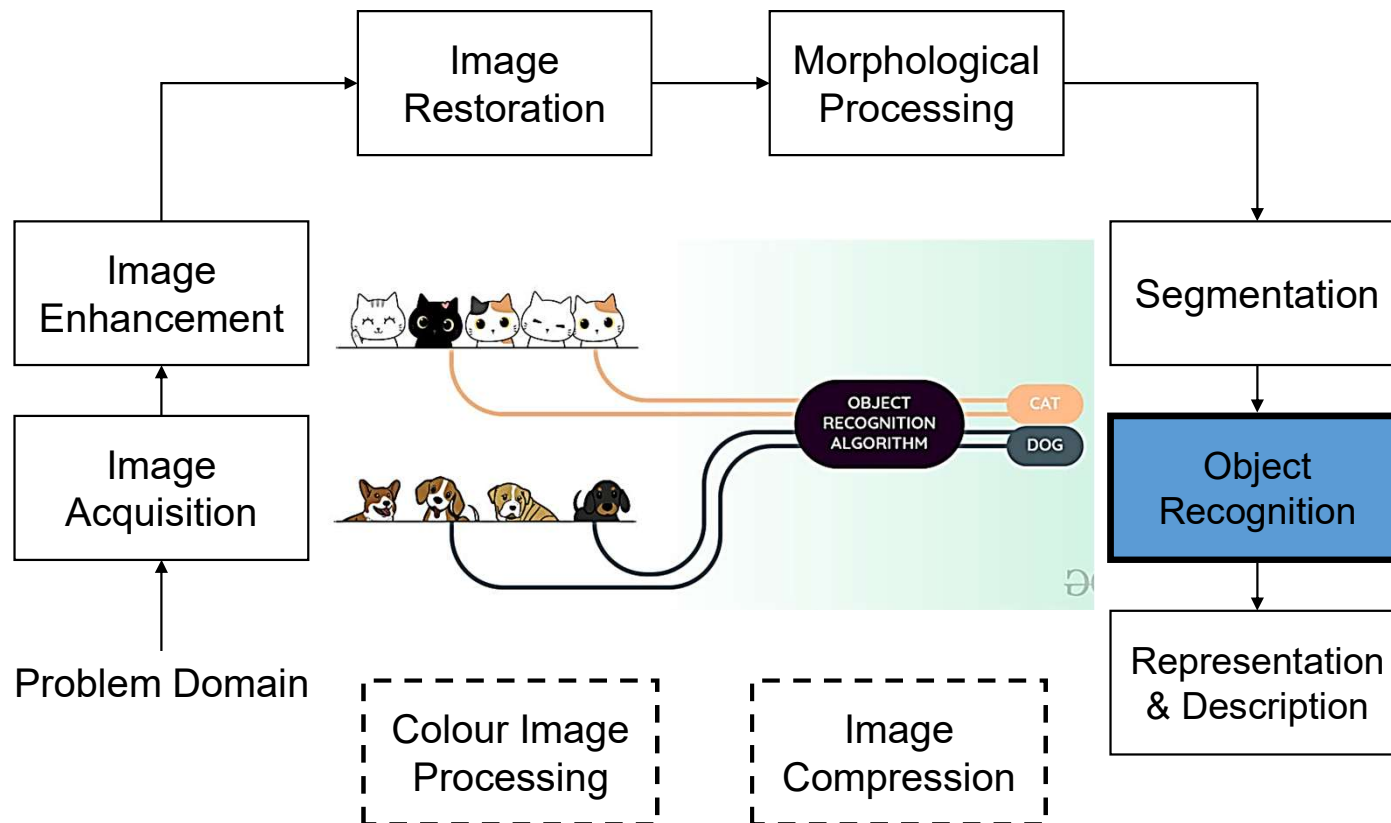
Key Stages in Digital Image Processing: Morphological Processing



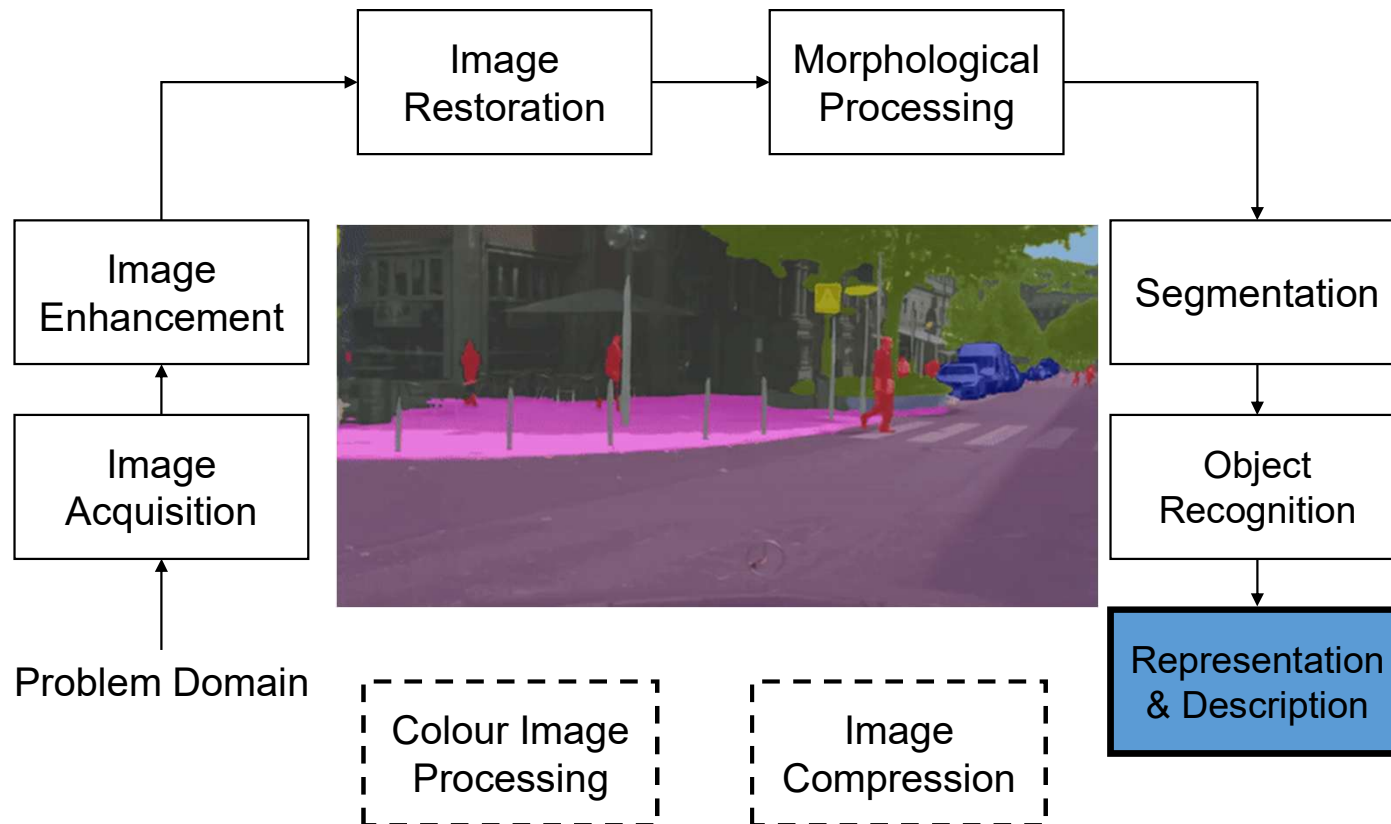
Key Stages in Digital Image Processing: Segmentation



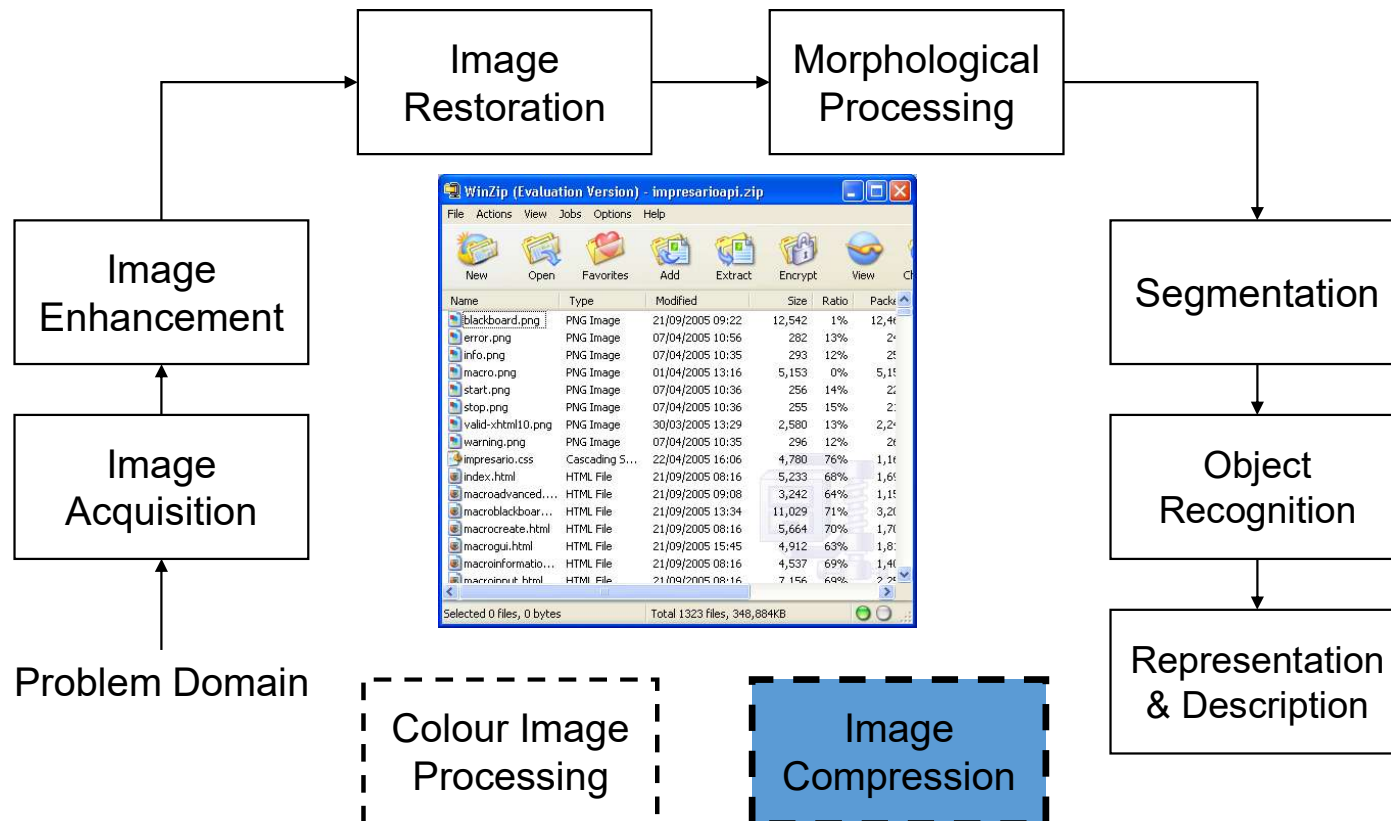
Key Stages in Digital Image Processing: Object Recognition



Key Stages in Digital Image Processing: Representation & Description



Key Stages in Digital Image Processing: Image Compression



Key Stages in Digital Image Processing: Colour Image Processing

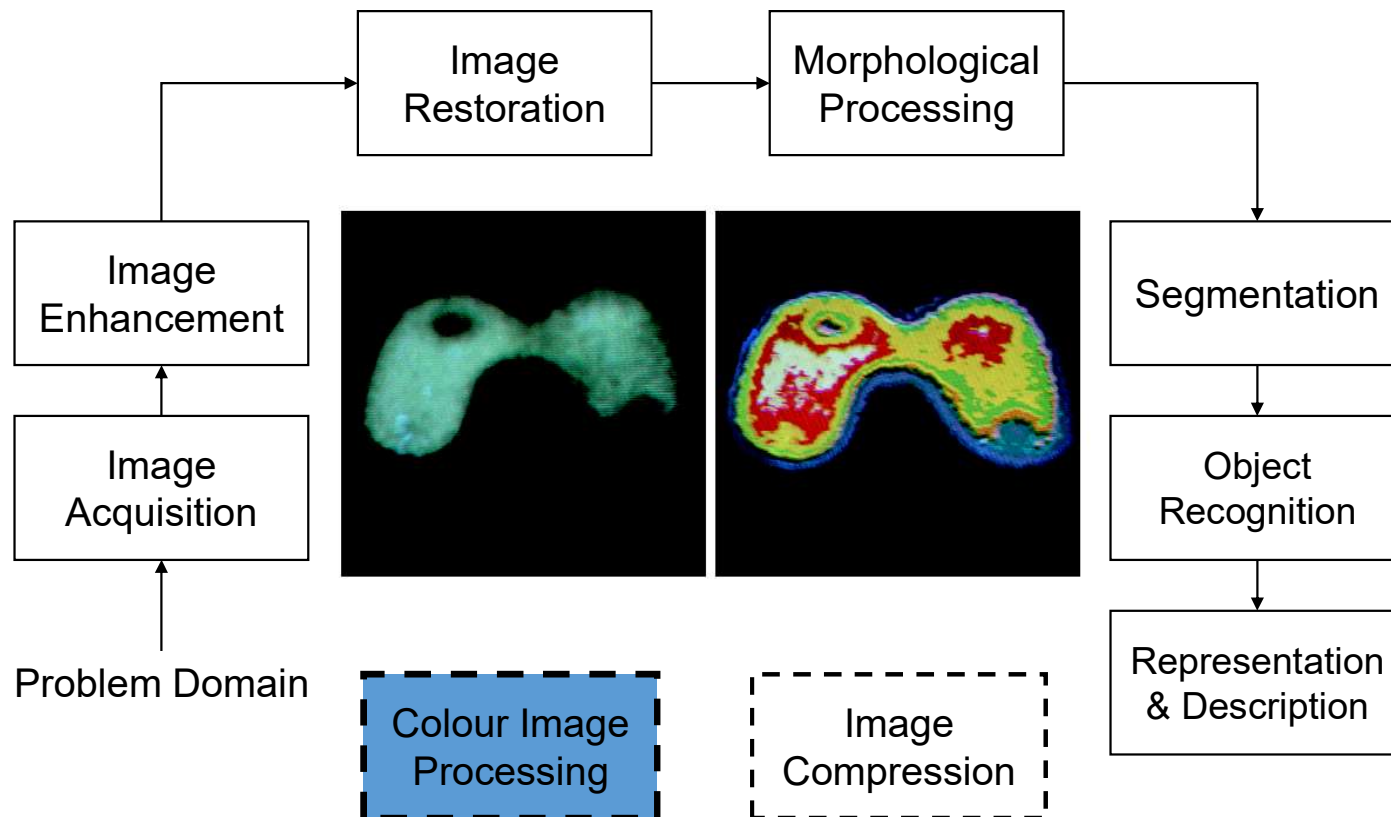


Image Enhancement

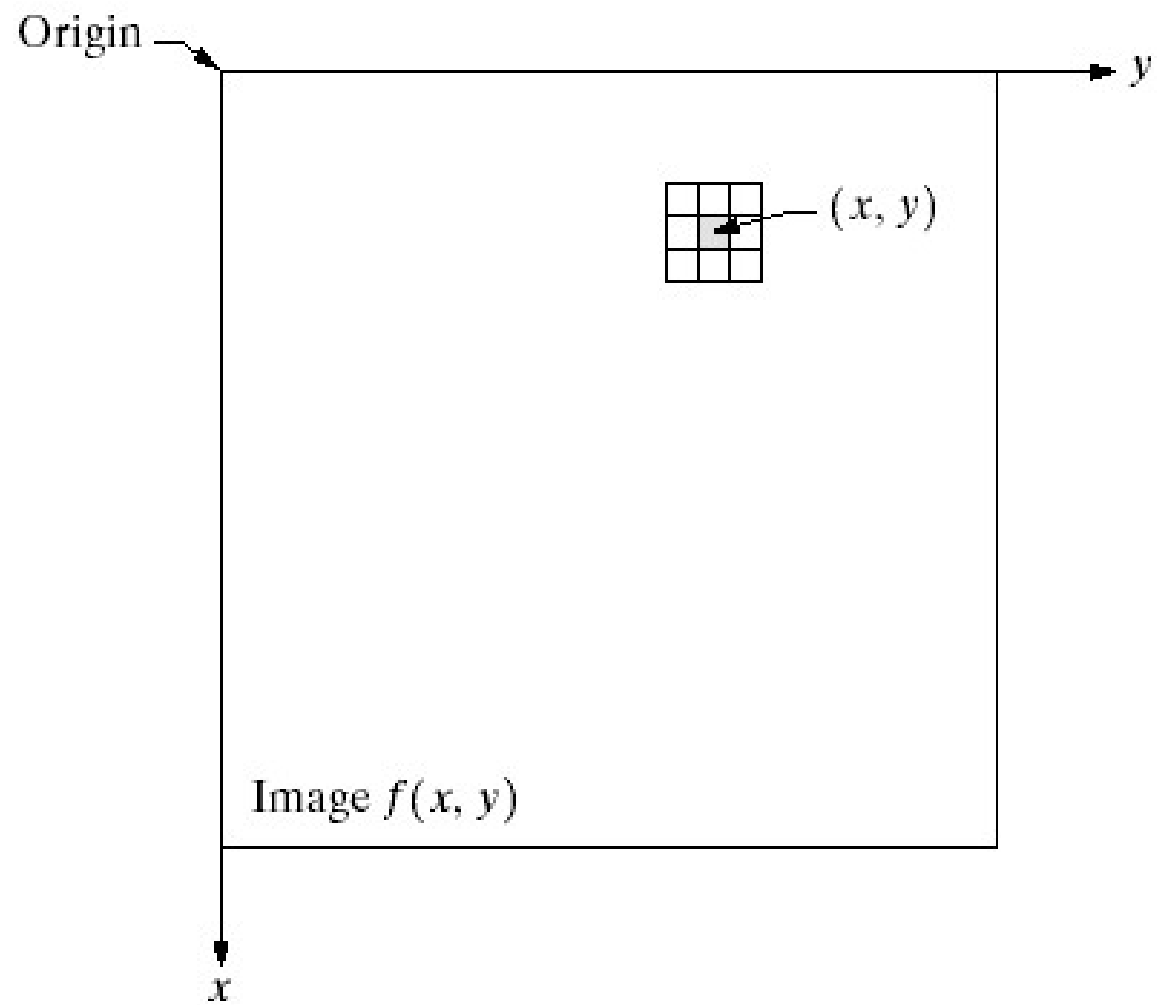
- To process an image so that the result is more suitable than the original image for a *specific* application.
- Spatial domain methods and frequency domain methods.

Spatial Domain Methods

- Procedures that operate directly on the aggregate of pixels composing an image

$$g(x, y) = T[f(x, y)]$$

- A neighborhood about (x, y) is defined by using a square (or rectangular) sub-image area centered at (x, y) .



Spatial Domain Methods

- When the neighborhood is 1 x 1 then g depends only on the value of f at (x,y) and T becomes a **gray-level transformation** (or mapping) **function**:

$$s=T(r)$$

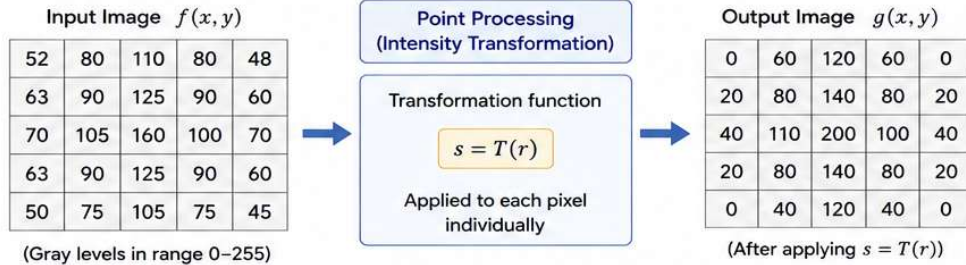
r,s : gray levels of $f(x,y)$ and $g(x,y)$ at (x,y)

- Point processing techniques (e.g. contrast stretching, thresholding)

POINT PROCESSING IN IMAGE

Point processing (intensity transformation) is an operation in which each output pixel value depends only on the corresponding input pixel value, **not on its neighbors**.

1. BASIC IDEA

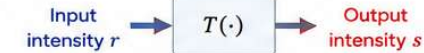


2. MATHEMATICAL MODEL

Let $f(x, y)$ be the input image and $g(x, y)$ be the output image. Point processing is defined as:

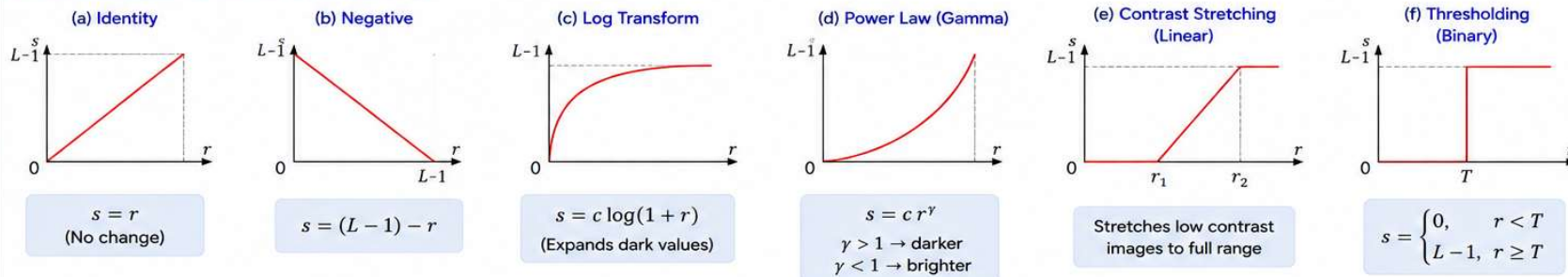
$$s = T(r) \quad \text{where } r = f(x, y) \quad \text{and} \quad s = g(x, y)$$

Thus, every output pixel s is obtained by applying the same transformation $T(\cdot)$ to the input pixel r .

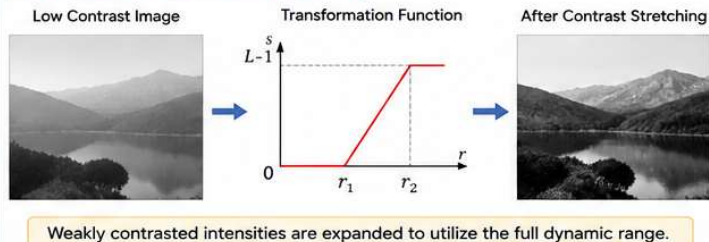


No neighborhood information is used. Only the value of the pixel itself matters.

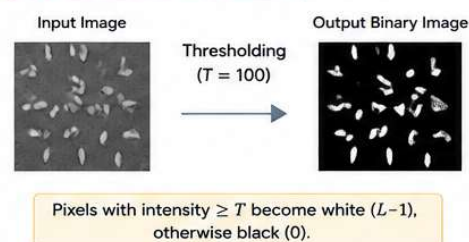
3. COMMON TRANSFORMATION FUNCTIONS



4. EXAMPLE: CONTRAST STRETCHING

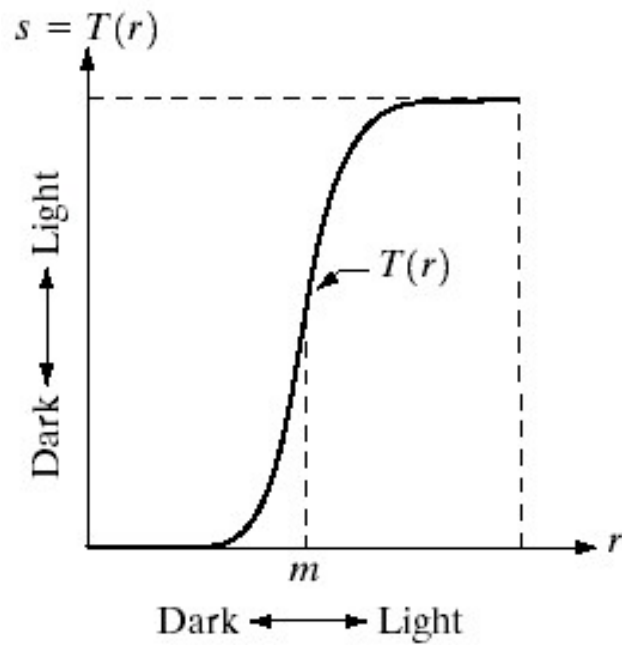


5. EXAMPLE: THRESHOLDING

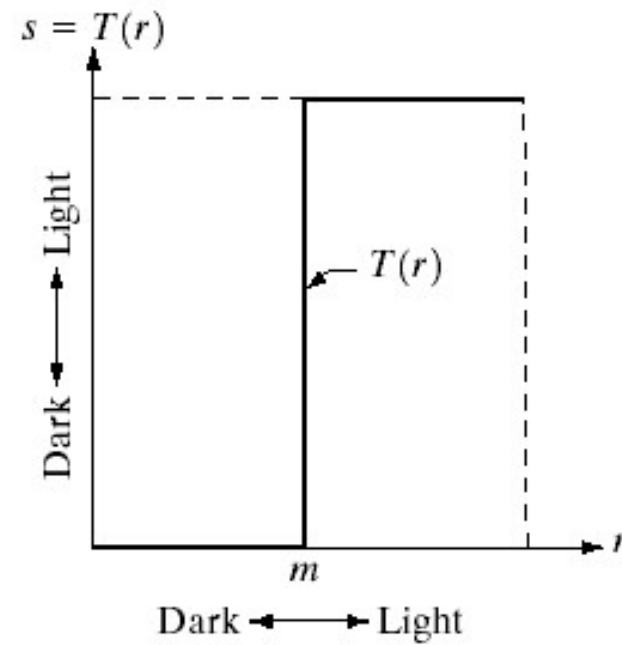


6. KEY POINTS

- Point processing transforms the intensity of each pixel independently.
- It is defined by a single-valued function $s = T(r)$.
- It does **NOT** use neighborhood information.
- Useful for enhancement: brightness adjustment, contrast stretching, thresholding, etc.
- Computationally simple and fast.



Contrast Stretching



Thresholding

Enhancement by Point Processing

- These are methods based only on the intensity of single pixels.
 - r denotes the pixel intensity **before** processing.
 - s denotes the pixel intensity **after** processing.

Some Simple Intensity Transformations

- Image negatives
- Piecewise-Linear Transformation Functions:
 - Contrast stretching
 - Gray-level slicing
 - Bit-plane slicing
- Implemented via Look-Up Tables (LUT) where values of T are stored in a 1-D array (for 8-bit, LUT will have 256 values)

Linear: Negative, Identity
 Logarithmic: Log, Inverse Log
 Power-Law: n th power, n th root

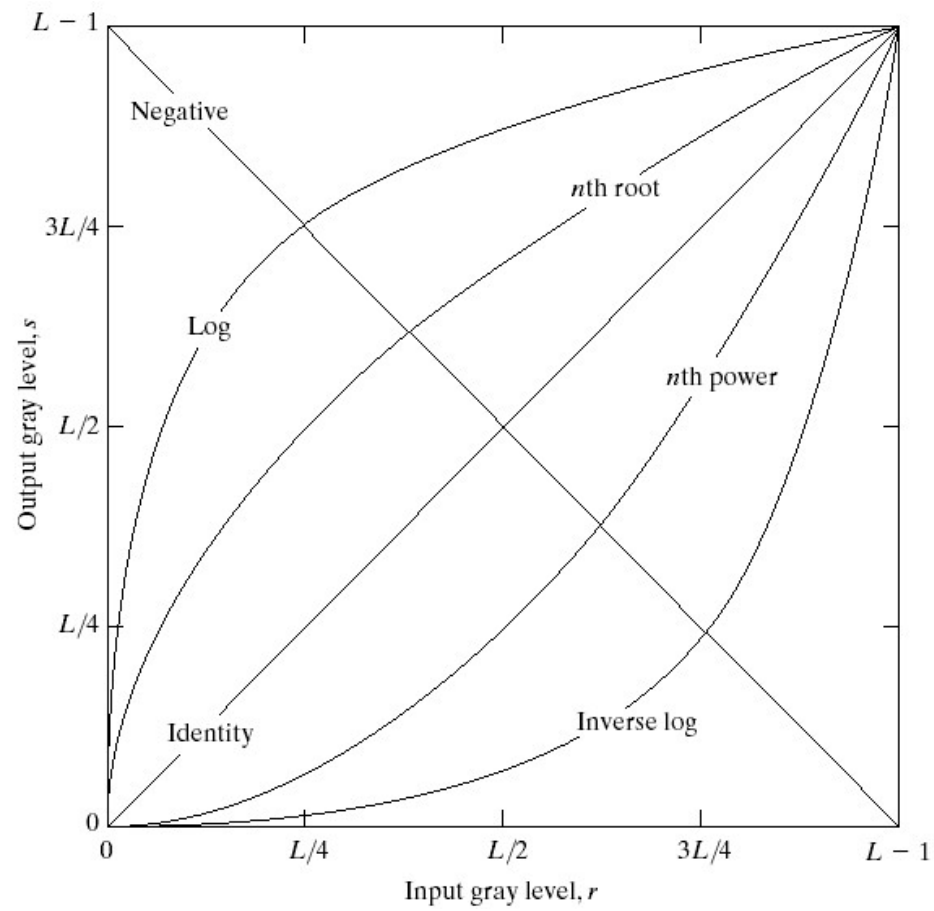
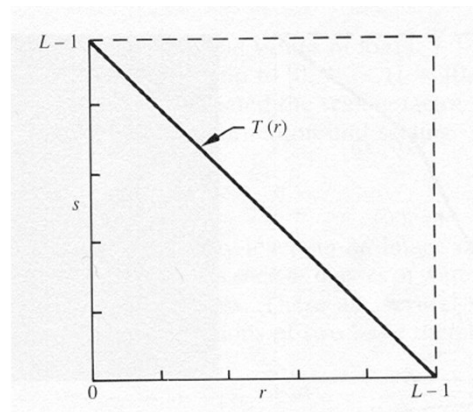


Image Negatives

- Are obtained by using the transformation function $s=T(r)$.



$[0, L-1]$ the range of gray levels
 $s = L-1-r$

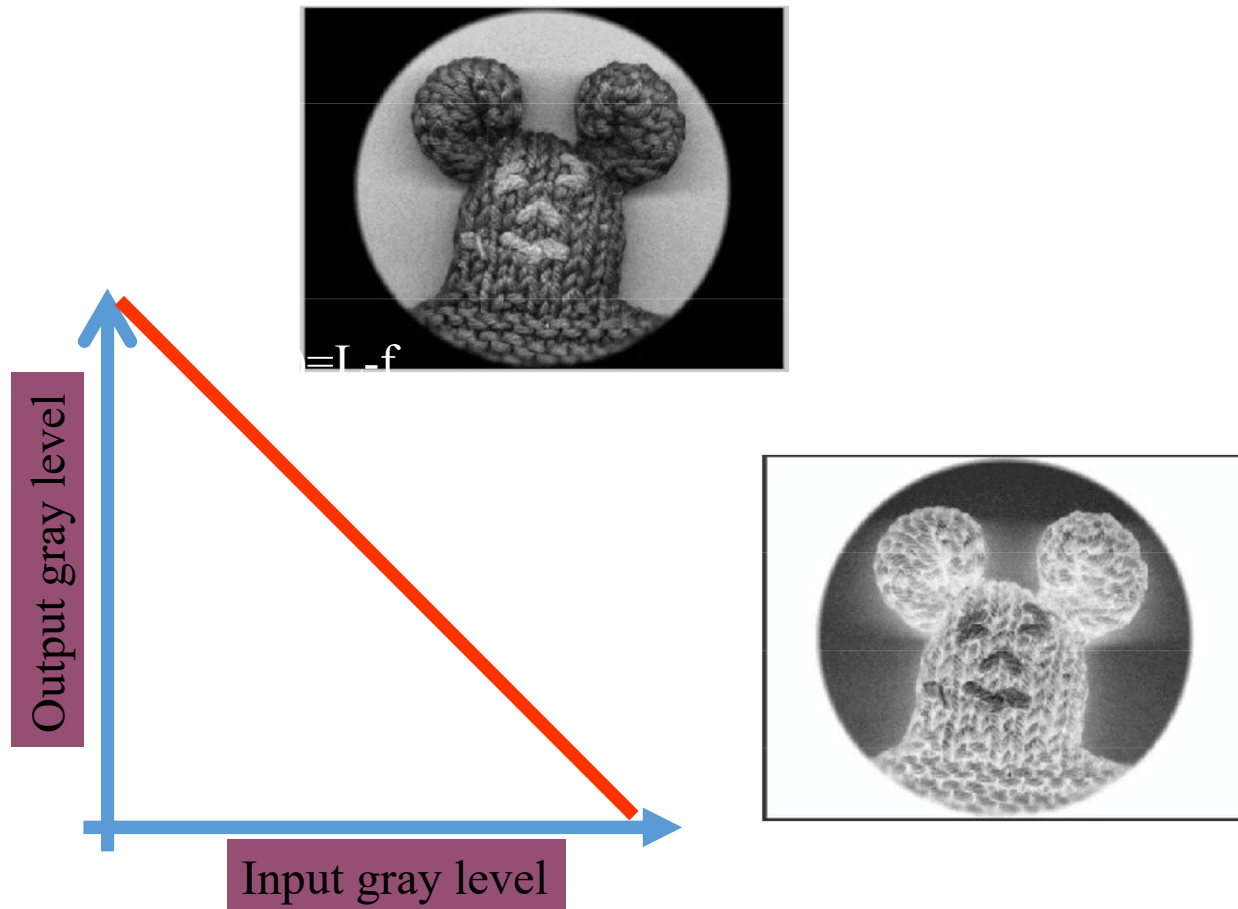


Image Negatives

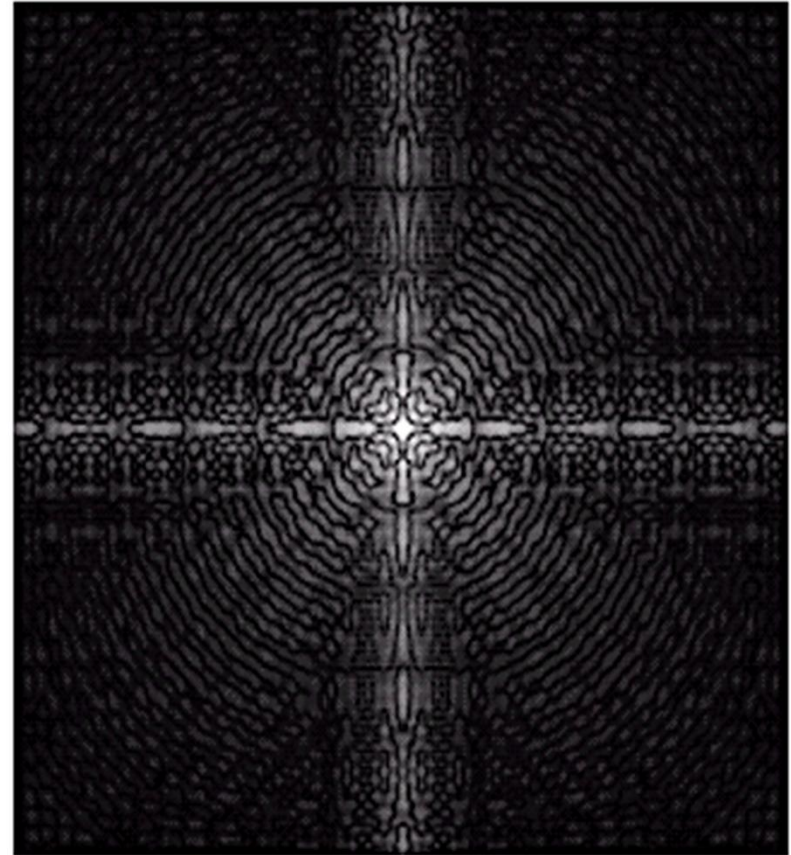
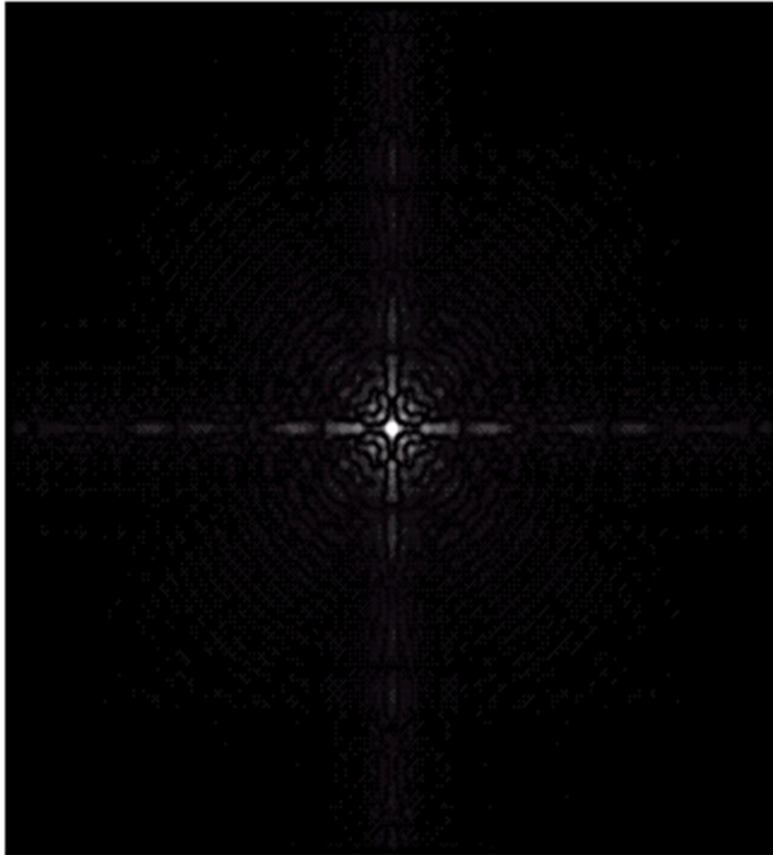
- Function reverses the order from black to white so that the intensity of the output image decreases as the intensity of the input increases.
- Used mainly in medical images and to produce slides of the screen.

Log Transformations

$$s = c \log(1+r)$$

c: constant

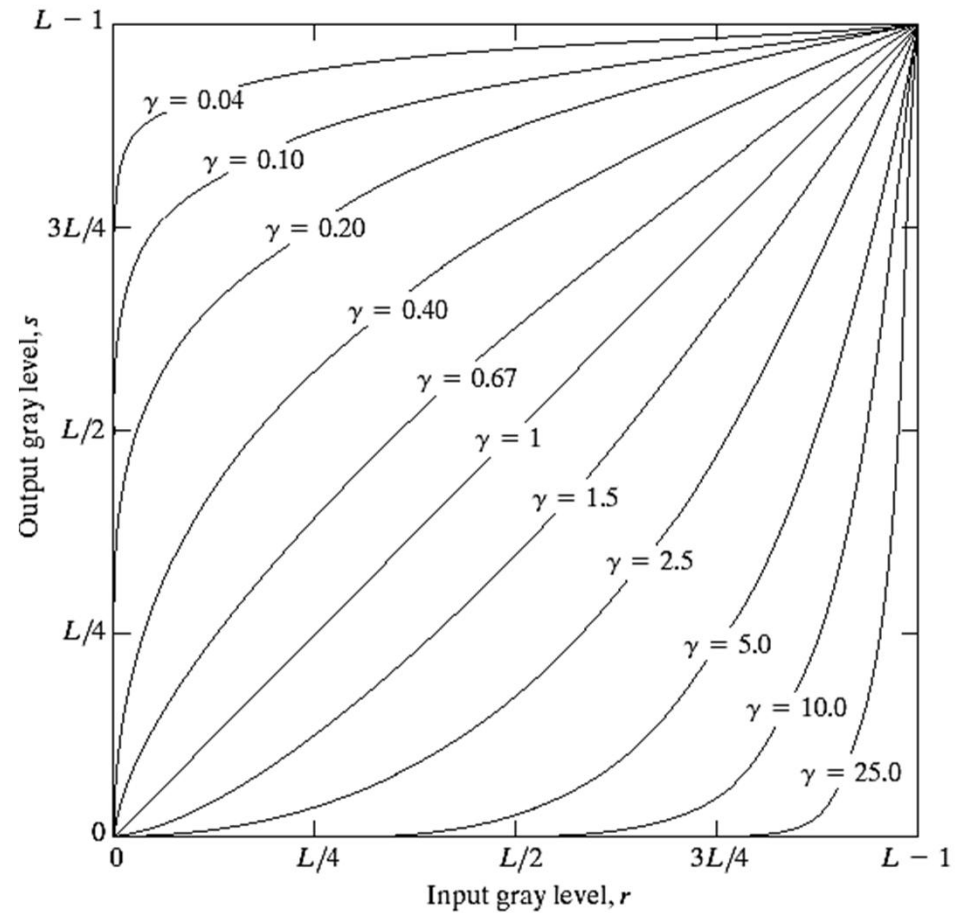
- Compresses the dynamic range of images with large variations in pixel values

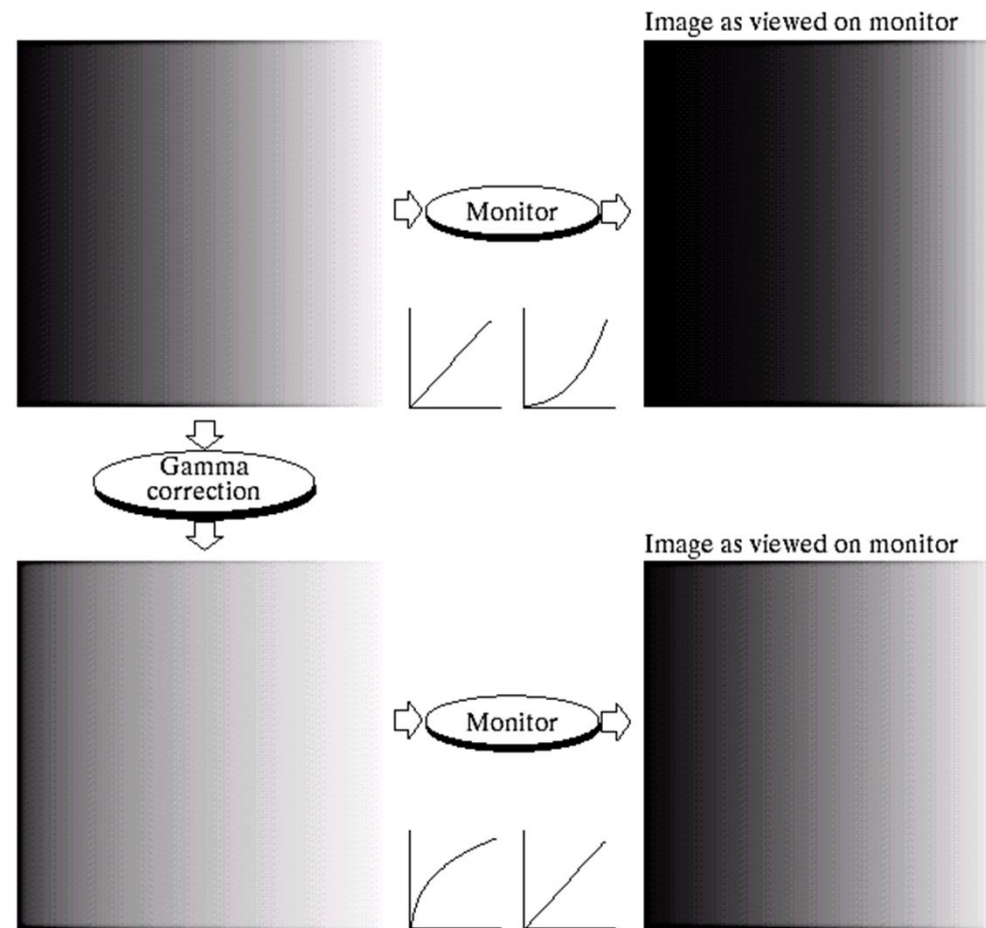


Power-Law Transformations

$$s = cr^\gamma$$

C, γ : positive constants



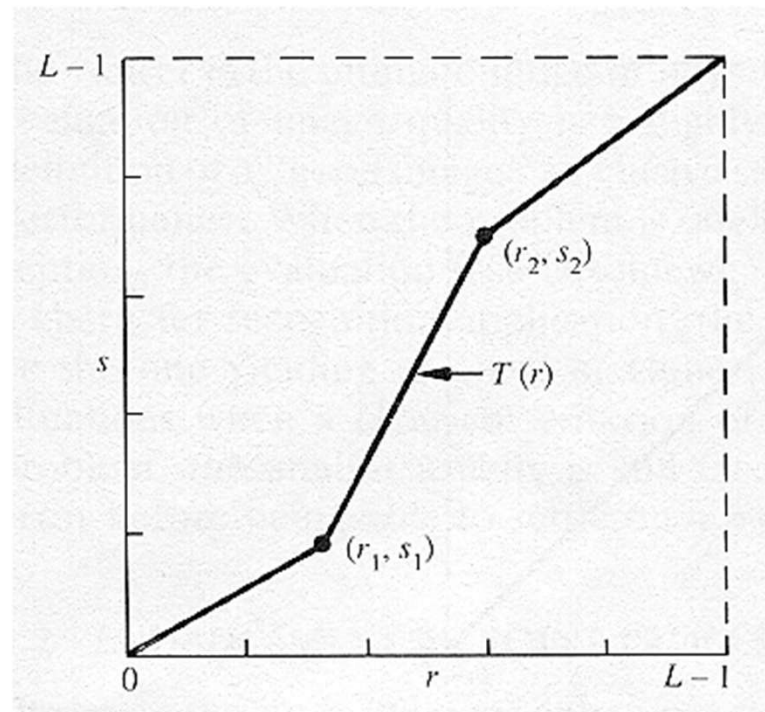




Piecewise-Linear Transformation Functions

Contrast Stretching

- To increase the dynamic range of the gray levels in the image being processed.

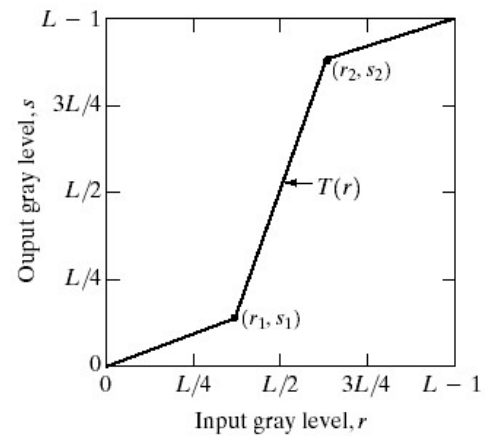


Contrast Stretching

- The locations of (r_1, s_1) and (r_2, s_2) control the shape of the transformation function.
 - If $r_1 = s_1$ and $r_2 = s_2$ the transformation is a linear function and produces no changes.
 - If $r_1 = r_2$, $s_1 = 0$ and $s_2 = L-1$, the transformation becomes a thresholding function that creates a binary image.

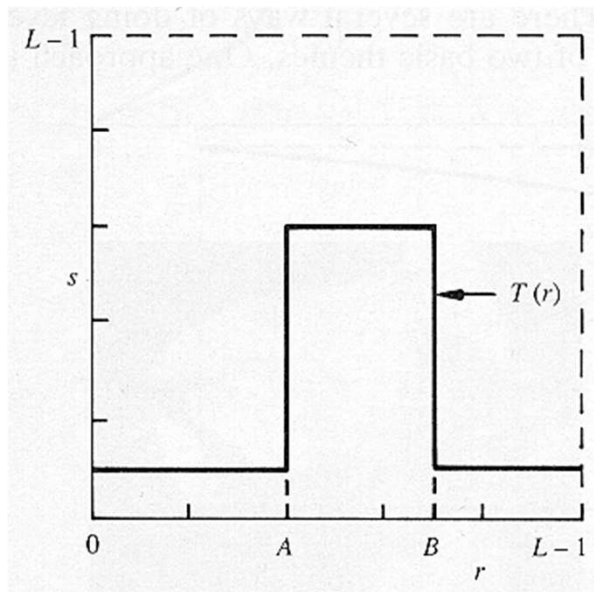
Contrast Stretching

- More on function shapes:
 - Intermediate values of (r_1, s_1) and (r_2, s_2) produce various degrees of spread in the gray levels of the output image, thus affecting its contrast.
 - Generally, $r_1 \leq r_2$ and $s_1 \leq s_2$ is assumed.



Gray-Level Slicing

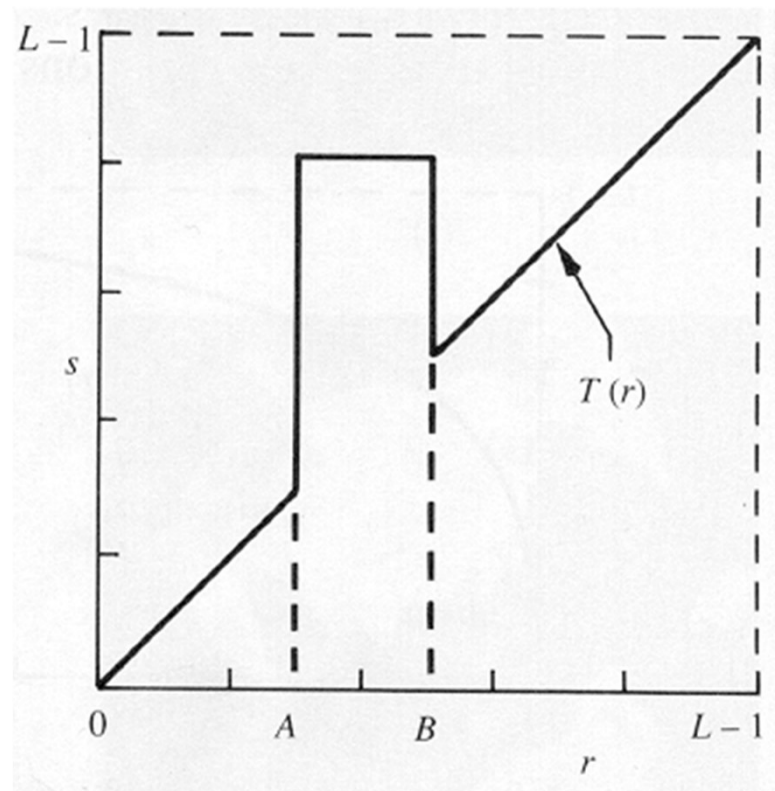
- To highlight a specific range of gray levels in an image (e.g. to enhance certain features).

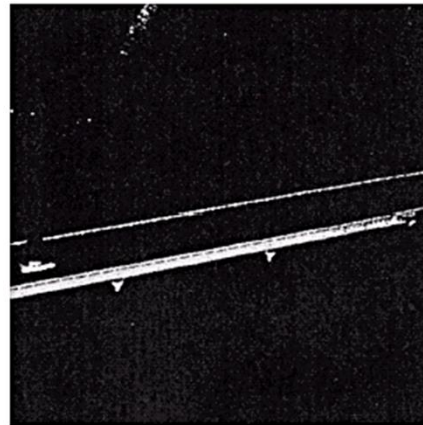
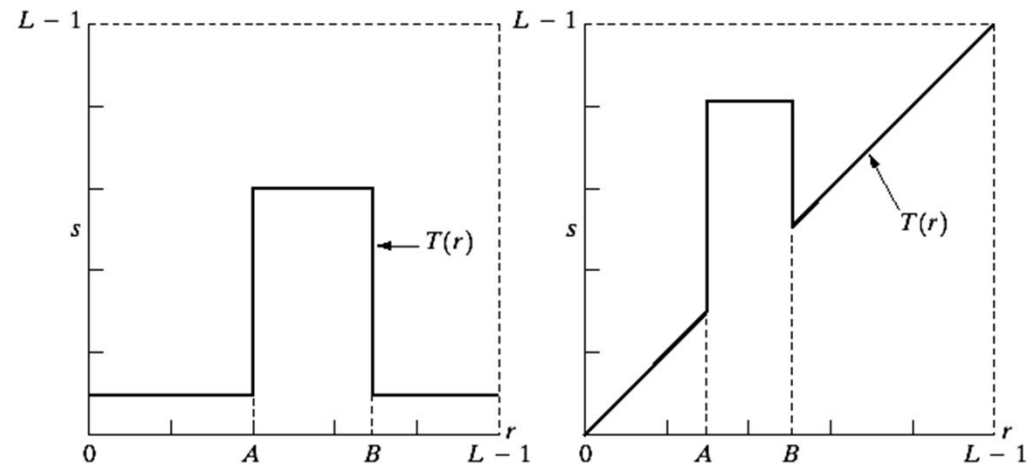


One way is to display a high value for all gray levels in the range of interest and a low value for all other gray levels (binary image).

Gray-Level Slicing

- The second approach is to brighten the desired range of gray levels but preserve the background and gray-level tonalities in the image:



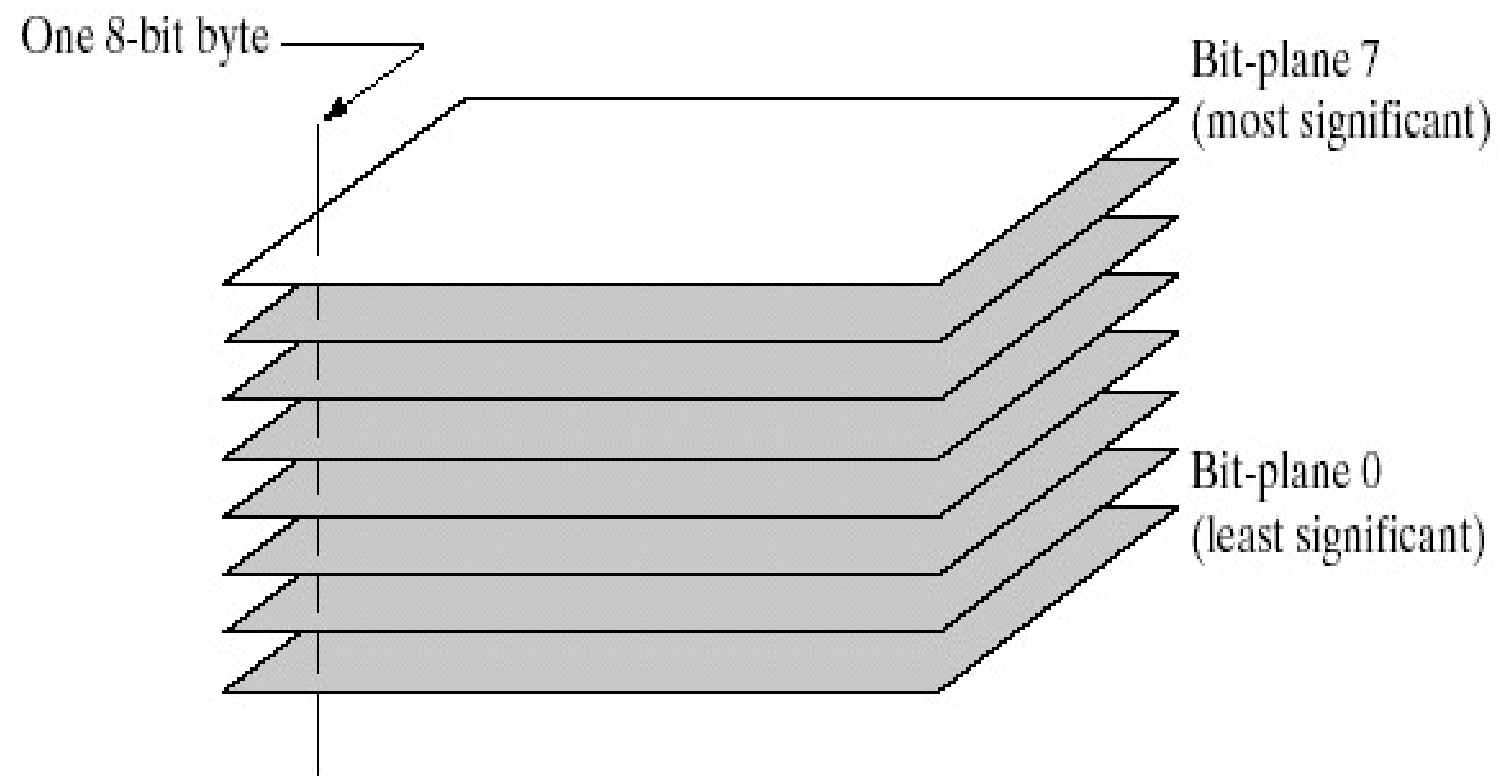


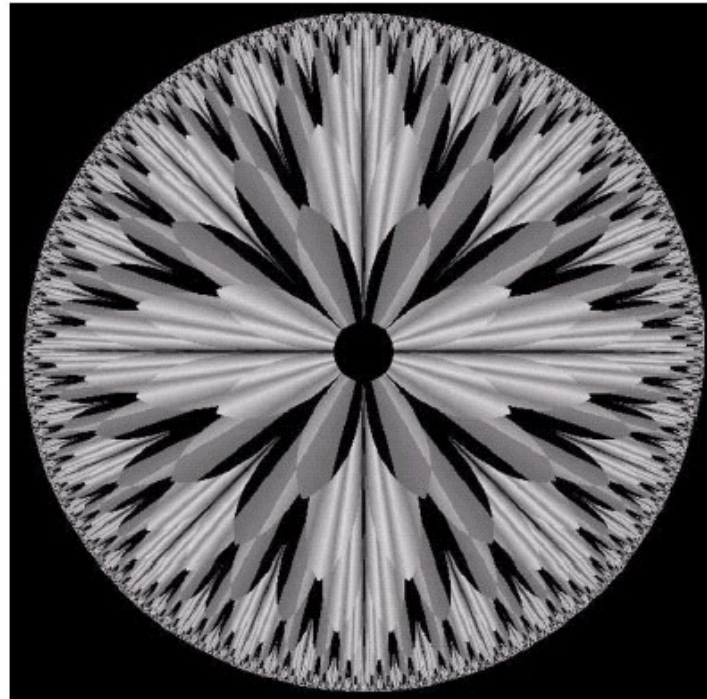
Bit-Plane Slicing

- To highlight the contribution made to the total image appearance by specific bits.
 - i.e. Assuming that each pixel is represented by 8 bits, the image is composed of 8 1-bit planes.
 - Plane 0 contains the least significant bit and plane 7 contains the most significant bit.

Bit-Plane Slicing

- More on bit planes:
 - Only the higher order bits (top four) contain visually significant data. The other bit planes contribute the more subtle details.
 - Plane 7 corresponds exactly with an image thresholded at gray level 128.





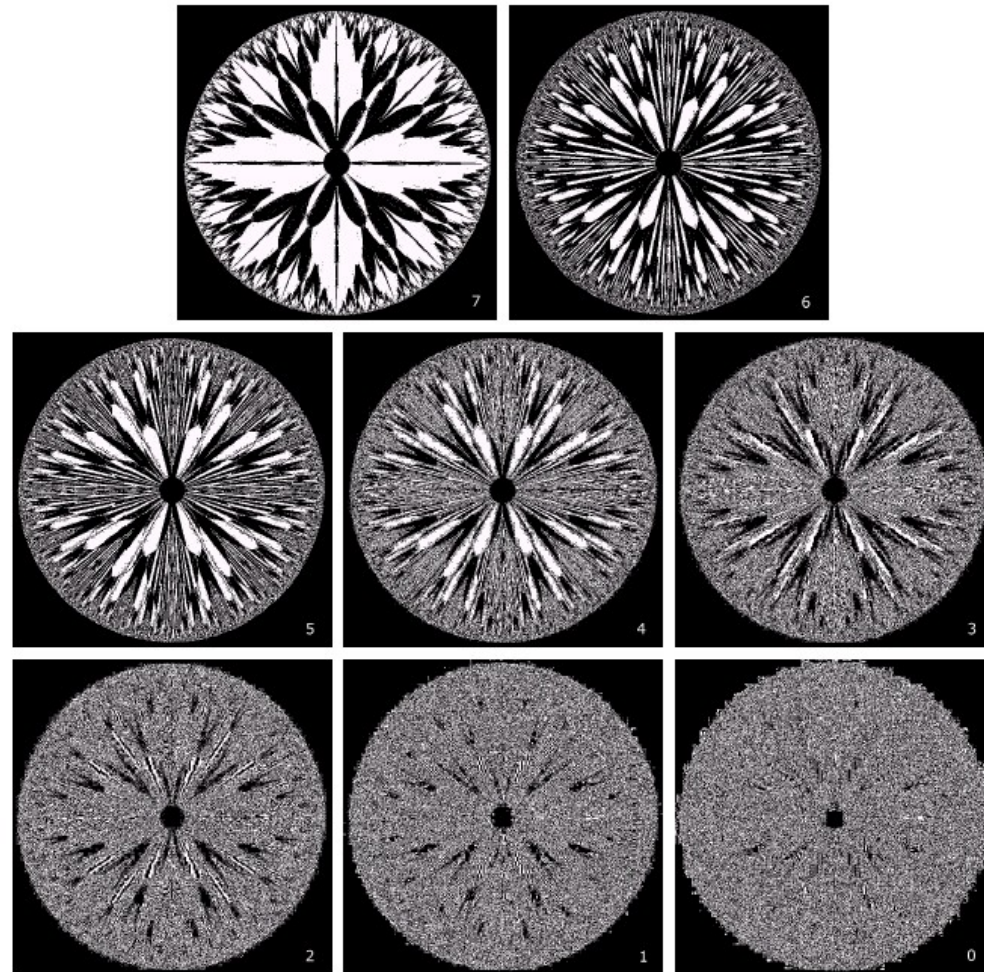
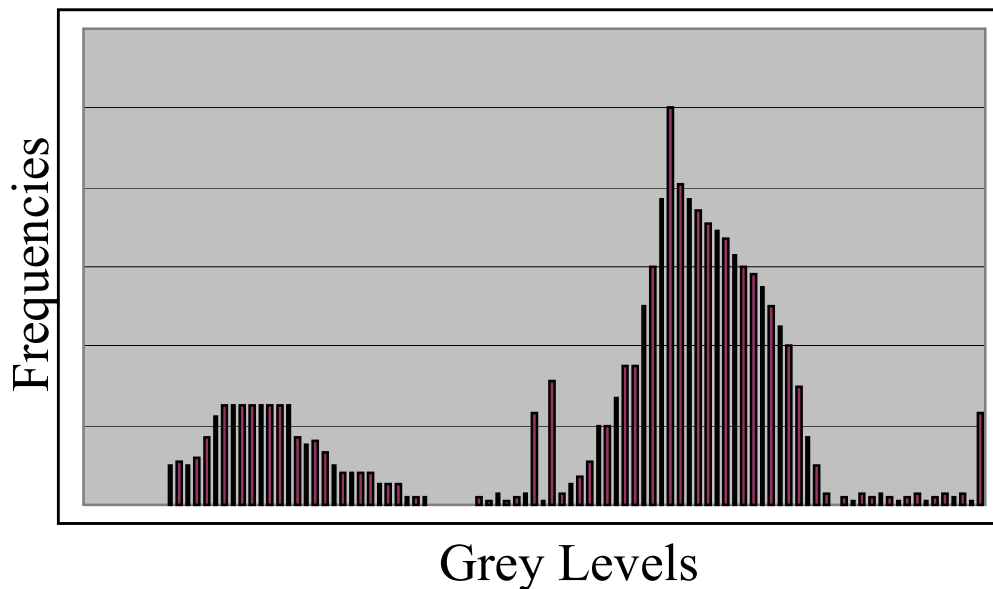


Image Histograms

The histogram of an image shows us the distribution of grey levels in the image

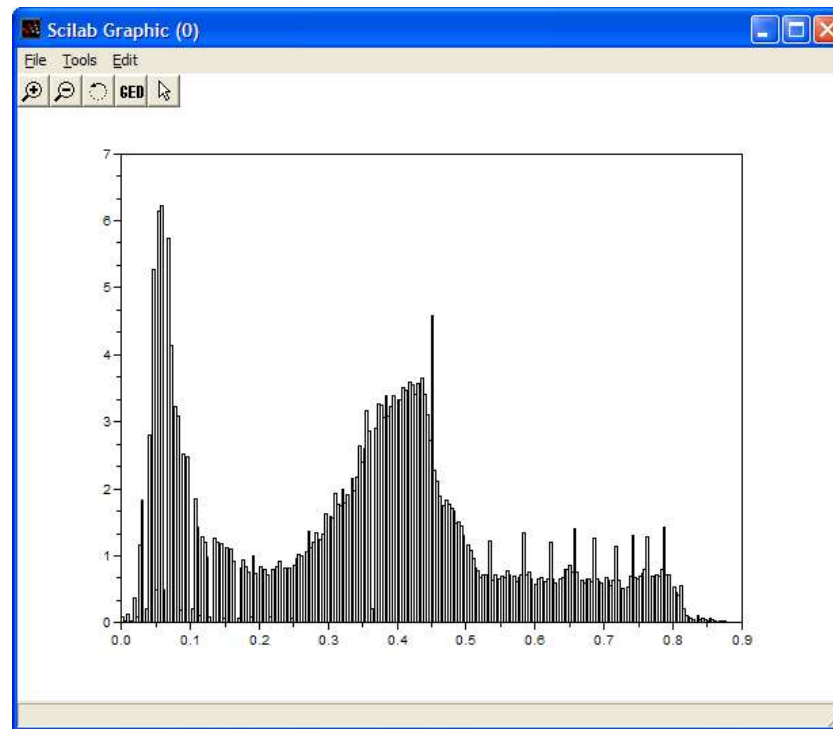
Massively useful in image processing, especially in segmentation



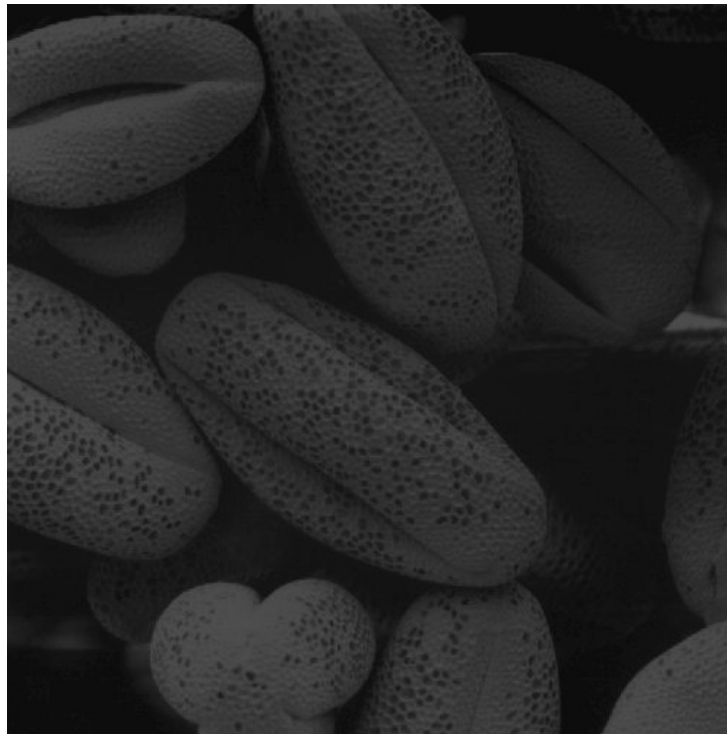
Histogram Examples



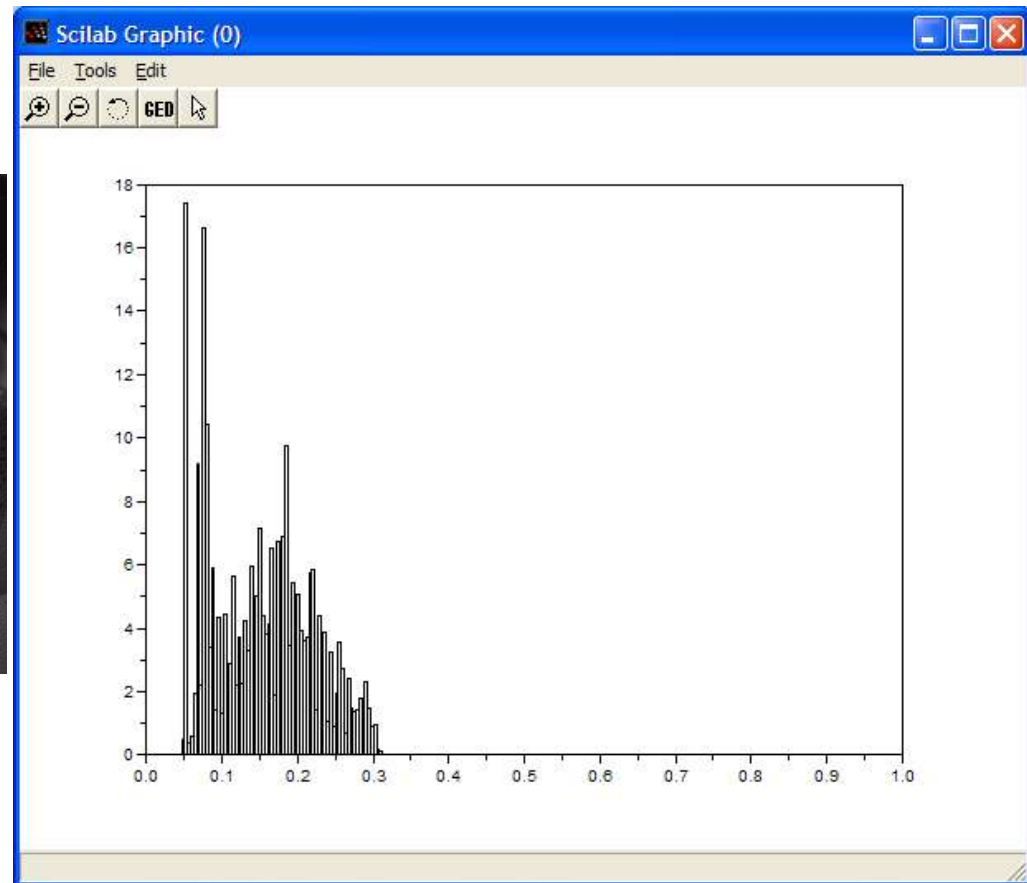
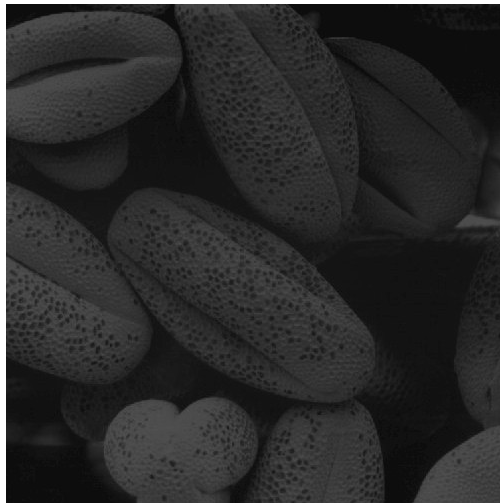
Histogram Examples (cont...)



Histogram Examples (cont...)



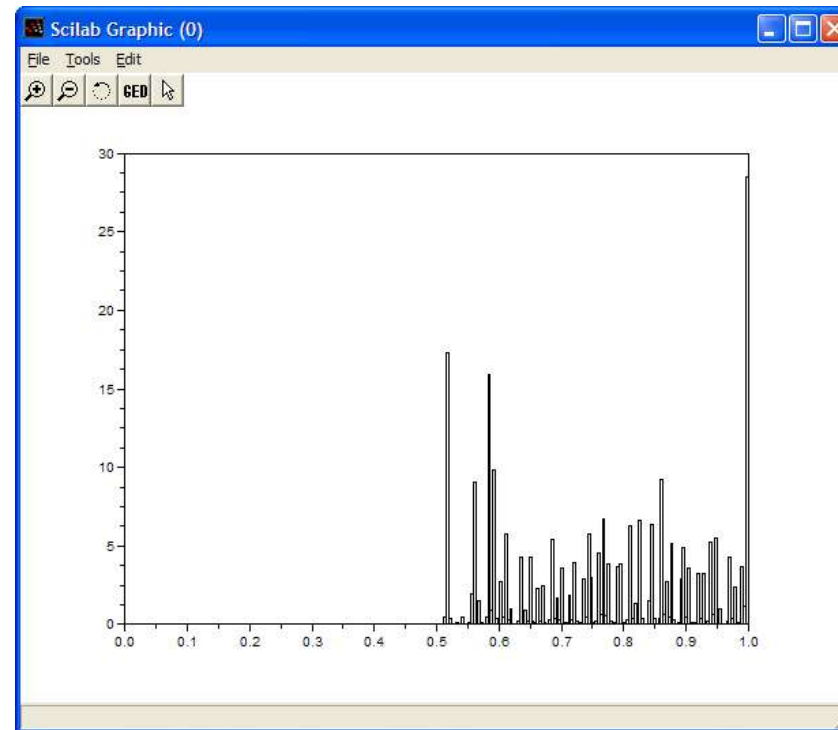
Histogram Examples (cont...)



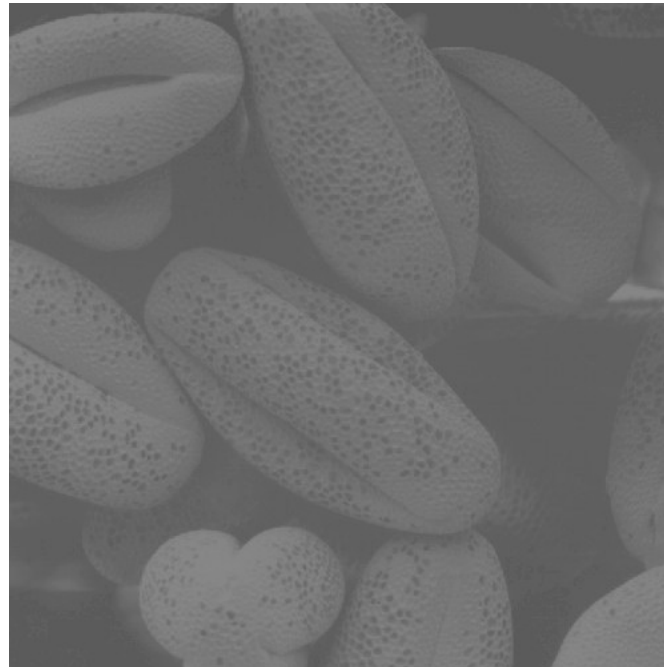
Histogram Examples (cont...)



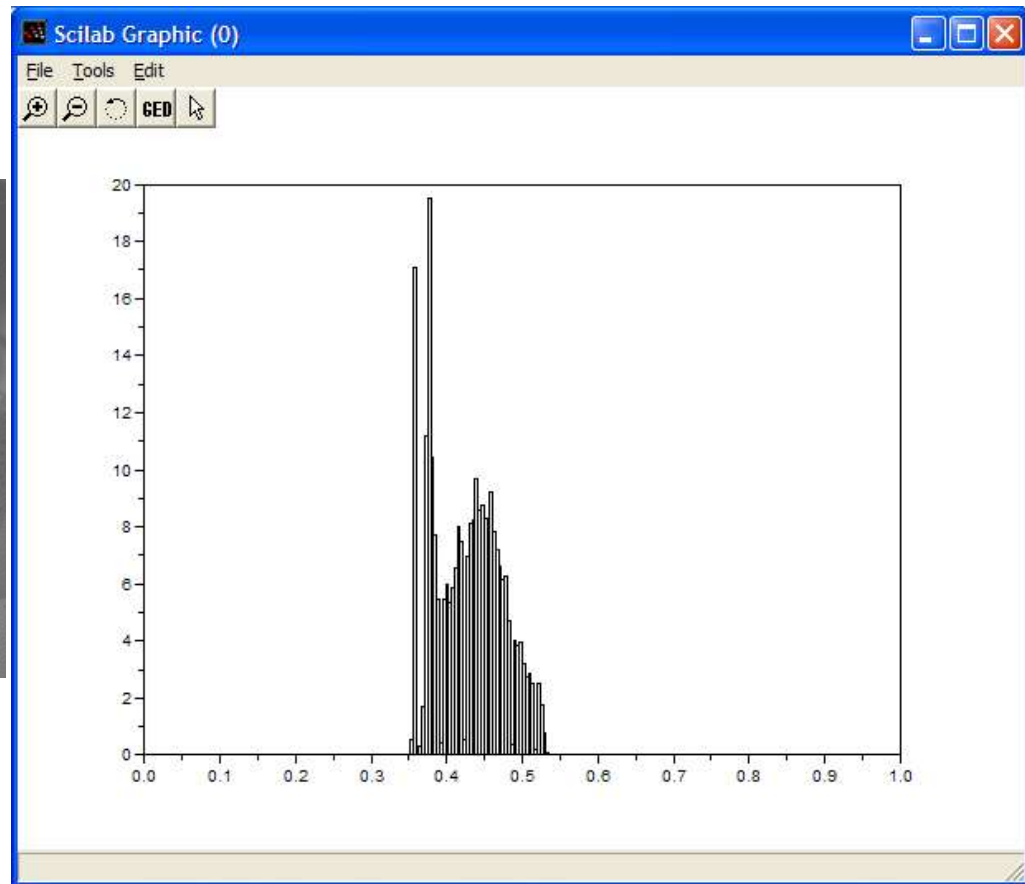
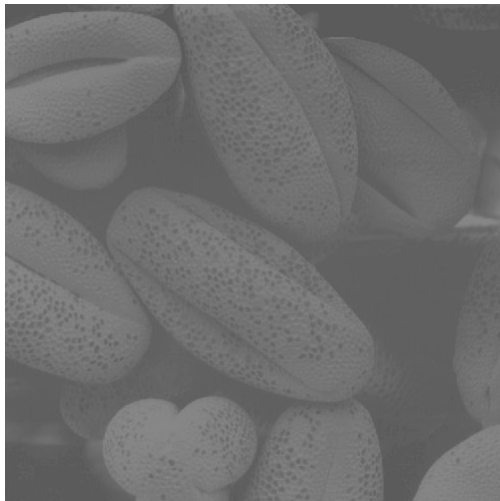
Histogram Examples (cont...)



Histogram Examples (cont...)



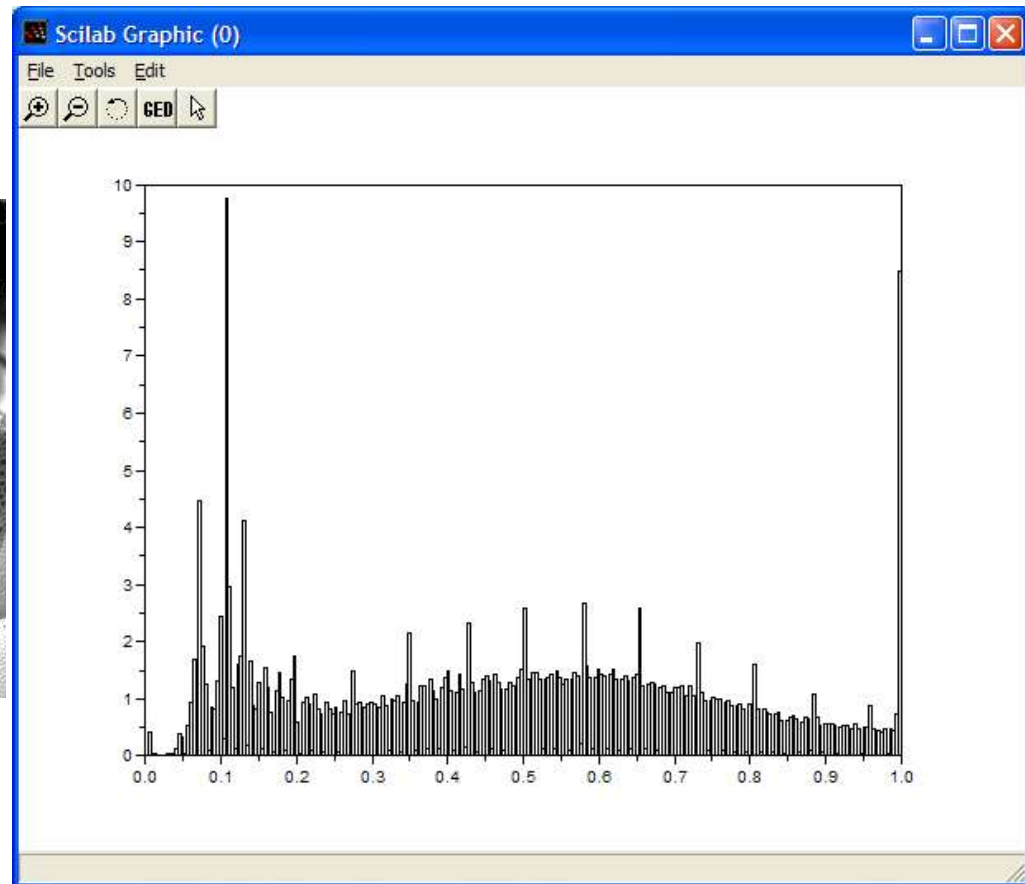
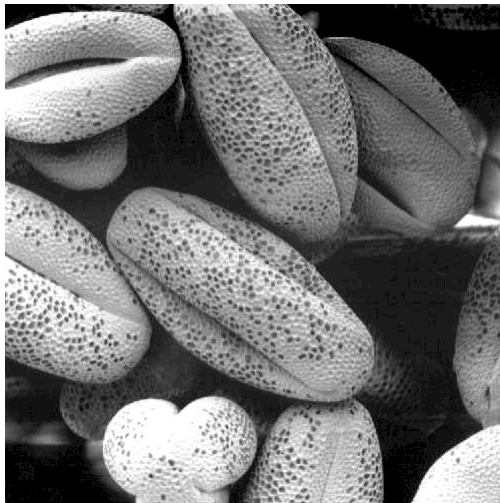
Histogram Examples (cont...)



Histogram Examples (cont...)



Histogram Examples (cont...)

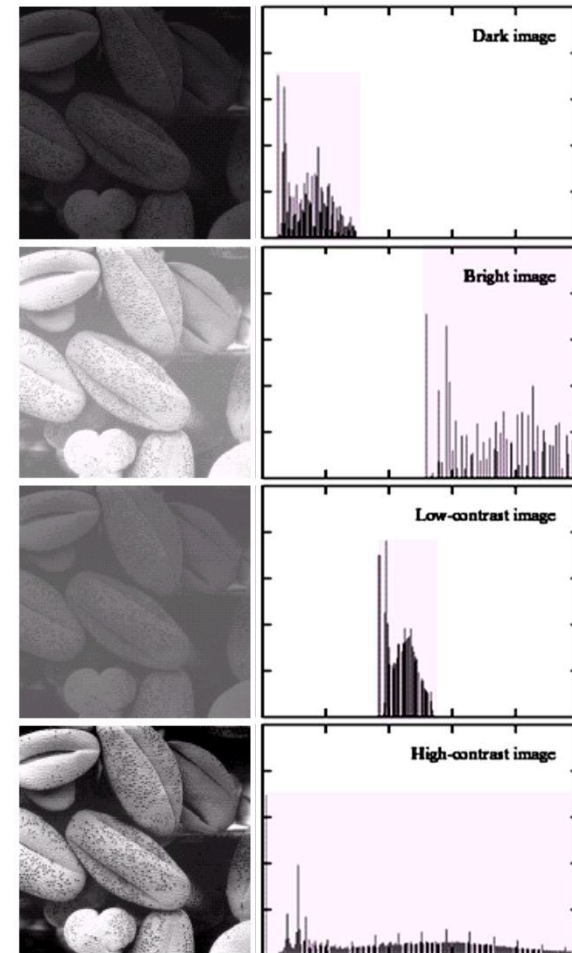


Histogram Examples (cont...)

A selection of images and their histograms

Notice the relationships between the images and their histograms

Note that the high contrast image has the most evenly spaced histogram



Histogram Equalisation

- Histogram Equalization is an image enhancement technique used to improve the contrast of an image by redistributing the intensity values so that the histogram becomes approximately uniform.
- It is particularly useful when the image is too dark or too bright.

Histogram Equalisation

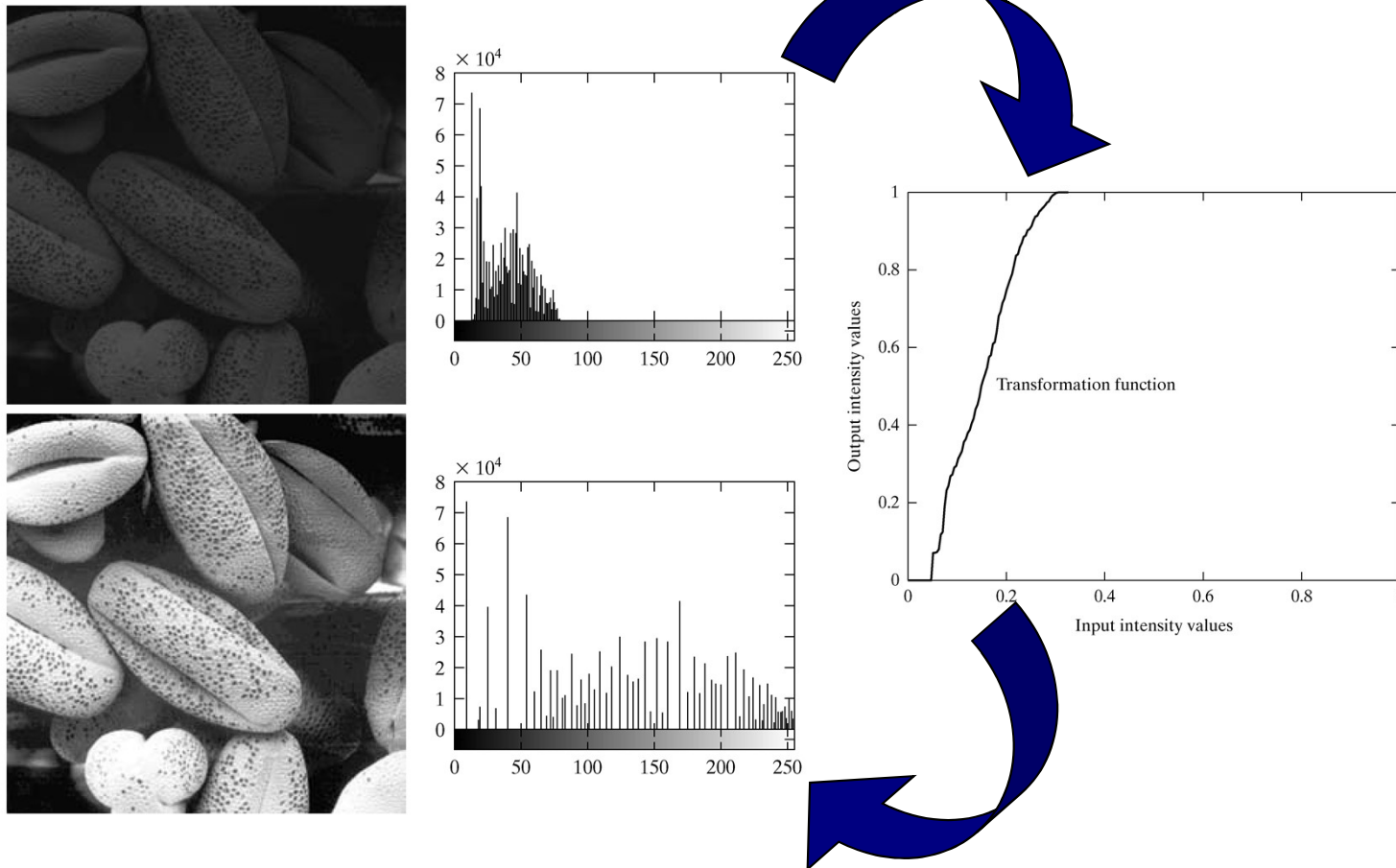
Spreading out the frequencies in an image (or equalising the image) is a simple way to improve dark or washed out images

The formula for histogram equalisation

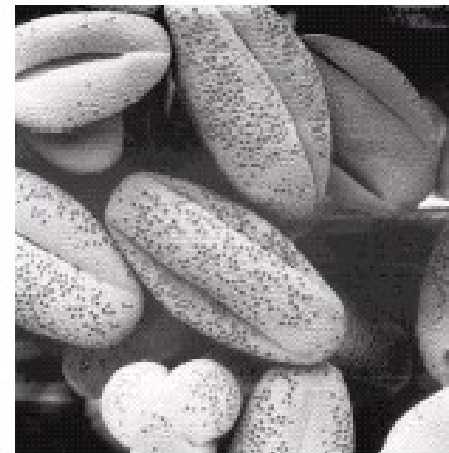
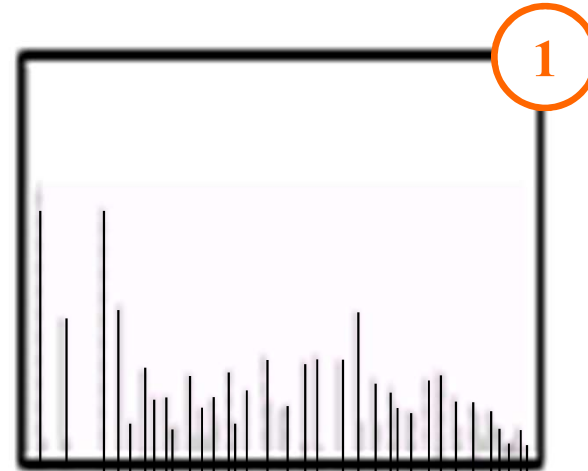
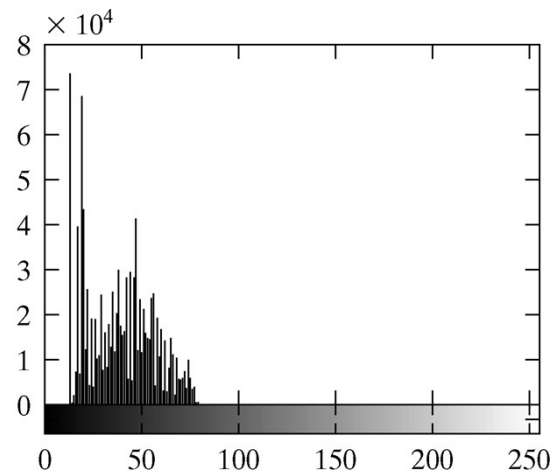
- r_k : input intensity
- s_k : processed intensity
- k : the intensity range (e.g 0.0 – 1.0)
- n_j : the frequency of intensity j
- n : the sum of all frequencies

$$\begin{aligned} s_k &= T(r_k) \\ &= \sum_{j=1}^k p_r(r_j) \\ &= \sum_{j=1}^k \frac{n_j}{n} \end{aligned}$$

Equalisation Transformation Function

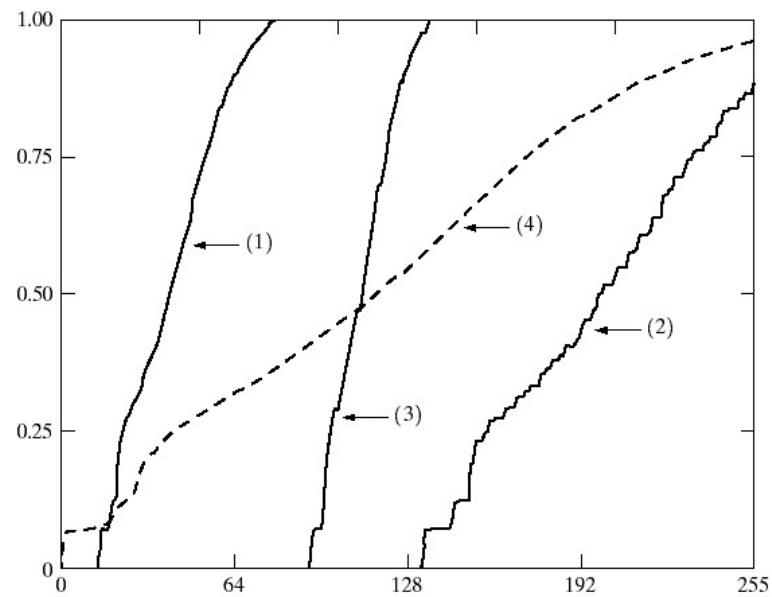


Equalisation Examples

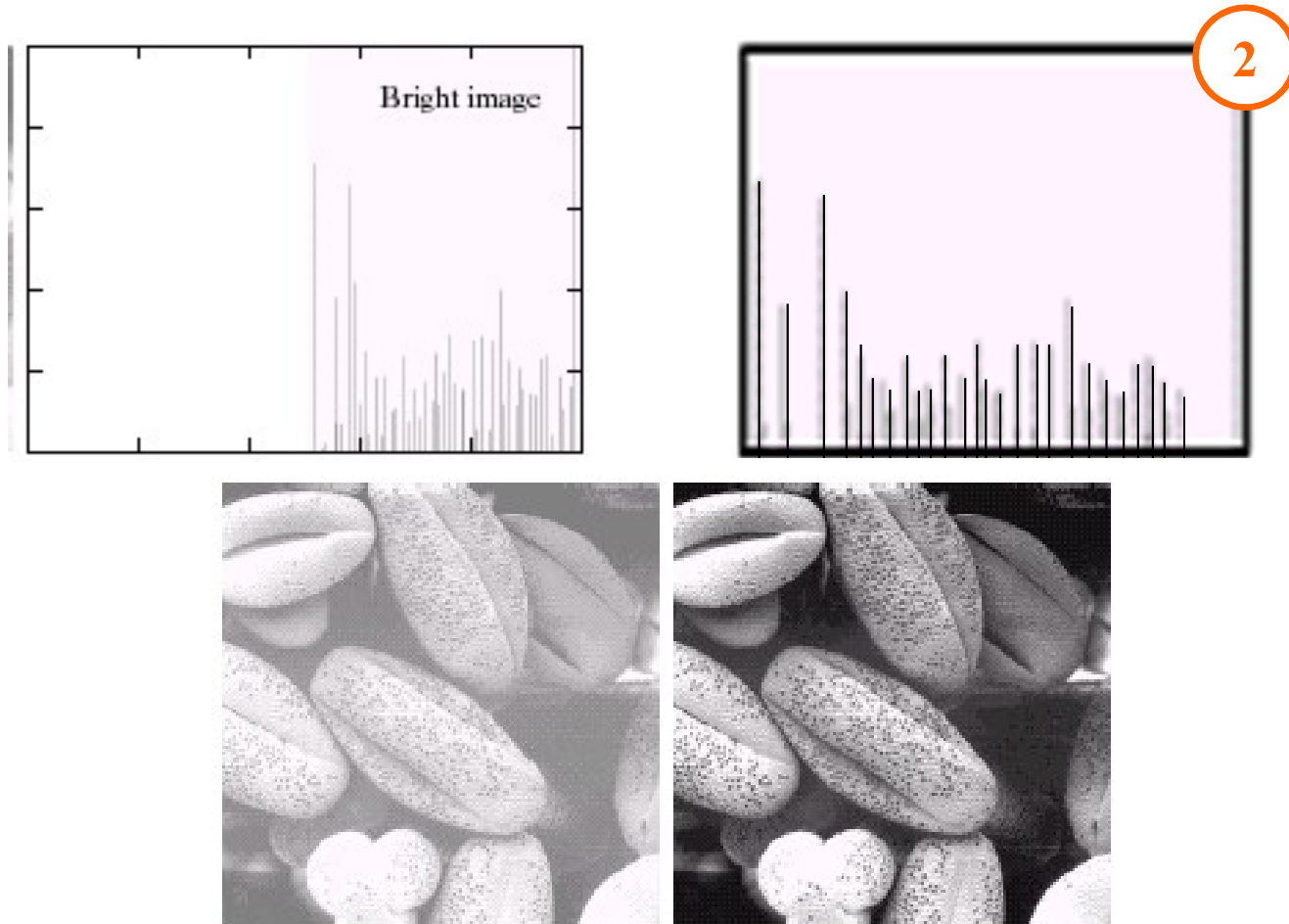


Equalisation Transformation Functions

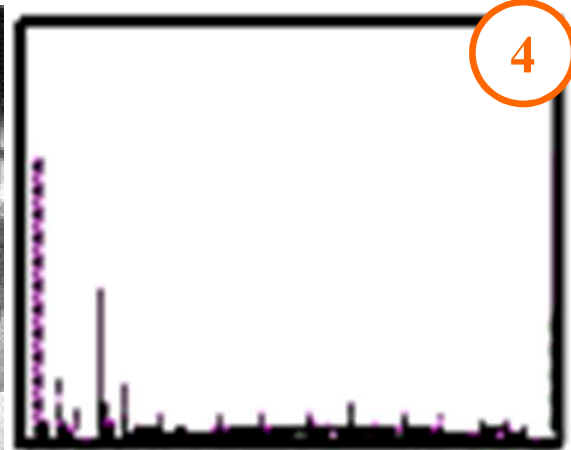
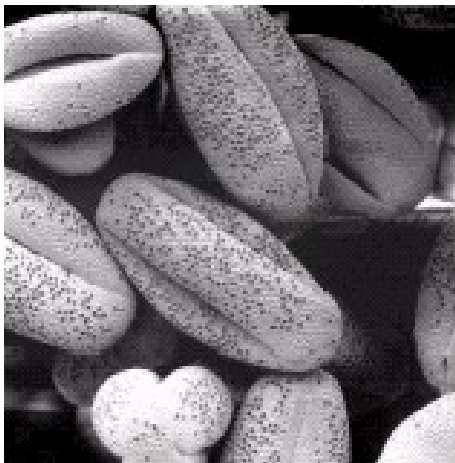
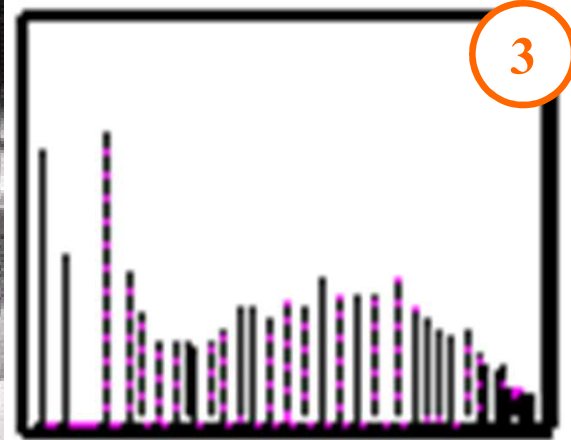
The functions used to equalise the images in the previous example



Equalisation Examples



Equalisation Examples (cont...)



Histogram Equalization – Steps with Example

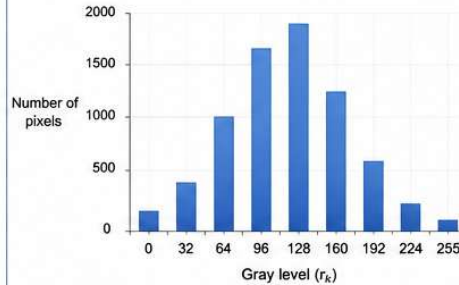
Goal: Improve the contrast of a grayscale image by spreading out the most frequent intensity values.

1 Input Grayscale Image

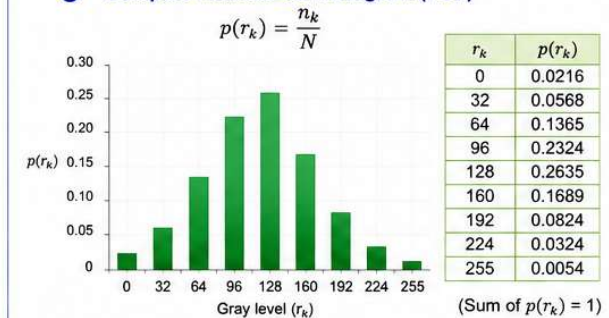


(Low contrast image)

2 Compute Histogram

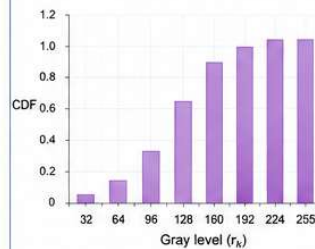


3 Compute Normalized Histogram (PDF)



4 Compute CDF

$$CDF(r_k) = \sum_{j=0}^k p(r_j)$$

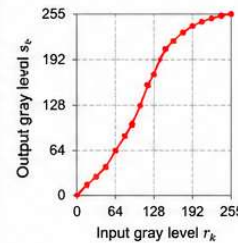


5 Compute Mapping (Transformation Function)

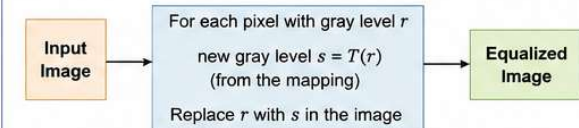
$$s_k = T(r_k) = \text{round}[(L - 1) \times CDF(r_k)]$$

where $L = 256$ (gray levels 0–255)

r_k	$CDF(r_k)$	$s_k = \text{round}(255 \times CDF(r_k))$
0	0.0216	6
32	0.0784	20
64	0.2149	55
96	0.4473	114
128	0.7108	181
160	0.8797	224
192	0.9622	245
224	0.9946	254
255	1.0000	255



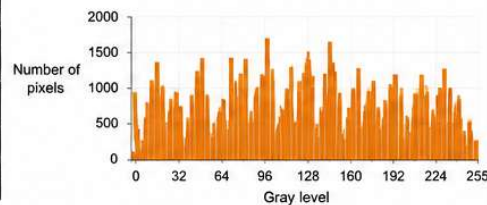
6 Apply Transformation



7 Result: Equalized Image



Histogram of Equalized Image



Summary of Steps

1. Input the grayscale image.
2. Compute the histogram n_k .
3. Compute the normalized histogram (PDF) $p(r_k) = \frac{n_k}{N}$.
4. Compute the cumulative distribution function (CDF).
5. Compute the mapping function $s_k = \text{round}[(L - 1) CDF(r_k)]$.
6. Apply the mapping to transform each pixel.
7. Obtain the equalized image with enhanced contrast.

Why it works?

The transformation spreads the most frequent intensity values over a wider range of gray levels, resulting in better contrast and more visible details.

Histogram Equalization: Example

For an 8-level image,

$$L = 8$$

so the possible intensity values are:

$$0, 1, 2, 3, 4, 5, 6, 7$$

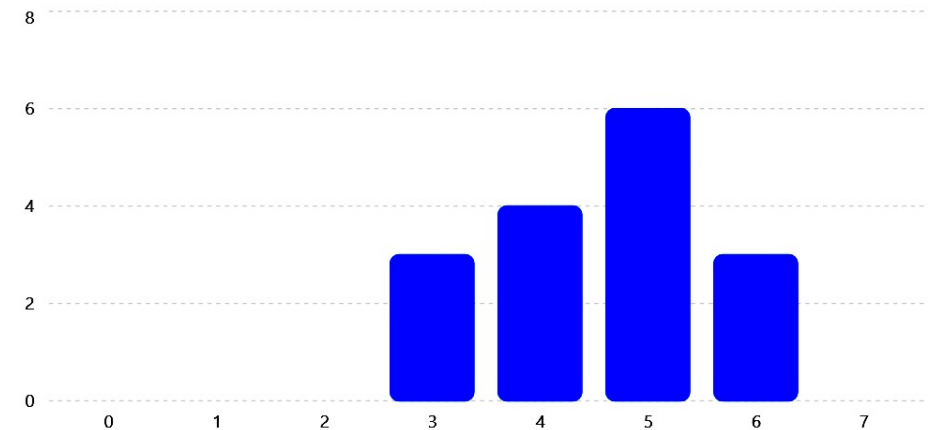
Consider the following 4×4 image:

$$I = \begin{bmatrix} 3 & 4 & 5 & 6 \\ 5 & 4 & 3 & 5 \\ 6 & 5 & 4 & 5 \\ 5 & 6 & 4 & 3 \end{bmatrix}$$

There are:

$$N = 4 \times 4 = 16$$

Intensity (r_k)	0	1	2	3	4	5	6	7
Number of pixels (n_k)	0	0	0	3	4	6	3	0



Gray levels 0, 1, 2 and 7 have zero occurrences.

Histogram Equalization:

Calculate the *probability distribution* and *Cumulative distribution*

r_k	n_k	$P(r_k) = n_k/16$	CDF
0	0	0	0
1	0	0	0
2	0	0	0
3	3	0.1875	0.1875
4	4	0.2500	0.4375
5	6	0.3750	0.8125
6	3	0.1875	1.0000
7	0	0	1.0000

Histogram Equalization

For histogram equalization, the transformation is

$$s_k = T(r_k)$$

and

$$s_k = \text{round}[(L - 1)CDF(r_k)]$$

Histogram Equalization

Input intensity (r)	CDF	(7\times CDF)	Output intensity (s)
0	0	0	0
1	0	0	0
2	0	0	0
3	0.1875	1.3125	1
4	0.4375	3.0625	3
5	0.8125	5.6875	6
6	1.0000	7.0000	7
7	1.0000	7.0000	7

$$3 \rightarrow 1$$

$$4 \rightarrow 3$$

$$5 \rightarrow 6$$

$$6 \rightarrow 7$$

Histogram Equalization

Our original image was:

$$\begin{bmatrix} 3 & 4 & 5 & 6 \\ 5 & 4 & 3 & 5 \\ 6 & 5 & 4 & 5 \\ 5 & 6 & 4 & 3 \end{bmatrix}$$

$3 \rightarrow 1, 4 \rightarrow 3, 5 \rightarrow 6, 6 \rightarrow 7$

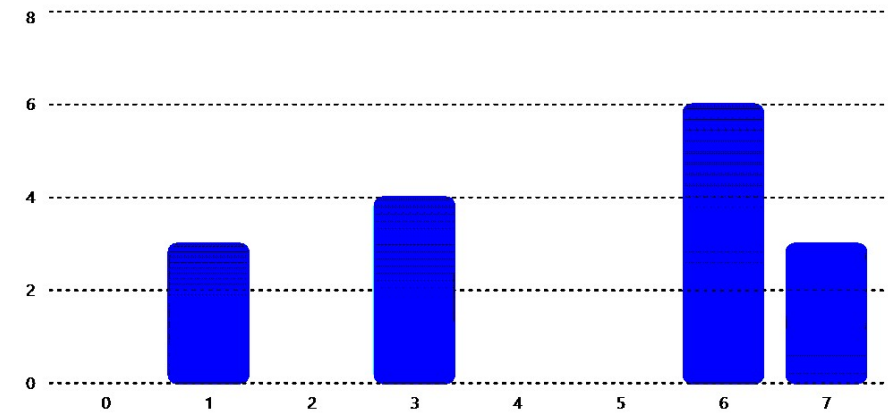
Equalized image is:

$$\begin{bmatrix} 1 & 3 & 6 & 7 \\ 6 & 3 & 1 & 6 \\ 7 & 6 & 3 & 6 \\ 6 & 7 & 3 & 1 \end{bmatrix}$$

Histogram Equalization

Output intensity (s)	0	1	2	3	4	5	6	7
Number of pixels	0	3	0	4	0	0	6	3

Important: Histogram equalization does not necessarily produce a perfectly flat histogram, especially for a small/discrete image. Its purpose is to redistribute intensities to improve the use of the available dynamic range and, consequently, improve contrast



The histogram is spread over a wider intensity range, increasing contrast.

Image Filtering in Spatial Domain

1. Flip the Kernel in both horizontal and vertical directions (center of the kernel must be provided)
2. Move over the array with kernel centered at interested point.
3. Multiply kernel data with overlapped area.
4. Sum or accumulate the output.

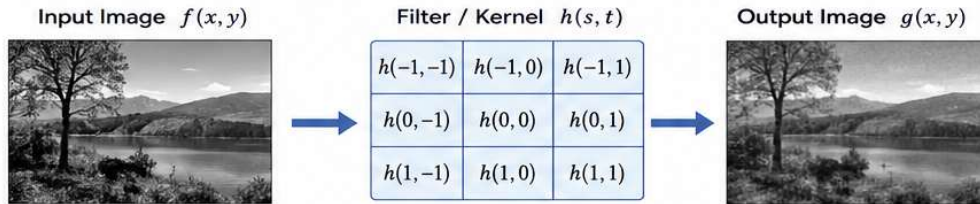
Note :

Size of output must be same size of input and we need to pad image for not defined pixels.

IMAGE FILTERING IN SPATIAL DOMAIN

Image filtering in spatial domain is performed directly on the pixels of an image using a small matrix called the filter or kernel (mask).

1. Basic Idea



2. Mathematical Model

The output image $g(x, y)$ is obtained by convolving the input image $f(x, y)$ with the filter $h(s, t)$.

2-D Convolution:

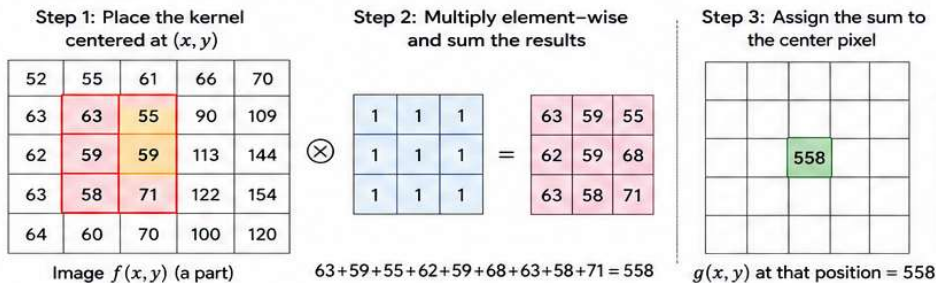
$$g(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b h(s, t) f(x-s, y-t)$$

$f(x, y)$: Input image

$h(s, t)$: Filter (kernel) of size $(2a+1) \times (2b+1)$

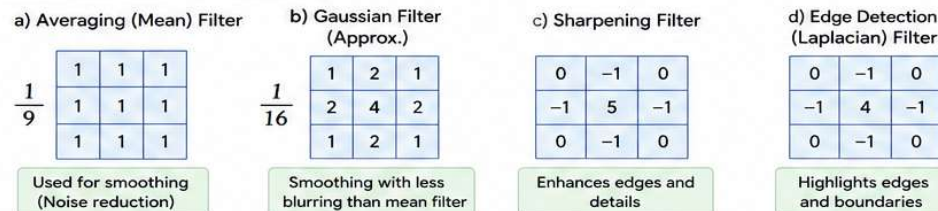
$g(x, y)$: Output (filtered) image

3. How It Works (Example with 3×3 Filter)



- Move the kernel to the next position.
- Repeat the process for all pixels.
- This produces the filtered image $g(x, y)$.

4. Common Filters (Kernels)



5. Effect of Filtering (Example)



Key Points

- Spatial domain filtering works directly on pixels.
- Uses a kernel (mask) that slides over the image.
- Each output pixel is a weighted sum of its neighbors.
- Useful for smoothing, sharpening, edge detection, etc.

1	2	3
4	5	6
7	8	9

Input

	m	-1	0	1
n	-1	-1	-2	-1
0	0	0	0	0
1	1	1	2	1

Kernel

-13	-20	-17
-18	-24	-18
13	20	17

Output

1	2	1		
0	0	1	2	3
-1	-2	-1	5	6
	7	8	9	

1	2	1		
0	1	0	2	3
-1	4	-2	5	6
	7	8	9	

	1	2	1	
1	0	2	0	3
4	-1	5	-2	6
7	8	9		

1	2	1		
0	0	4	5	6
-1	-2	7	8	9

1	1	2	1	3
0	4	0	5	6
-1	7	-2	8	9

Various filter kernels:

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Mean Operator

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Gaussian Kernel

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Prewitt Operator

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Sobel Operator

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Laplacian Operator

Image Restoration

- The purpose of image restoration is to restore a degraded/distorted image to its original content and quality.
- Distinctions to Image Enhancement
 - Image restoration assumes a degradation model that is known or can be estimated.
 - Original content and quality $\neq$ Good looking

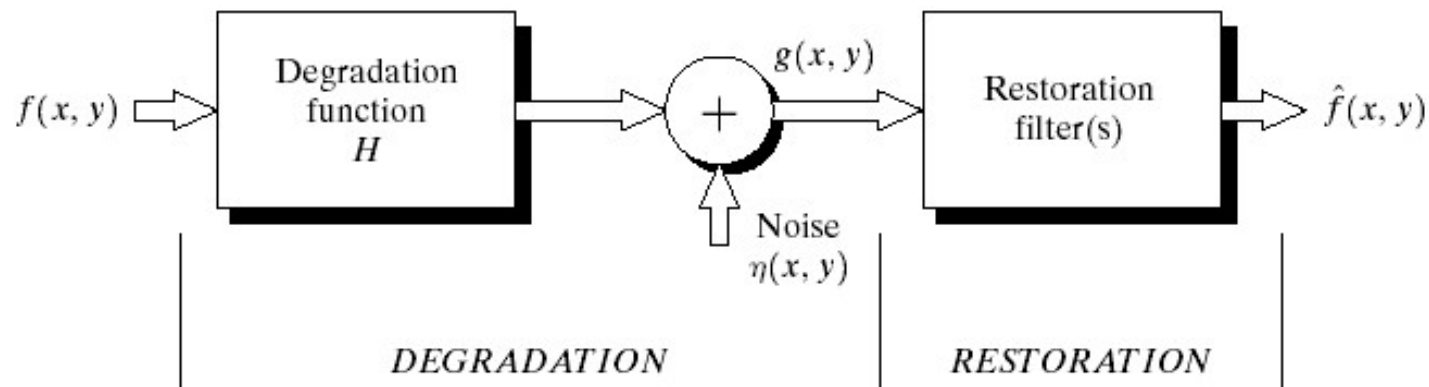


Image Degradation

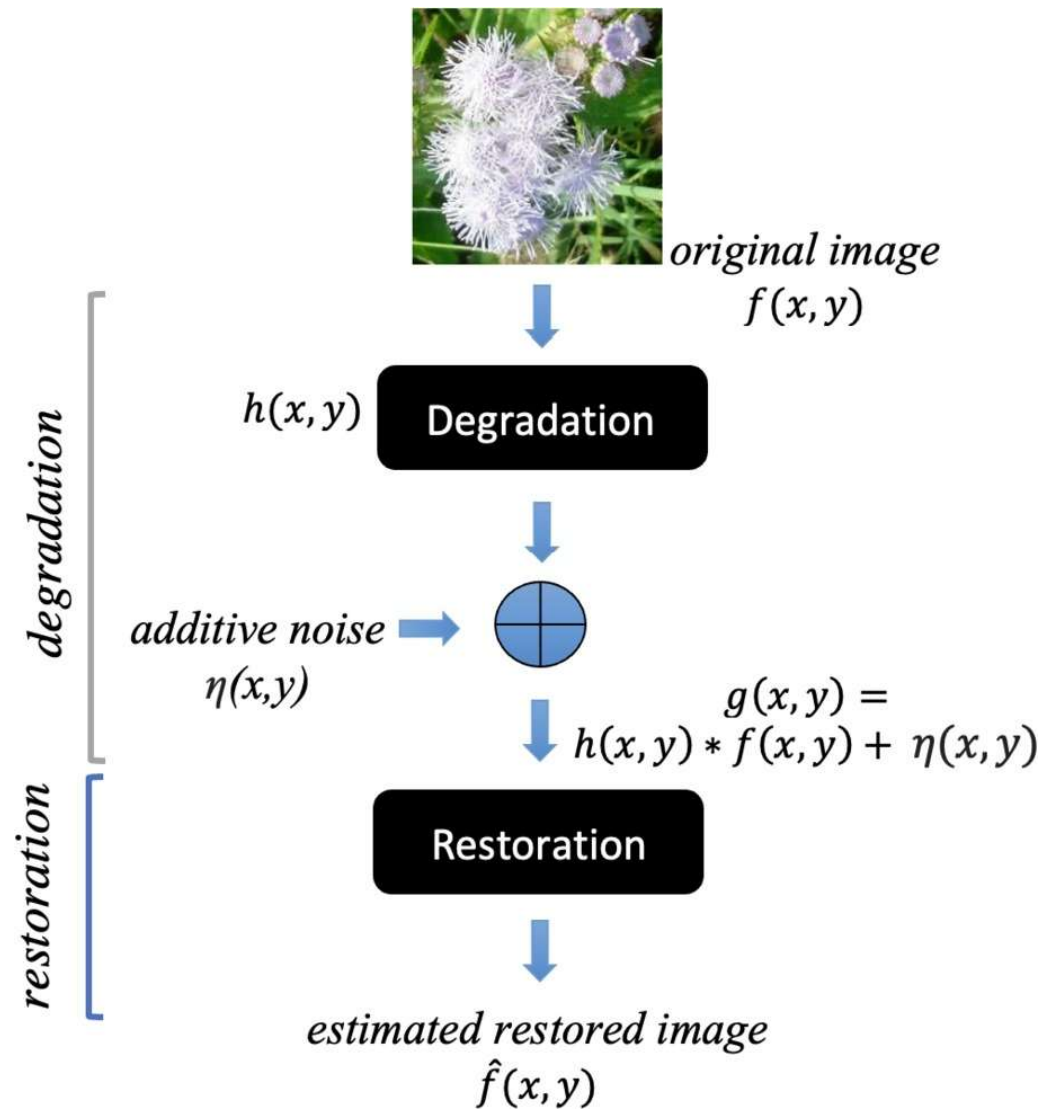


Image Degradation

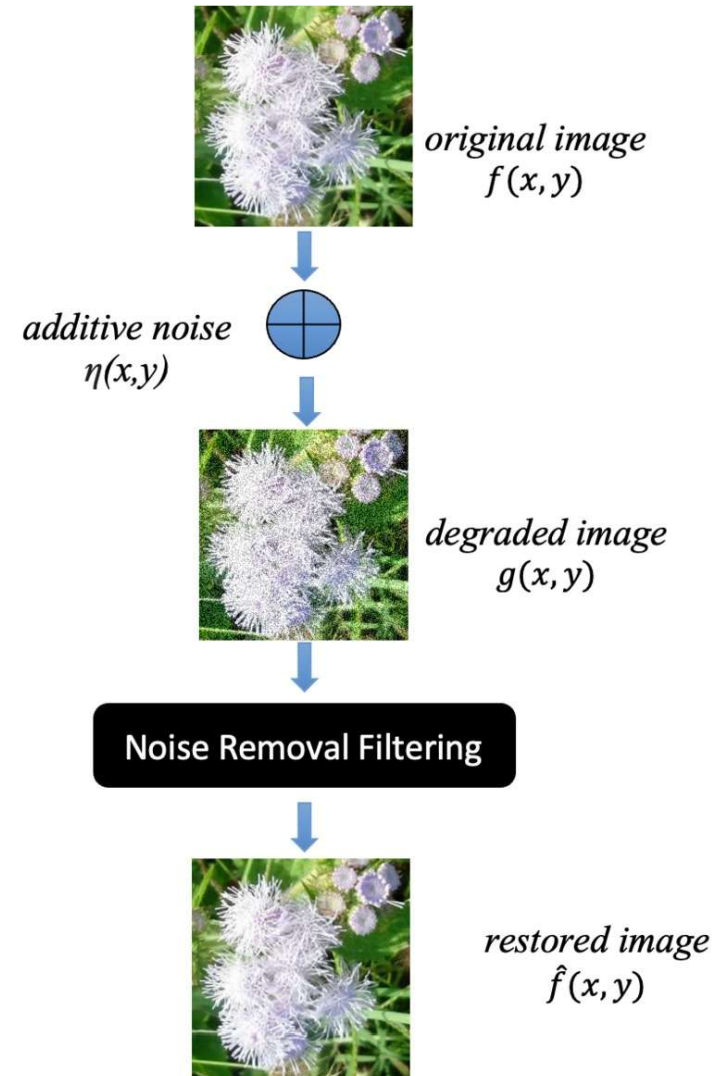
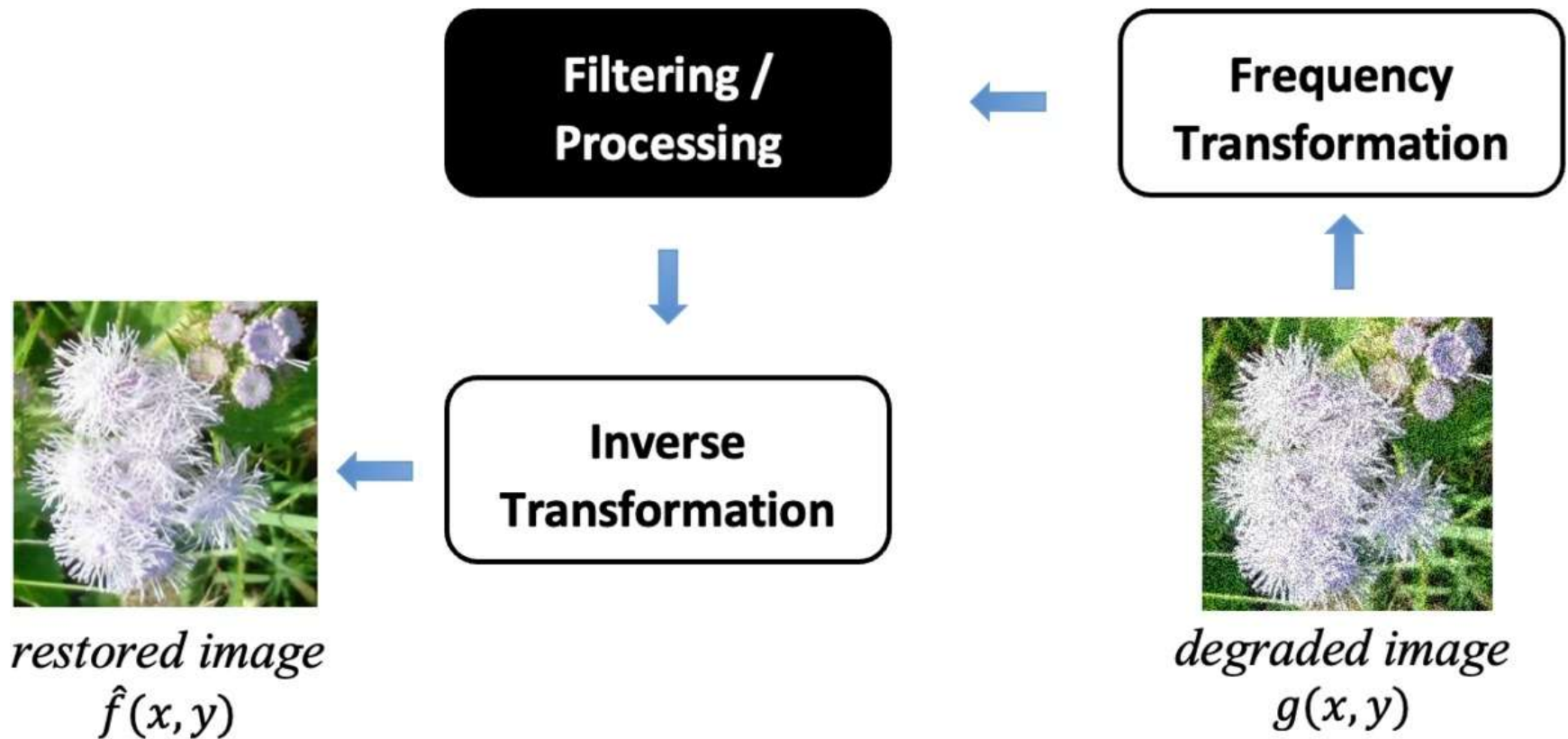
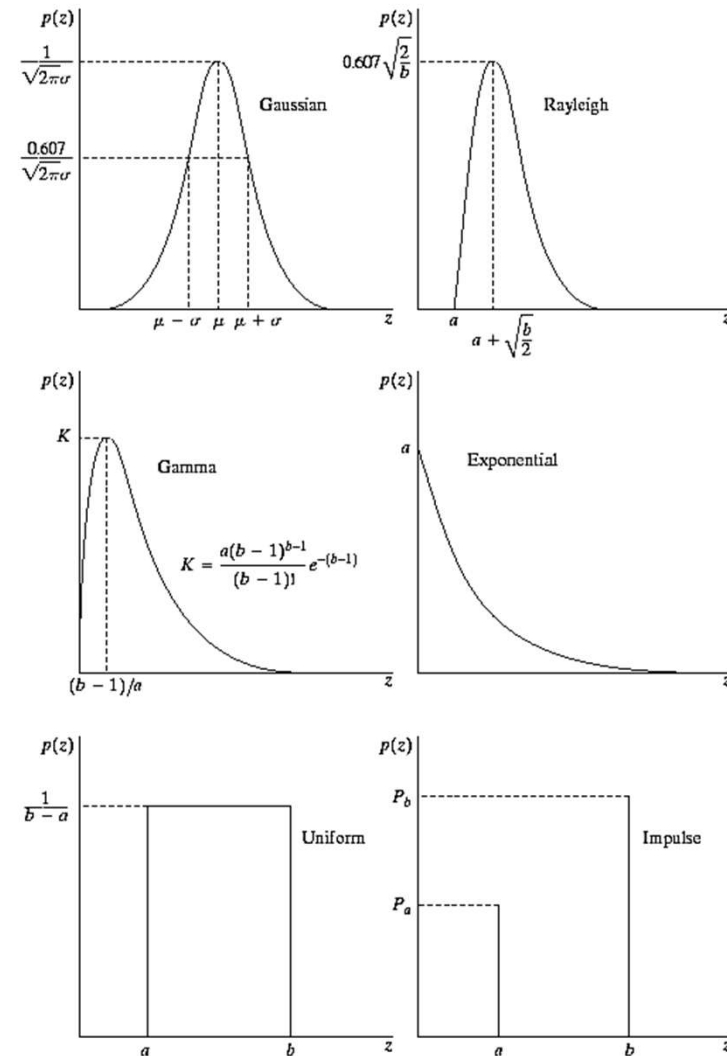


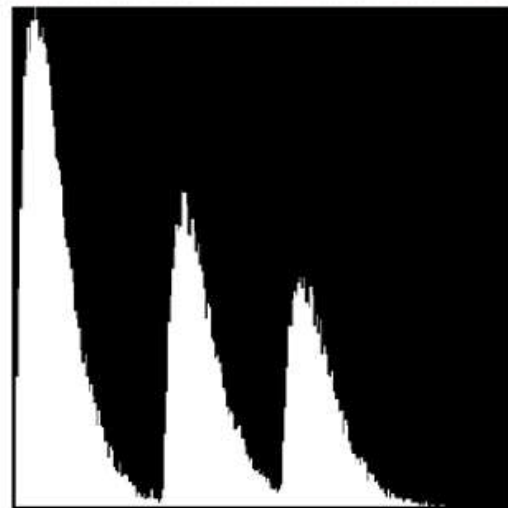
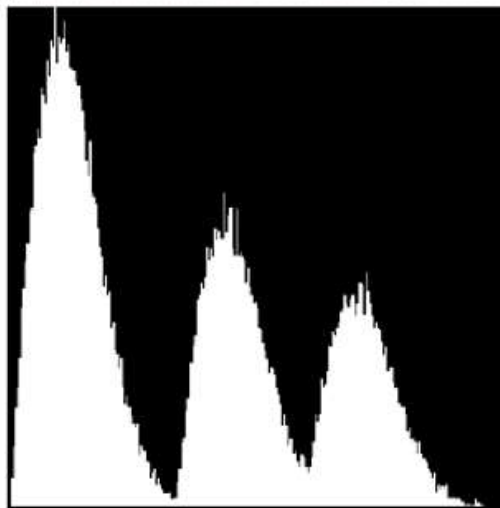
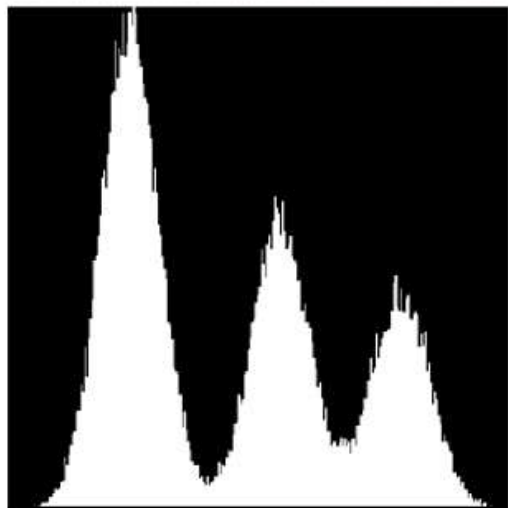
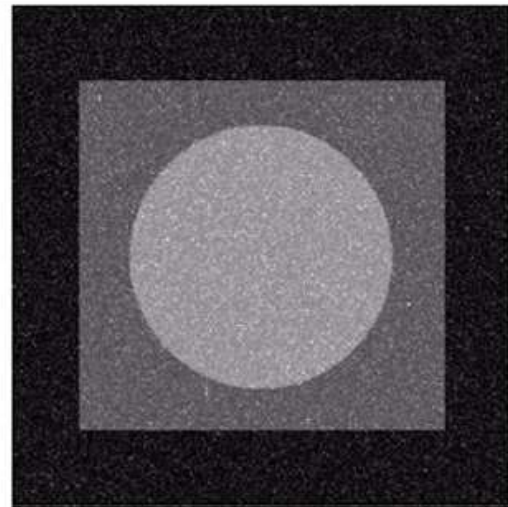
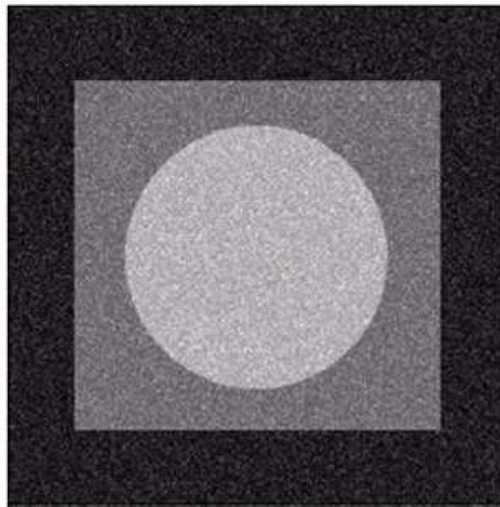
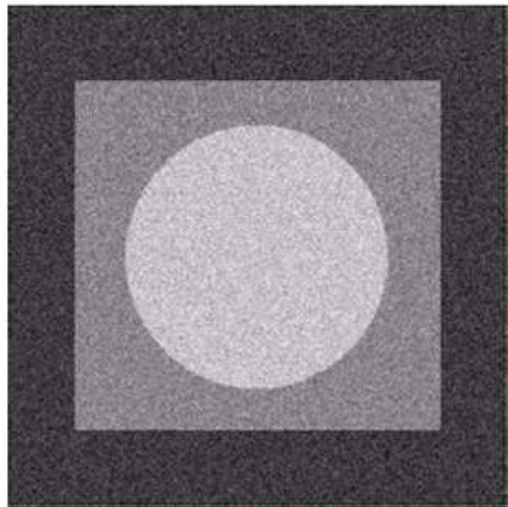
Image Degradation



Noise Models

- Most types of noise are modeled as known probability density functions
- Noise model is decided based on understanding of the physics of the sources of noise.
 - Gaussian: poor illumination
 - Rayleigh: range image
 - Gamma, exp: laser imaging
 - Impulse: faulty switch during imaging,
 - Uniform is least used.
- Parameters can be estimated based on histogram on small flat area of an image

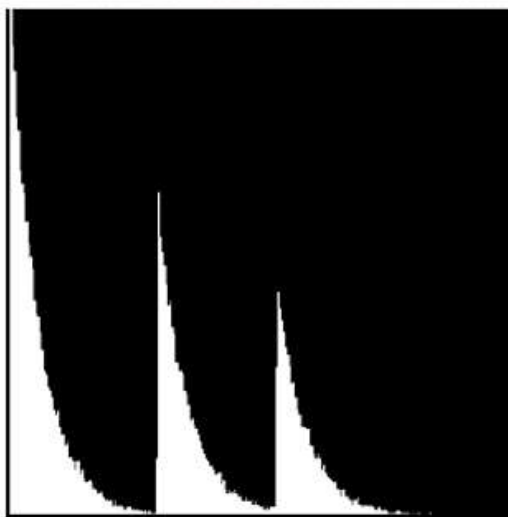
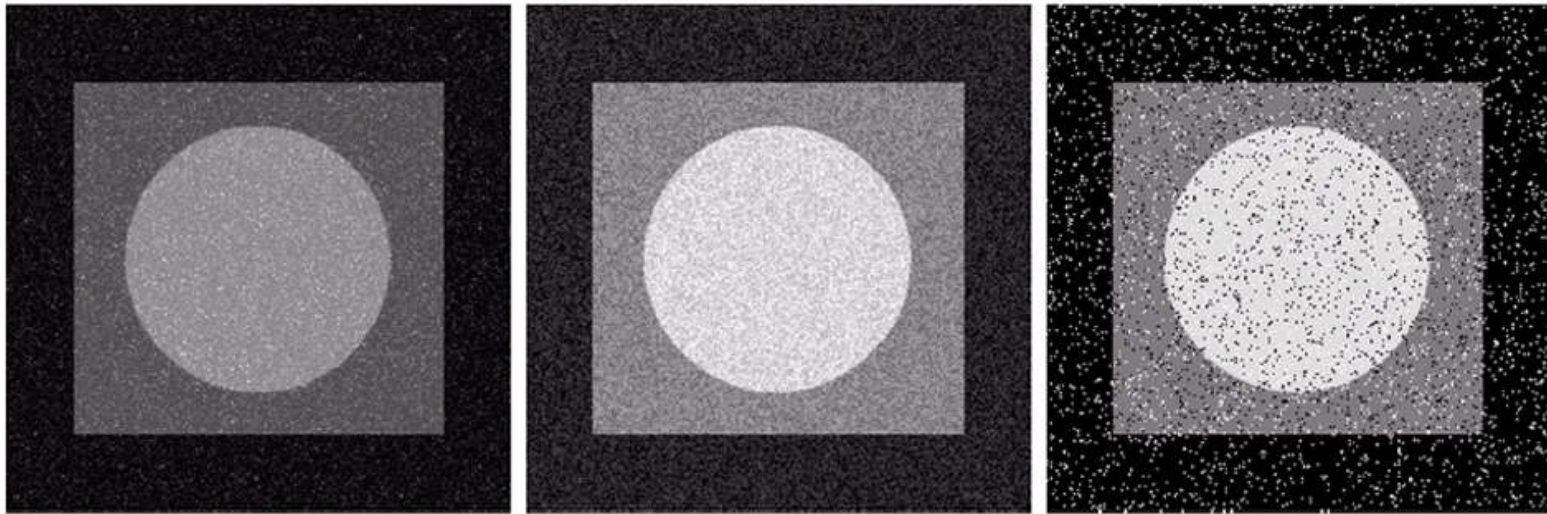




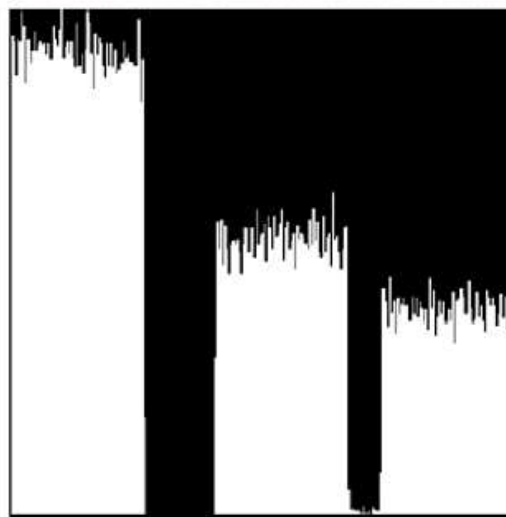
Gaussian

Rayleigh

Gamma



Exponential



Uniform



Salt & Pepper

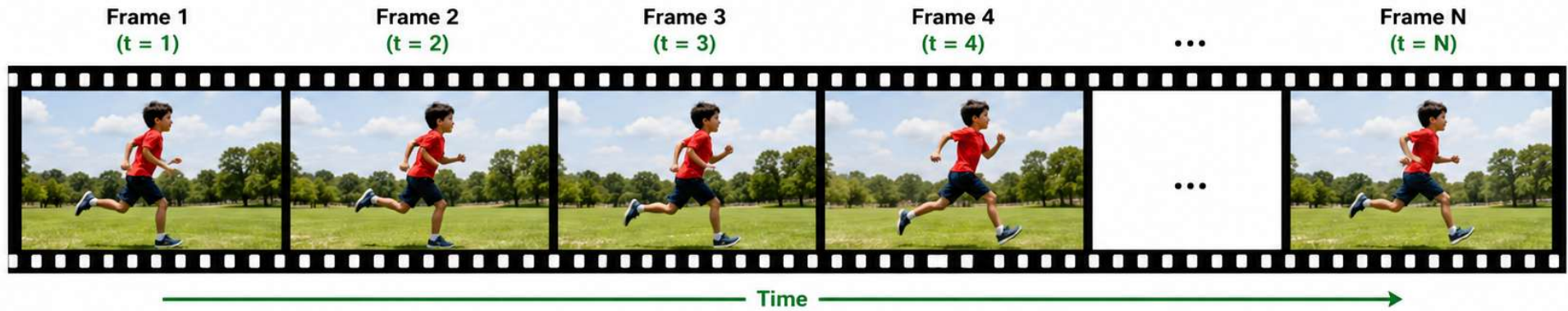
Video Analysis

Digital Video

- It is a continuous arrangement of image frames that represents **movement of object(s), background or movement of both** over time.
- A **digital video is fundamentally a 3D signal** if we consider its two spatial dimensions (x, and y), and time (t).
- **Important distinction:** A video is often called a *sequence of 2D signals/images*, but mathematically, when the temporal dimension is included, it is a **3D spatiotemporal signal**.

Video = Sequence of Frames

A video is a series of images (**frames**) displayed one after another over **time**.



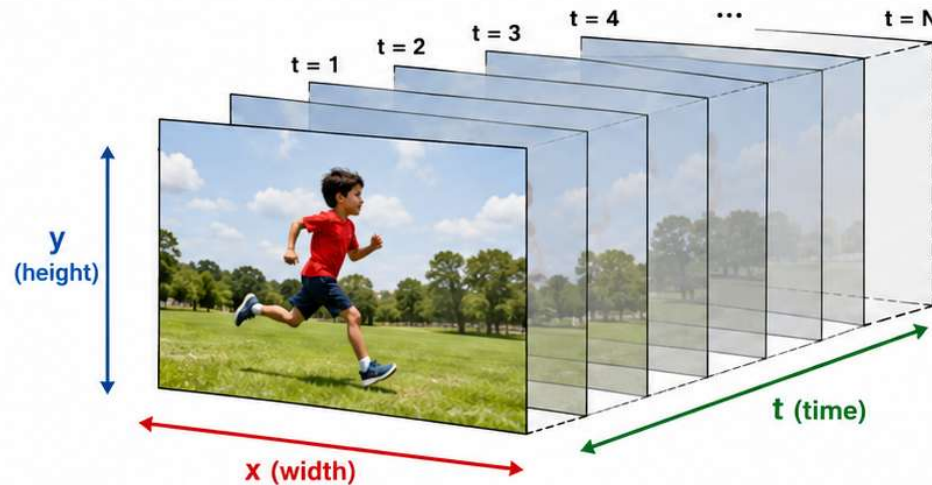
Key Idea

A video $V(x, y, t)$ has two spatial dimensions (x, y) and one time dimension (t).

x : horizontal (width)

y : vertical (height)

t : time (frame index)



Mathematical Representation

Video: $V(x, y, t)$

where

$x = 1, 2, \dots, W$ (**width**)

$y = 1, 2, \dots, H$ (**height**)

$t = 1, 2, \dots, N$ (**time/frame number**)

Example

A video of resolution 1920×1080 with 100 frames is a 3D signal of size $1920 \times 1080 \times 100$.

VIDEO PROCESSING

Video processing involves manipulating video frames (images) over time to extract information, improve quality, or derive meaningful insights.

1. VIDEO ACQUISITION

Capture video using camera / sensor



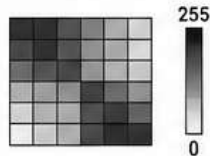
2. SAMPLING

Convert continuous video into discrete frames (spatial sampling)



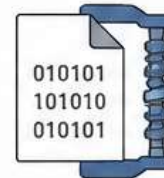
3. QUANTIZATION

Convert pixel values to finite levels (intensity quantization)



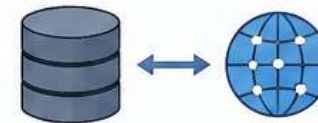
4. COMPRESSION

Reduce data for storage and transmission



5. STORAGE / TRANSMISSION

Store in memory / disk or transmit over network



6. DISPLAY / PLAYBACK

Reconstruct and display video to the user



KEY VIDEO PROCESSING OPERATIONS

SPATIAL DOMAIN PROCESSING

Operate on individual frames (images)

- Filtering / Smoothing
- Enhancement
- Edge Detection

Original Frame



Filtered Frame



TEMPORAL DOMAIN PROCESSING

Operate over time by analyzing pixel changes across frames

- Frame Differencing
- Motion Estimation

Frame t



Frame t+1



Frame Difference

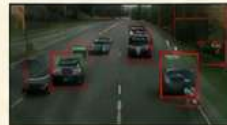


MOTION ANALYSIS

Detect, track and analyze moving objects

- Object Detection
- Object Tracking

Detection



Tracking



VIDEO ENHANCEMENT

Improve visual quality of video

- Denoising
- Contrast Enhancement
- Super-resolution

Low Quality Frame



Enhanced Frame



VIDEO SEGMENTATION

Partition video into meaningful regions or objects

- Foreground / Background
- Semantic Segmentation

Original Frame



Segmented Output



VIDEO COMPRESSION

Reduce redundancy to save storage/bandwidth

- Intra-frame (Spatial)
- Inter-frame (Temporal)

Original Frames



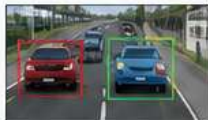
Compressed Bitstream

01010101011010101010...

APPLICATIONS



Surveillance & Security



Autonomous Vehicles (ADAS)



Telemedicine



Video Editing & Post-production



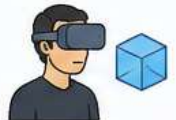
Sports Analytics



Traffic Monitoring



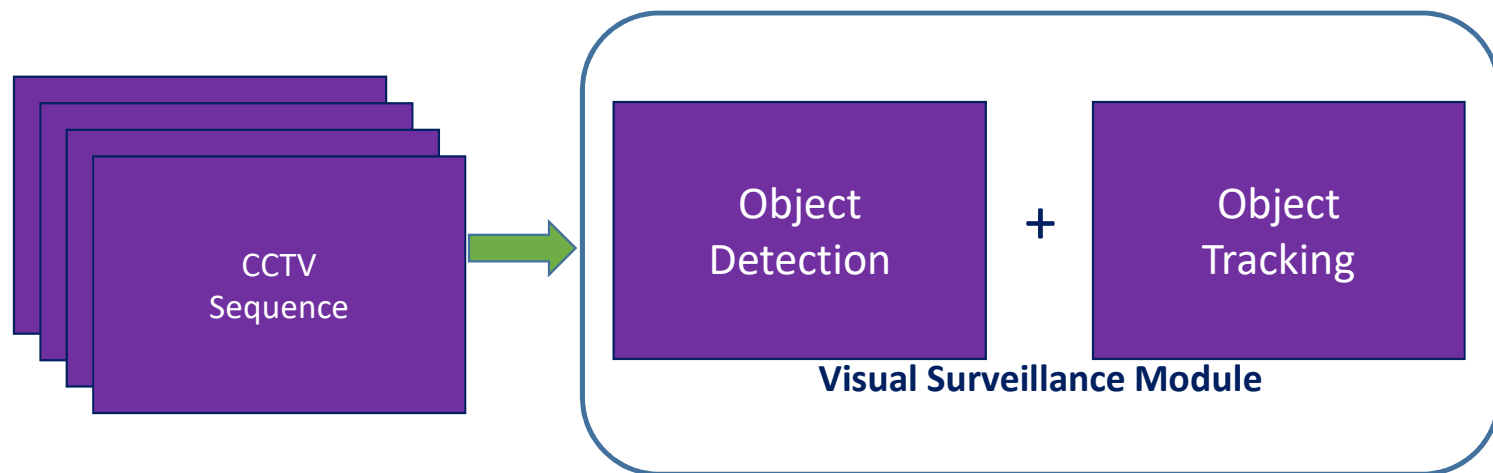
Media Streaming



VR / AR / Entertainment

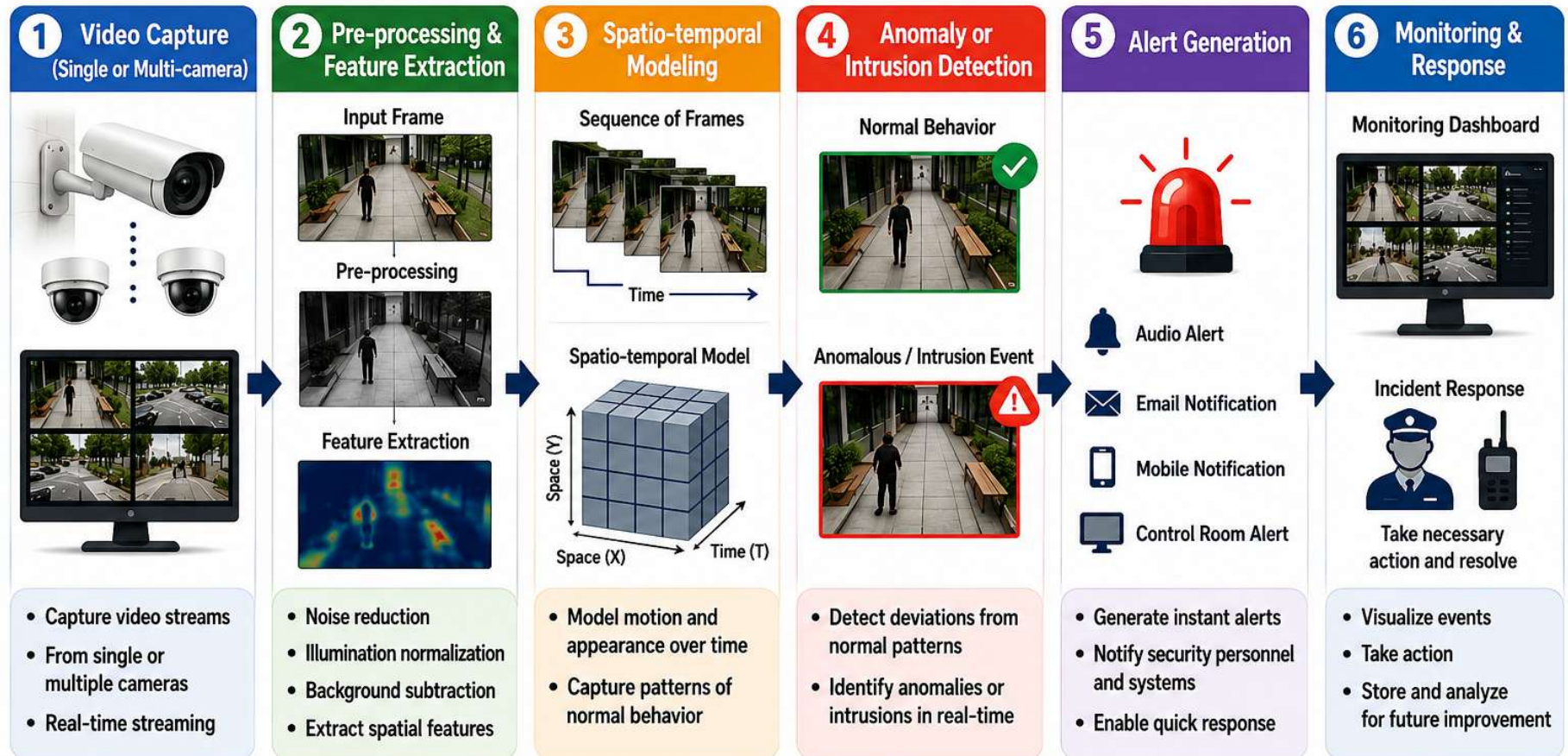
Aspect	Video Processing	Video Analytics
Purpose	Improve, transform, or manipulate video	Extract meaningful information and make decisions
Focus	Pixels and frames	Objects, events, and behavior
Typical input	Raw video	Raw or processed video
Typical operations	Filtering, enhancement, compression, frame extraction, stabilization	Object detection, tracking, classification, activity recognition
Output	Enhanced/modified video or processed frames	Information, alerts, events, statistics, decisions
AI requirement	Not necessarily	Frequently uses AI/ML/DL
Example	Remove noise from a video	Detect a person entering a restricted area
Another example	Resize a 4K video to HD	Count vehicles on a road
Main question	"How can I process/modify this video?"	"What is happening in this video?"

Visual Surveillance System



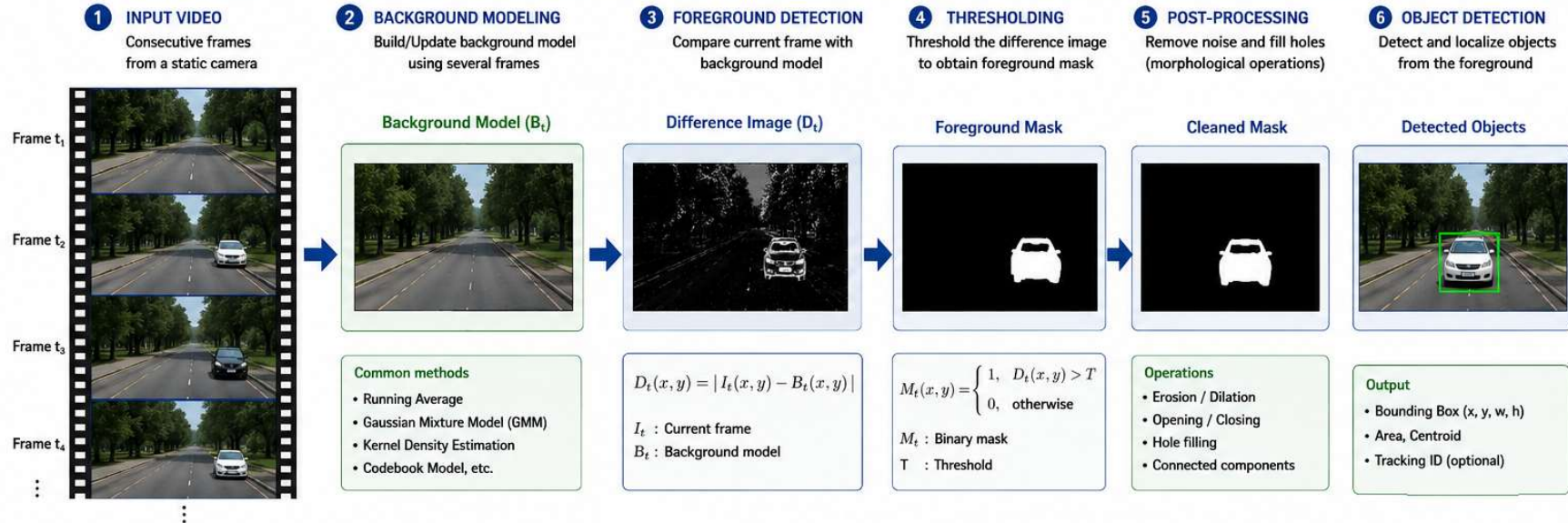
Intelligent Surveillance Pipeline

From video capture to intelligent alert generation



OBJECT DETECTION BY BACKGROUND MODELING

Build a model of the static background and detect objects as regions that differ from the background.



BACKGROUND MODELING (Example: Running Average)

Update rule:

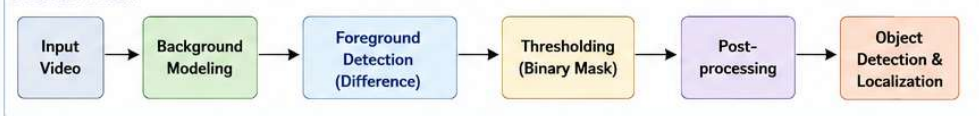
$$B_t(x, y) = (1 - \alpha)B_{t-1}(x, y) + \alpha I_t(x, y),$$

where $\alpha \in (0, 1]$ is the learning rate.

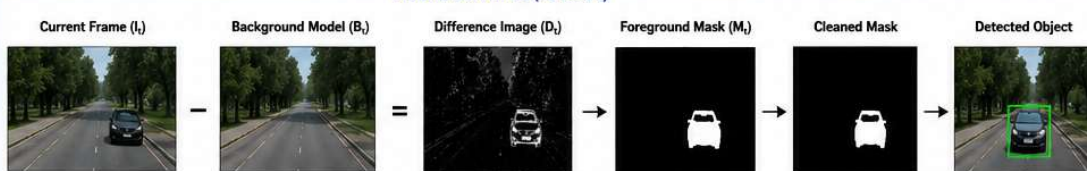
Benefits

- Adapts slowly to gradual changes
- Robust to illumination variations
- Works in real-time

FLOW DIAGRAM



VISUAL EXAMPLE (At time t)

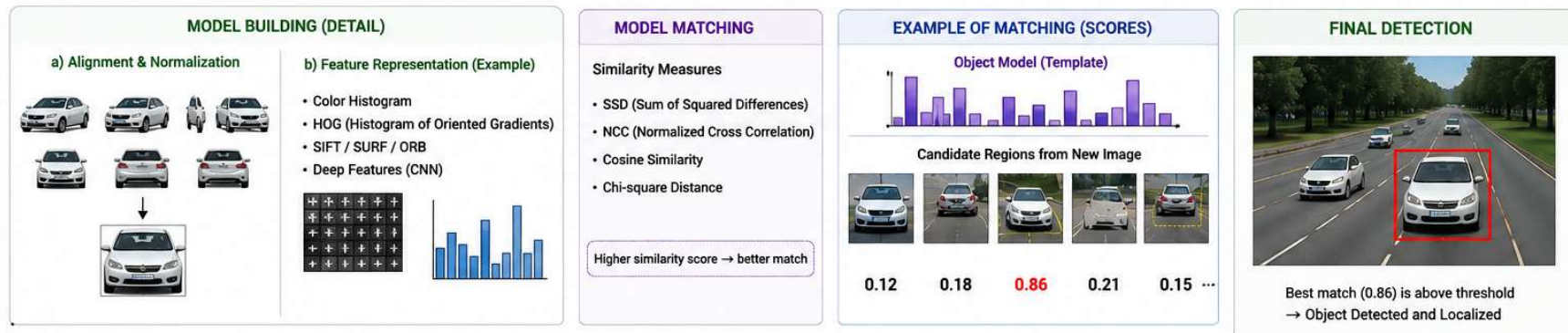
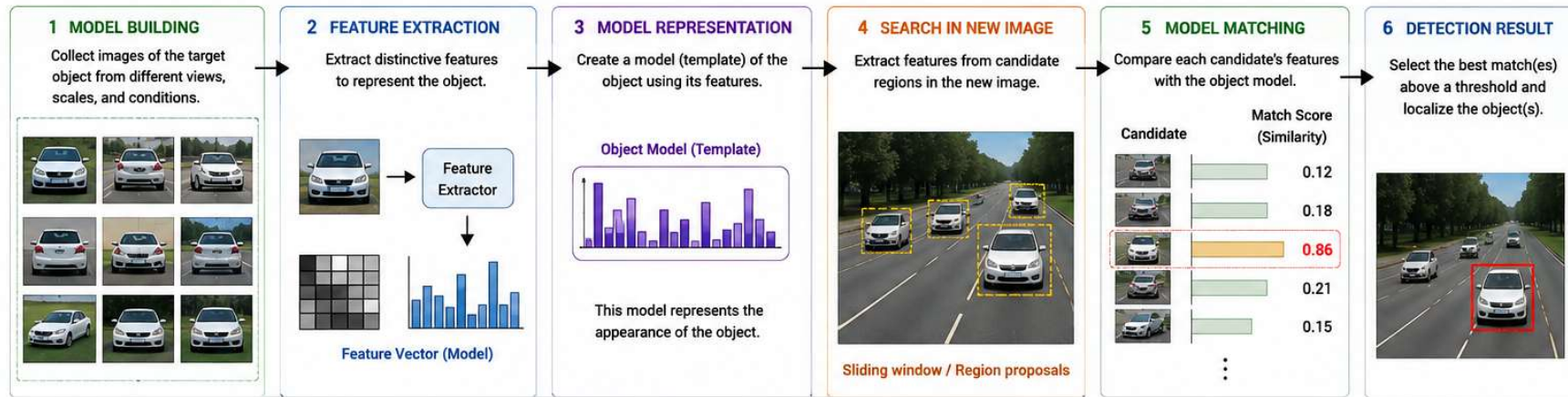


APPLICATIONS



OBJECT DETECTION BY OBJECT MODELING

Object detection is performed by creating a model (representation) of the target object and searching for the best match of this model in the new image.



KEY POINTS



Build a model of the target object using training images.



Represent the object with discriminative features.



Scan the new image and compare each region with the model.

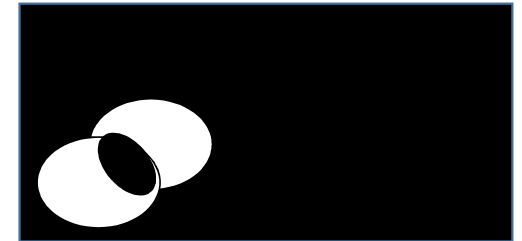
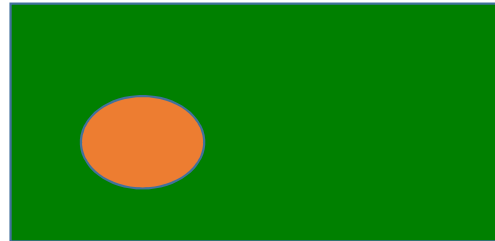
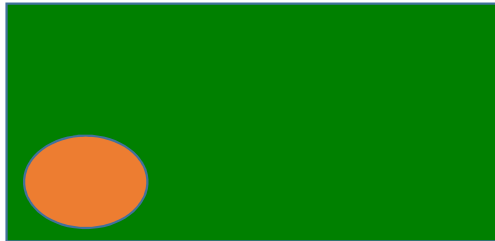
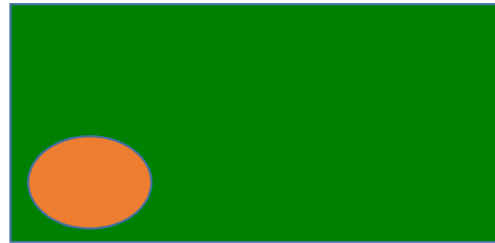


Pick regions with high similarity scores.



Provides localization and can detect multiple objects.

Object Detection



MATHEMATICAL FORMULATION

Let $I_t(x, y)$ be the intensity of pixel (x, y) at time t .

1. Frame Difference:

$$D_t(x, y) = | I_t(x, y) - I_{t-1}(x, y) |$$

2. Thresholding:

$$B_t(x, y) = \begin{cases} 1, & \text{if } D_t(x, y) \geq T \\ 0, & \text{otherwise} \end{cases}$$

where T is the threshold.

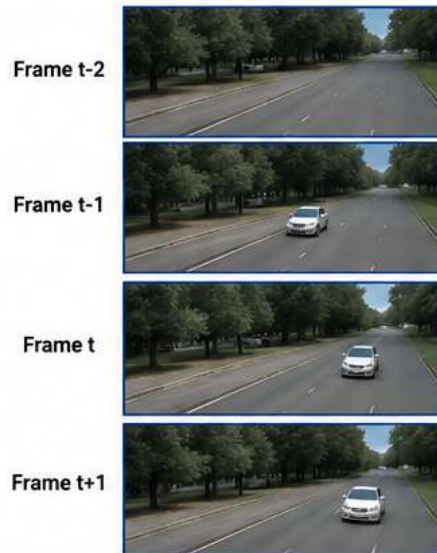
3. Post-processing: Apply morphological operations (opening/closing) to remove noise.

FRAME DIFFERENCING FOR OBJECT DETECTION

Frame differencing detects moving objects by subtracting consecutive frames.

1 Input Video Frames

Consecutive frames from a static camera.



2 Frame Differencing

Subtract current frame from previous frame.

$$\text{Diff}(t) = | \text{Frame } t - \text{Frame } t-1 |$$



$$\text{Diff}(t+1) = | \text{Frame } t+1 - \text{Frame } t |$$



3 Thresholding

Apply a threshold to obtain binary (foreground) image.

$$\text{Binary}(t)$$



$$\text{Binary}(t+1)$$



4 Post-processing

Remove noise and fill holes (morphological operations).

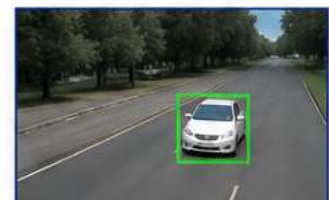
$$\text{Cleaned Foreground}$$



5 Object Detection

Detect and localize moving objects.

$$\text{Detected Object } (t)$$



$$\text{Detected Object } (t+1)$$



FLOW DIAGRAM



KEY POINTS

- Works well for static background and moving objects.
- Sensitive to illumination changes and camera noise.
- Simple, fast and effective for real-time applications.
- Often used as a preprocessing step for advanced methods.

APPLICATIONS



Surveillance



Traffic Monitoring



Pedestrian
Detection

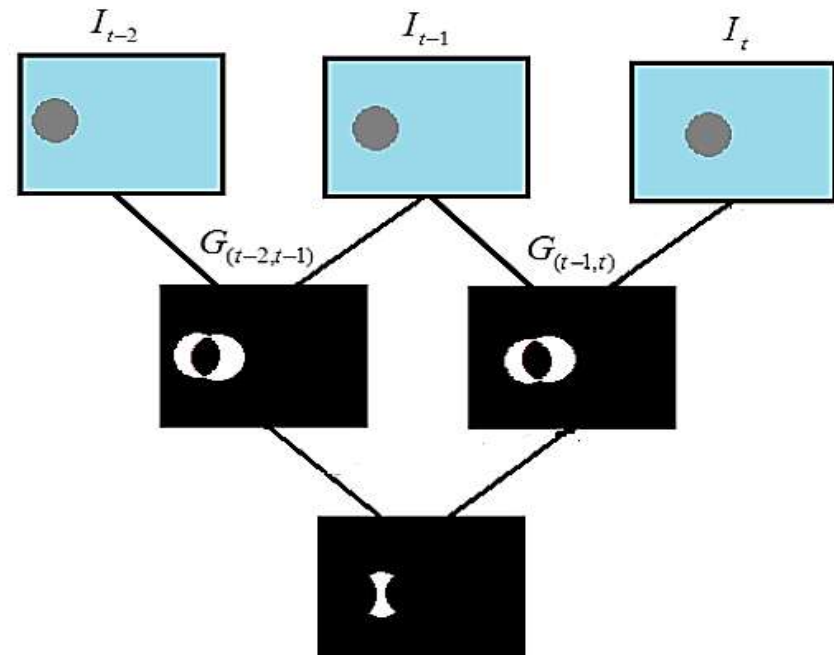
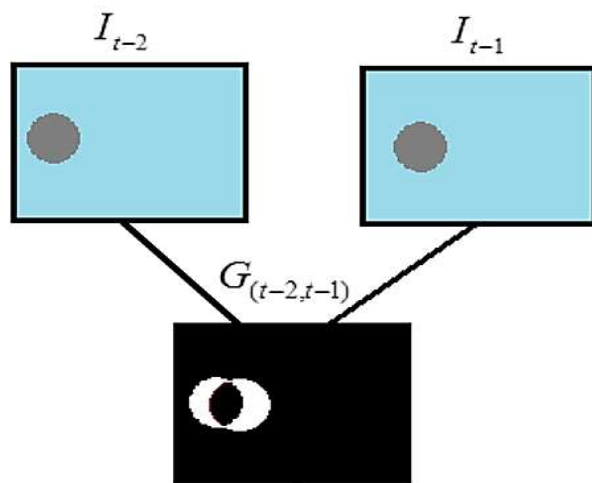


Industrial
Automation

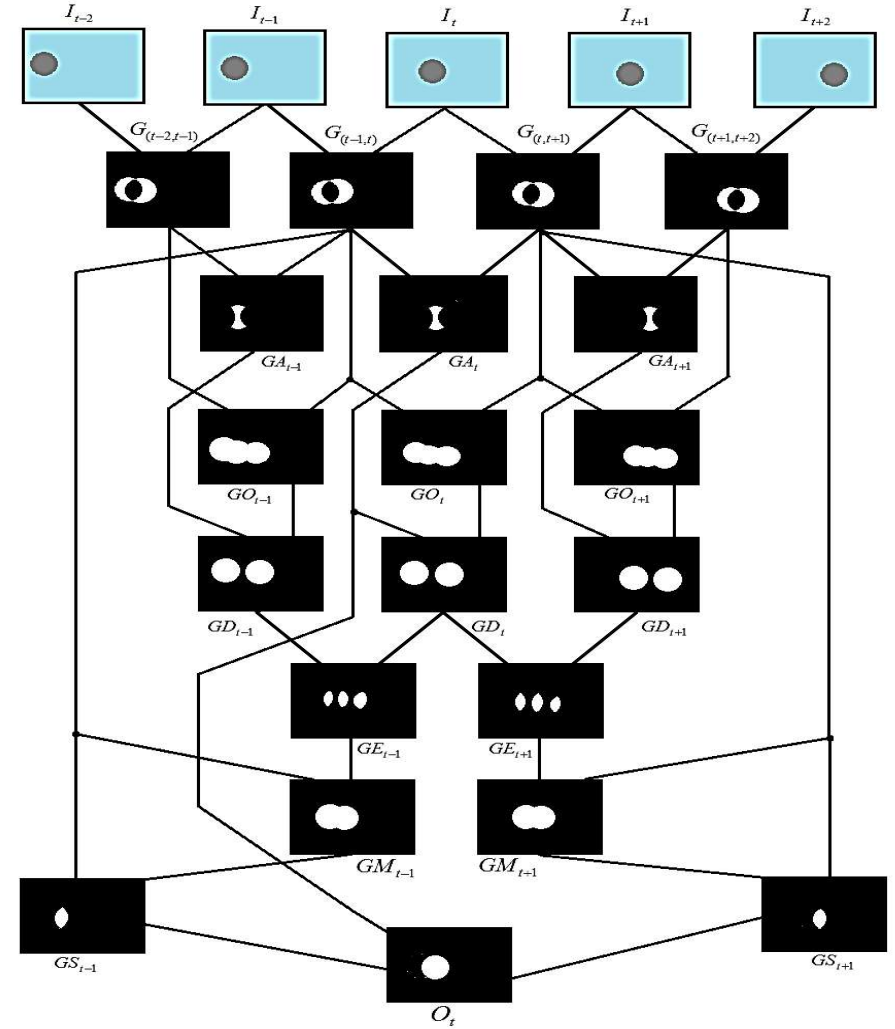


Smart
Environments

Detection of the Intruder

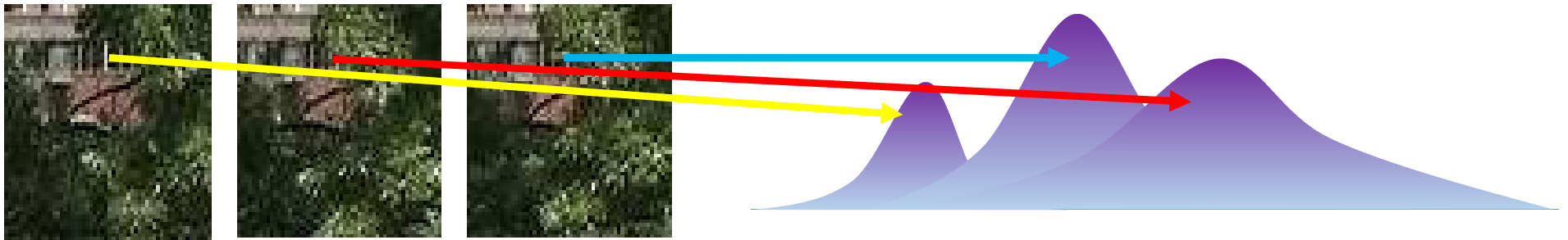


1. Take 5 consecutive frames, say $I_{t-2}, I_{t-1}, I_t, I_{t+1}$ and I_{t+2} .
2. Compute $D_{(t-2,t-1)}, D_{(t-1,t)}, D_{(t,t+1)}$, and $D_{(t+1,t+2)}$ using (7).
3. (1) Compute $G_{(t-2,t-1)}, G_{(t-1,t)}, G_{(t,t+1)}$, and $G_{(t+1,t+2)}$ using (8).
(2) Compute $GA_{t-1} = G_{(t-2,t-1)} \cdot G_{(t-1,t)}$,
 $GA_t = G_{(t-1,t)} \cdot G_{(t,t+1)}$,
and $GA_{t+1} = G_{(t,t+1)} \cdot G_{(t+1,t+2)}$.
5. Compute $GO_{t-1} = G_{(t-2,t-1)} + G_{(t-1,t)}$,
 $GO_t = G_{(t-1,t)} + G_{(t,t+1)}$,
and $GO_{t+1} = G_{(t,t+1)} + G_{(t+1,t+2)}$.
6. Compute $GD_{t-1} = |GO_{t-1} - GA_{t-1}|$,
 $GD_t = |GO_t - GA_t|$, and
 $GD_{t+1} = |GO_{t+1} - GA_{t+1}|$.
7. Compute $GE_{t-1} = GD_{t-1} \cdot GD_t$,
and $GE_{t+1} = GD_t \cdot GD_{t+1}$.
8. Compute $GM_{t-1} = G_{(t-1,t)} + GE_{t-1}$, and
 $GM_{t+1} = G_{(t,t+1)} + GE_{t+1}$.
9. Compute $GS_{t-1} = |GM_{t-1} - G_{(t-1,t)}|$, and
 $GS_{t+1} = |GM_{t+1} - G_{(t,t+1)}|$.
10. Compute $O_t = GS_{t-1} + GA_t + GS_{t+1}$, where O_t is the change detection mask for the I_t frame.



D. K. Rout, P. G. Bhat, T. Veerakumar, B. N. Subudhi, and S. Chaudhury. "A novel five-frame difference scheme for local change detection in underwater video." In *IEEE International Conference on Image Information Processing*, pp. 1-6. 2017.

Gaussian Mixture Model for Background subtraction

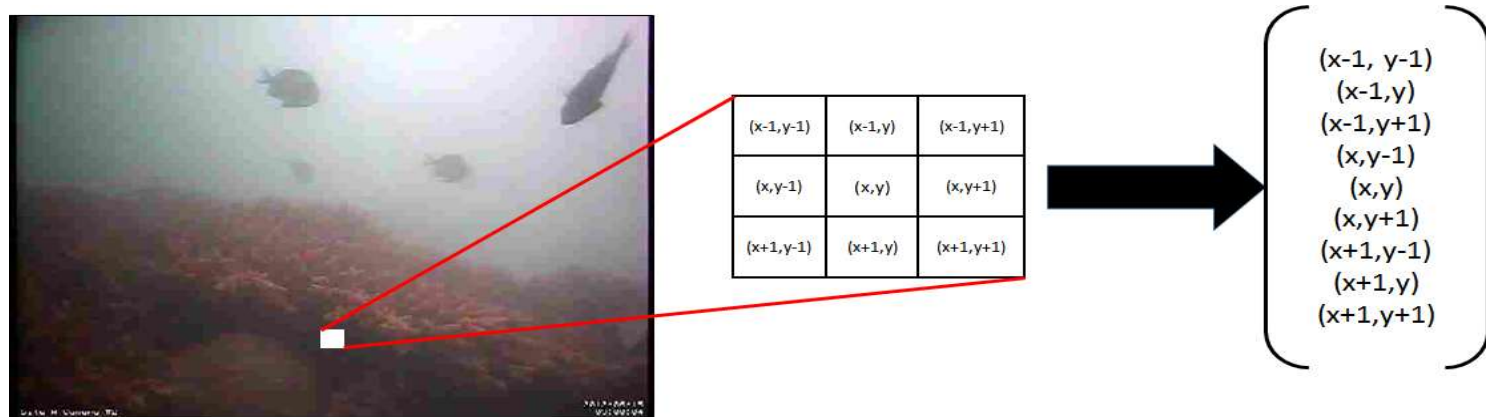


- Model the particular pixel as a mixture of Gaussians to represent multiple background.
- Multiple adaptive Gaussians are necessary to cope with non-static background, illumination variation, underwater vegetation etc.
- Pixel values that do not fit the background distributions are considered foreground.

D. K. Rout, B. N. Subudhi, T. Veerakumar, & S. Chaudhury, "Spatio-contextual Gaussian mixture model for local change detection in underwater video." *Expert Systems with Applications*, 97, 117-136, 2018

Wronskian Change Detection Model

Wronskian change detector is a vector image model to detect changes from one frame to another.



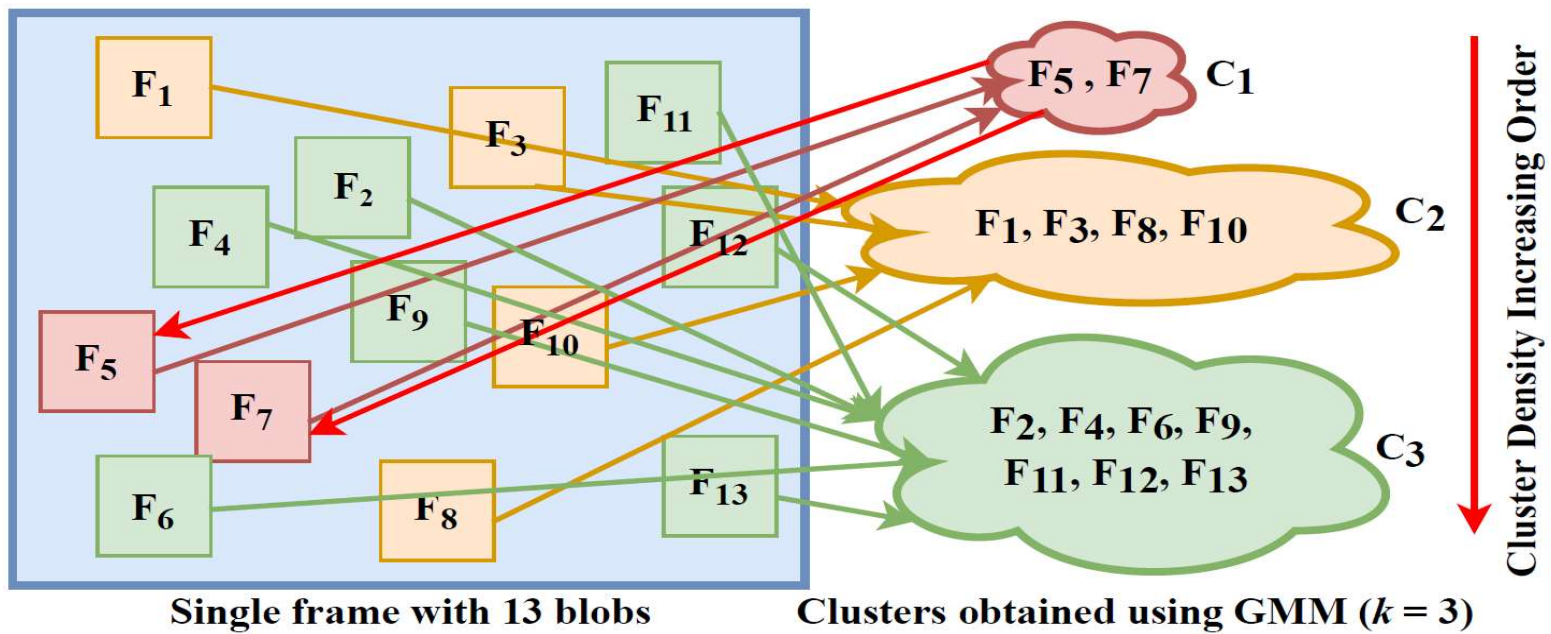
We can incorporate the spatio-contextual property of a pixel in Wronskian change detection model by including the region of support.

$$|W| = \frac{1}{n} \sum_{i=1}^n \left(\left(\frac{\vec{F}_t(x, y)_i}{\vec{F}_{t-1}(x, y)_i} \right)^2 - \frac{\vec{F}_t(x, y)_i}{\vec{F}_{t-1}(x, y)_i} \right)$$

- E. Durucan, and T. Ebrahimi, "Change detection and background extraction by linear algebra," *Proceedings of the IEEE (Special Issue on Advanced Video Surveillance)*, 89(10), 1368–1381, 2001.
- D. K. Rout, B. N. Subudhi, T. Veerakumar, and S. Chaudhury, "Spatio-contextual Gaussian mixture model for local change detection in underwater video." *Expert Systems with Applications*, 97, 117-136, 2018

Intruder Detection and Tracking in Crowd

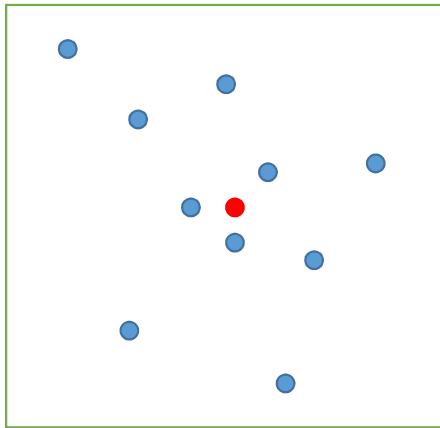
Clustering and reassigning step:



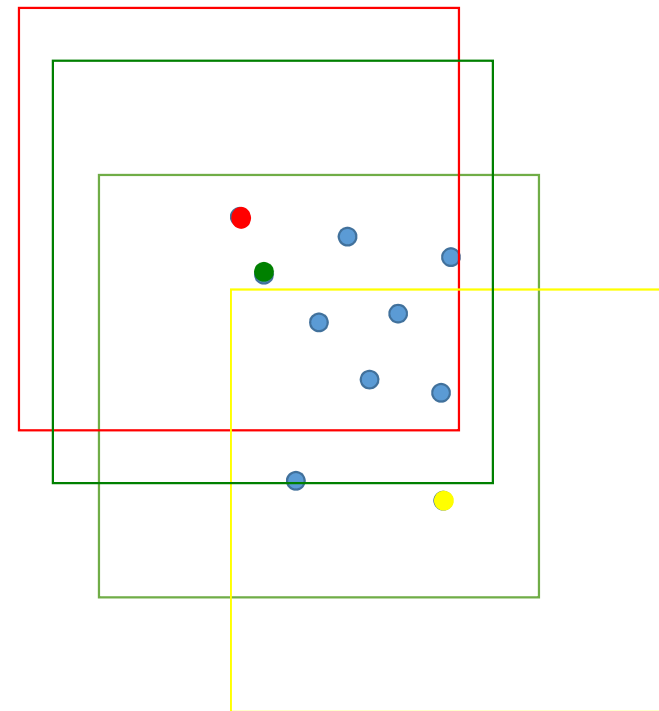
D. K. Rout, B. N. Subudhi, T. Veerakumar, and S. Chaudhury, "Prominent object detection in underwater environment using a dual-feature framework". In *Global IEEE Oceans 2020: Singapore-US Gulf Coast*, pp. 1-5, 2020

Particle filter Framework for Tracking

$$S_{t-1} = \{x_c, y_c, v_x, v_y, a_x, a_y\}$$



Time instant 't-1'



Time instant 't'

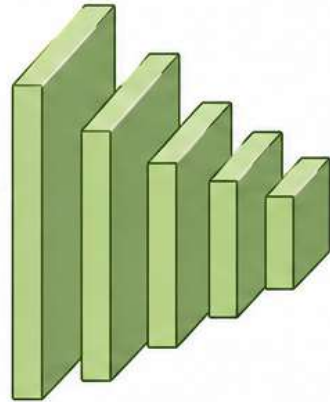
Rout, D. K., Subudhi, B. N., Veerakumar, T., & Chaudhury, S. Walsh–Hadamard-kernel-based features in particle filter framework for underwater object tracking. *IEEE Transactions on Industrial Informatics*, 16(9), 5712-5722, 2019

General Deep Learning Architecture for Object Detection

1. Input Image

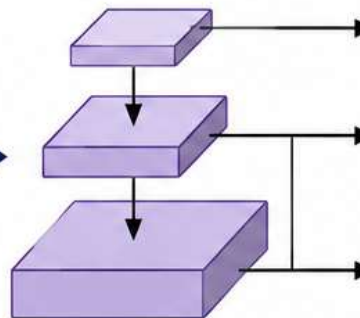


2. Backbone
(Feature Extraction)



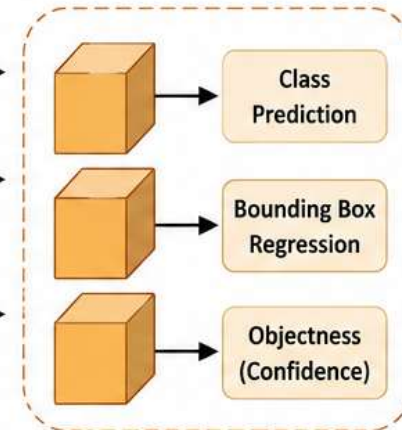
CNN (e.g., ResNet, CSPDarknet, EfficientNet) extracts rich hierarchical features from the input image.

3. Neck
(Feature Aggregation)



FPN / PAN / BiFPN aggregates multi-scale features to detect objects of different sizes.

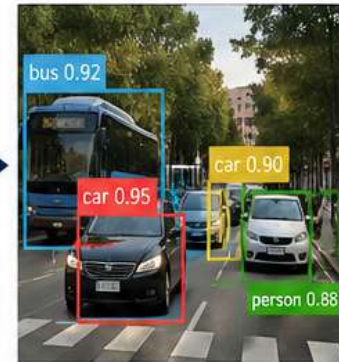
4. Detection Head
(Prediction)



For each spatial location / anchor / query, predicts:

- Object class (softmax)
- Bounding box (x, y, w, h)
- Objectness score (confidence)

5. Output
(Detections)



FPN: Feature Pyramid Network, PAN: Path Aggregation Network, BiFPN: Bidirectional Feature Pyramid Network

AI Models Used

- Object detection: YOLO, Faster R-CNN
- Tracking: DeepSORT
- Temporal modeling: 3D CNNs, LSTM, Transformers
- Anomaly detection: Autoencoders, GANs

How it Works

1. **Input Image:** The raw image is resized and fed to the network.
2. **Backbone:** Convolutional layers extract hierarchical feature maps.
3. **Neck:** Combines low-level and high-level features to retain spatial information and strong semantics.
4. **Detection Head:** For each location/anchor/query, the model predicts class, bounding box and objectness score.
5. **Output:** Predictions are filtered using confidence threshold and Non-Maximum Suppression (NMS) to produce the final set of detections.

Acronym	Full form	Main purpose
FPN	Feature Pyramid Network	Combines high-level semantic features with lower-level spatial features to detect objects at different scales.
PAN	Path Aggregation Network	Enhances FPN by adding a bottom-up path, improving the flow of localization and low-level information.
BiFPN	Bidirectional Feature Pyramid Network	Uses repeated top-down and bottom-up feature fusion with learnable weights for efficient multi-scale feature aggregation.

WORKING OF A CNN MODEL

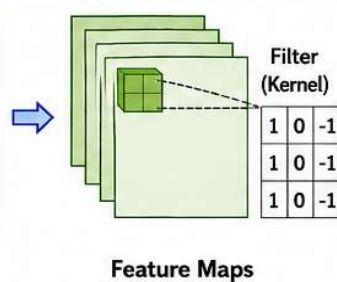
A Convolutional Neural Network (CNN) automatically learns hierarchical features from an input image and uses them to classify the image.

1. INPUT IMAGE

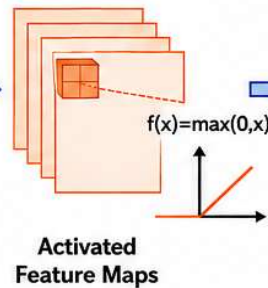


Input
(128 × 128 × 1)

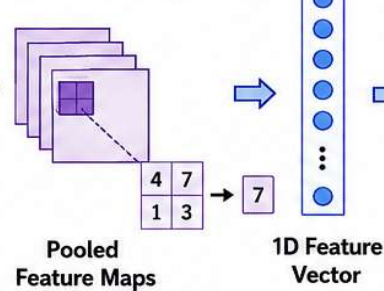
2. CONVOLUTION



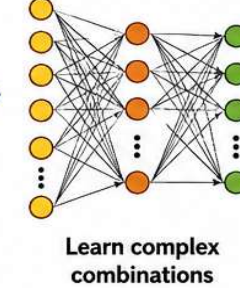
3. ACTIVATION (ReLU)



4. POOLING (Max Pooling)



6. FULLY CONNECTED (LAYERS)



7. OUTPUT (SOFTMAX)

Class Probabilities

Fractured	0.92
Non-Fractured	0.08

(Sum = 1)

HOW IT WORKS

1. Input Image

The model takes an input image (e.g., X-ray of hand) and converts it into a numerical matrix.

0	0	1	...
0	1	2	...
1	2	3	...
...	...	...	...

2. Convolution

A small filter (kernel) slides over the image and performs element-wise multiplication and summation to produce a feature map that detects patterns like edges, lines, textures.

$$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

3. Activation (ReLU)

ReLU adds non-linearity by replacing all negative values in the feature map with zero. This helps the model learn more complex patterns.

$$f(x) = \max(0, x)$$

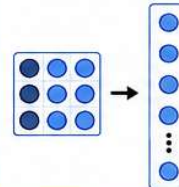
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$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \rightarrow 4$$

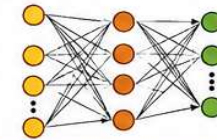
5. Flatten

The 2D feature maps are converted into a 1D vector so that it can be fed into the fully connected layers.



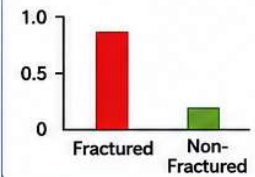
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Key Point: CNNs automatically learn hierarchical features:

Edges → Textures → Parts → Objects → Classification

Benefits of CNN

- ✓ Automatic feature learning (no manual feature extraction)
- ✓ Works well with images and visual data
- ✓ High accuracy for complex tasks

ReLU : Rectified Linear Unit

CNN : Convolutional Neural Network

Softmax : Converts scores to probabilities

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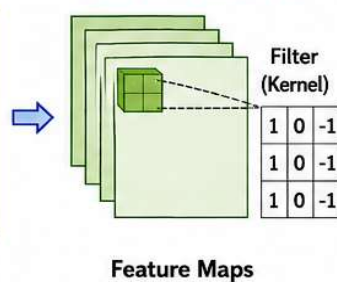
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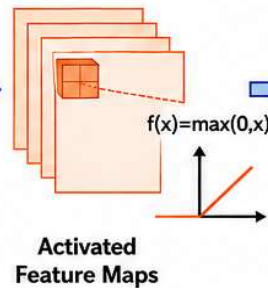


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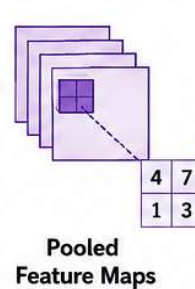
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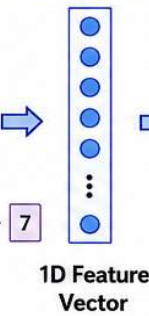
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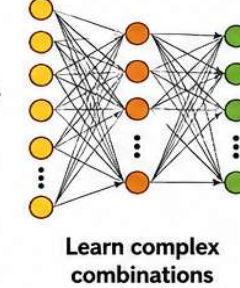
4. POOLING (Max Pooling)



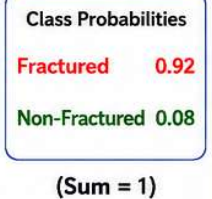
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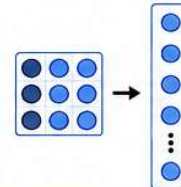
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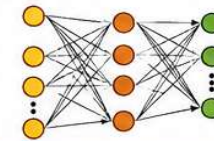
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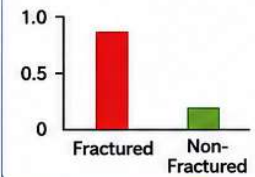
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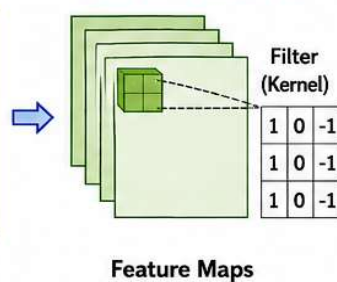
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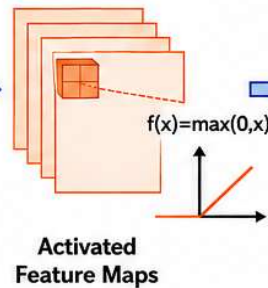


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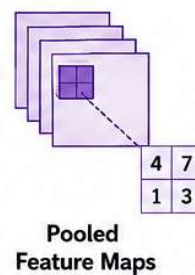
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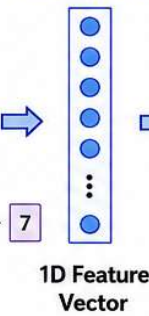
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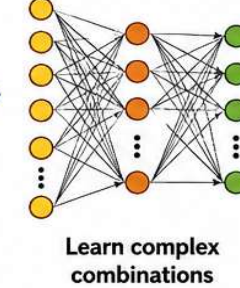
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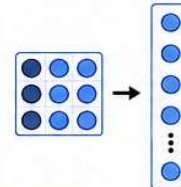
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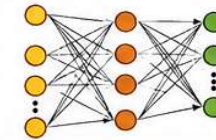
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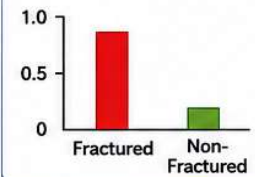
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- S. Jayaraman, S. Esakkirajan, and T. Veerakumar. *“Digital Image Processing”*, TMH Publication, New Delhi, 2011
- Rafael C. Gonzalez, and Richard E. Woods. *“Digital Image Processing”*. Pearson Education India, New Delhi, 2009.
- Anil K. Jain. *“Fundamentals of Digital Image Processing”*. Prentice-Hall, Inc. India, 1989.

Thank You